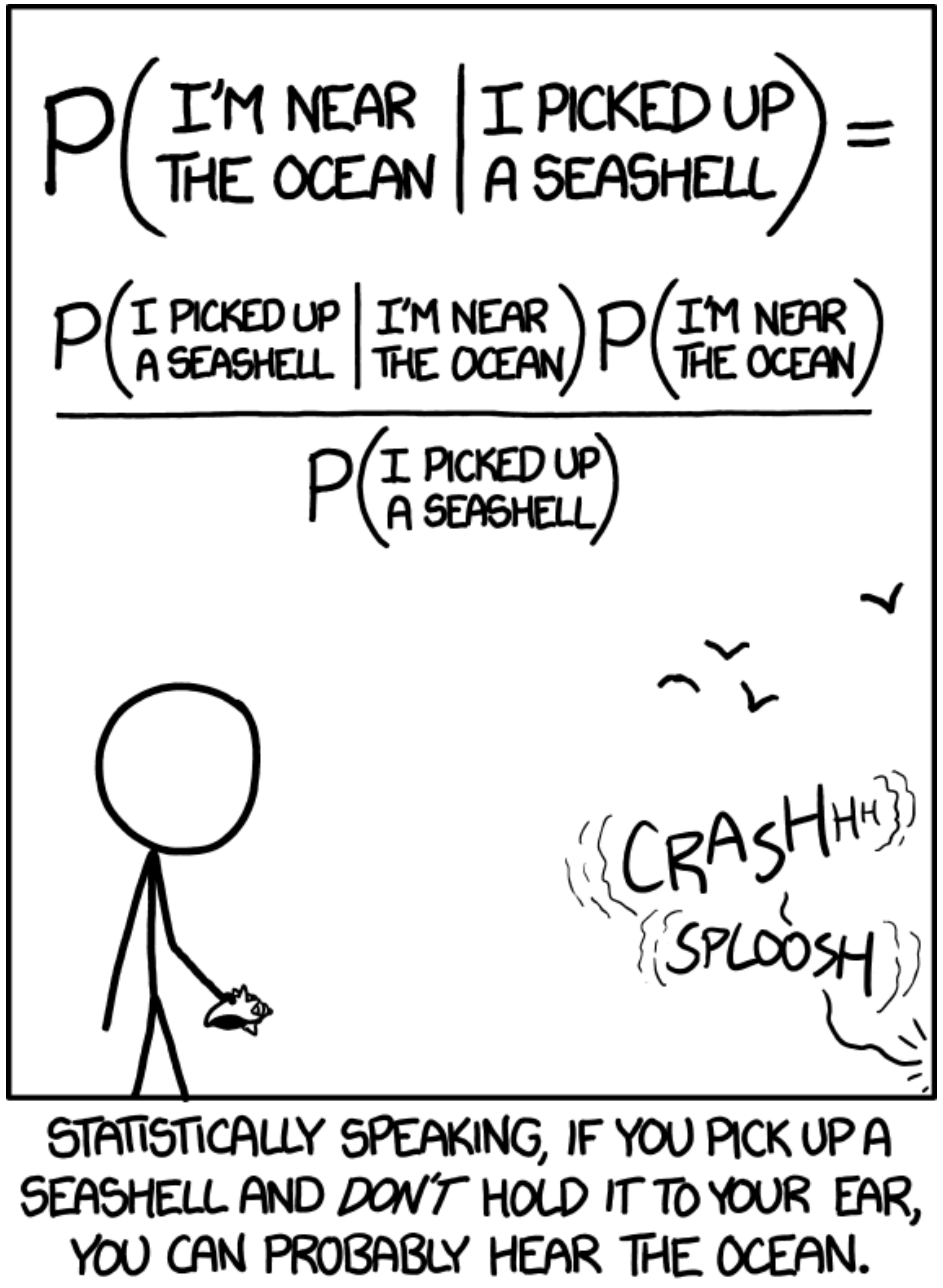
Bayesian data analysis 3

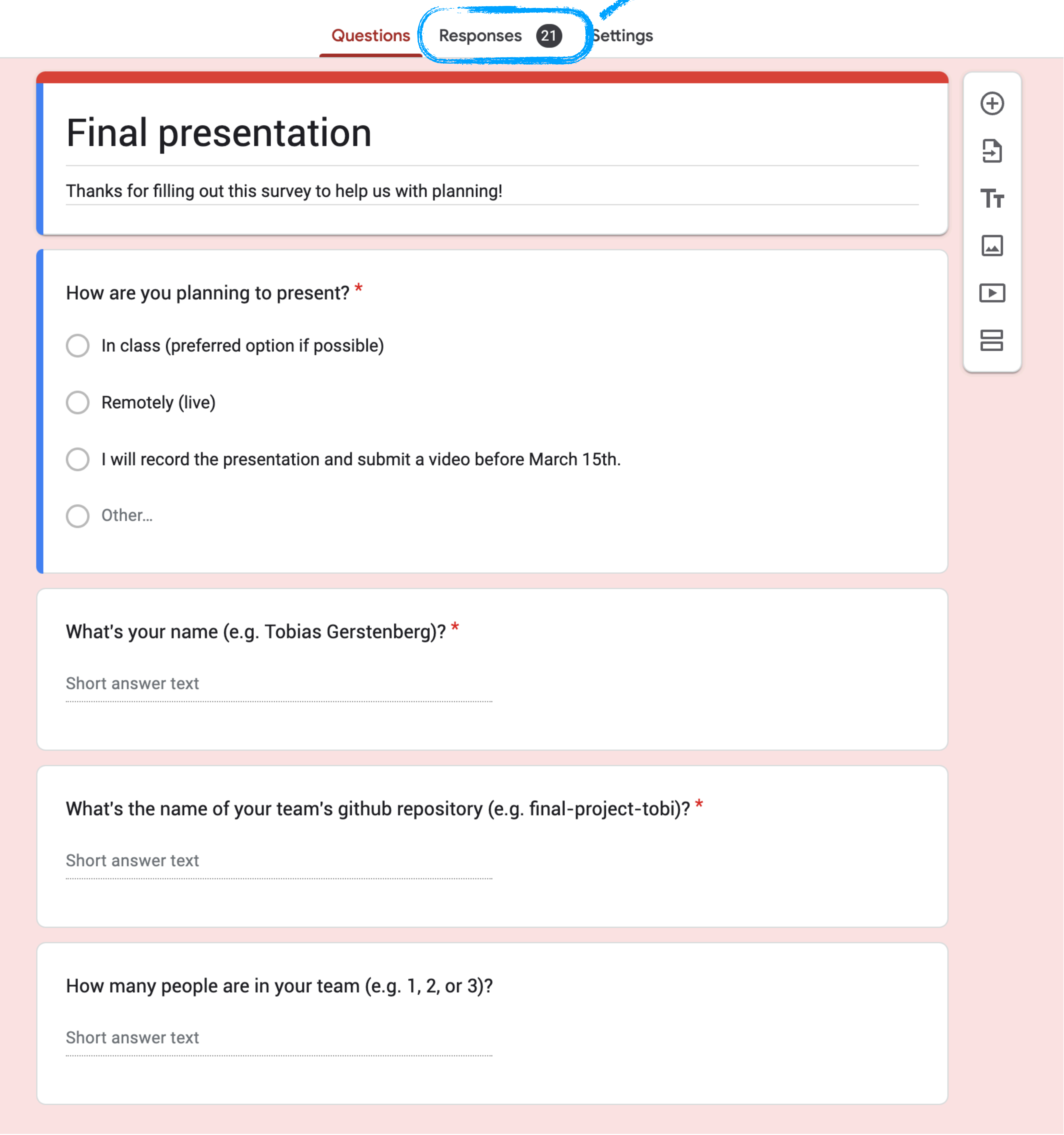


03/03/2026

Logistics

Final presentation survey

goal: everyone!


Questions
Responses 21
Settings

Final presentation

Thanks for filling out this survey to help us with planning!

How are you planning to present? *

- In class (preferred option if possible)
- Remotely (live)
- I will record the presentation and submit a video before March 15th.
- Other...

What's your name (e.g. Tobias Gerstenberg)? *

Short answer text

What's the name of your team's github repository (e.g. final-project-tobi)? *

Short answer text

How many people are in your team (e.g. 1, 2, or 3)?

Short answer text

https://tinyurl.com/psych252presentation26

Plan for today

- Quick recap
- Doing Bayesian data analysis **with BRMS**
  - Testing hypotheses
  - Model evaluation
  - Reporting results
- Some more examples
  - Sleep data
  - Titanic data
- Going beyond

Quick recap

Quick recap: What affects the posterior?

What affects the posterior?

1. the prior over hypotheses
2. the likelihood of the data given each hypothesis

$$\underbrace{p(H|D)}_{\textbf{Posterior}} = \frac{\overbrace{p(D|H)}^{\textbf{Likelihood}} \cdot \overbrace{p(H)}^{\textbf{Prior}}}{\underbrace{p(D)}_{\textbf{Normalizing constant}}}$$

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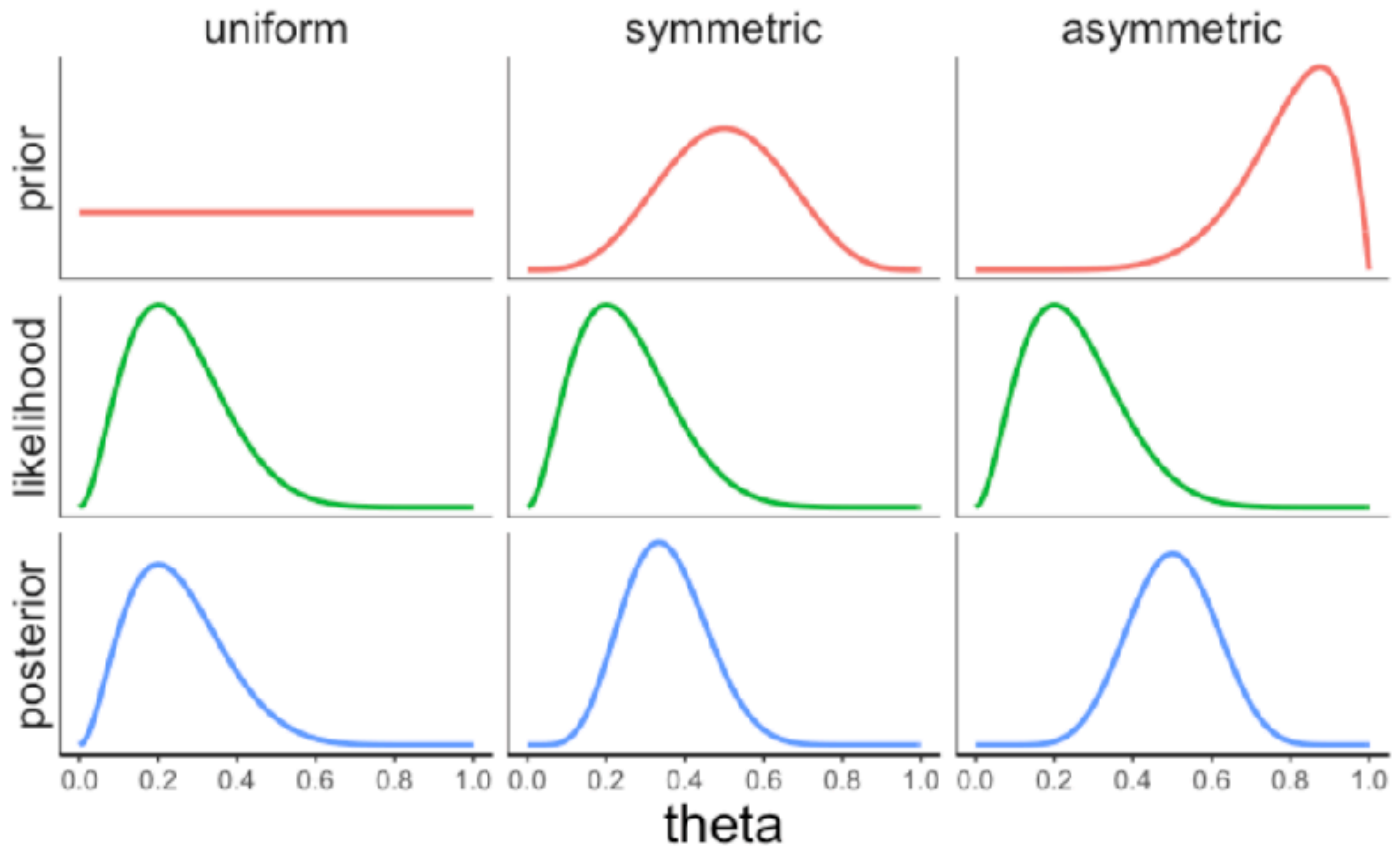
The effect of the **prior**

same data, different priors



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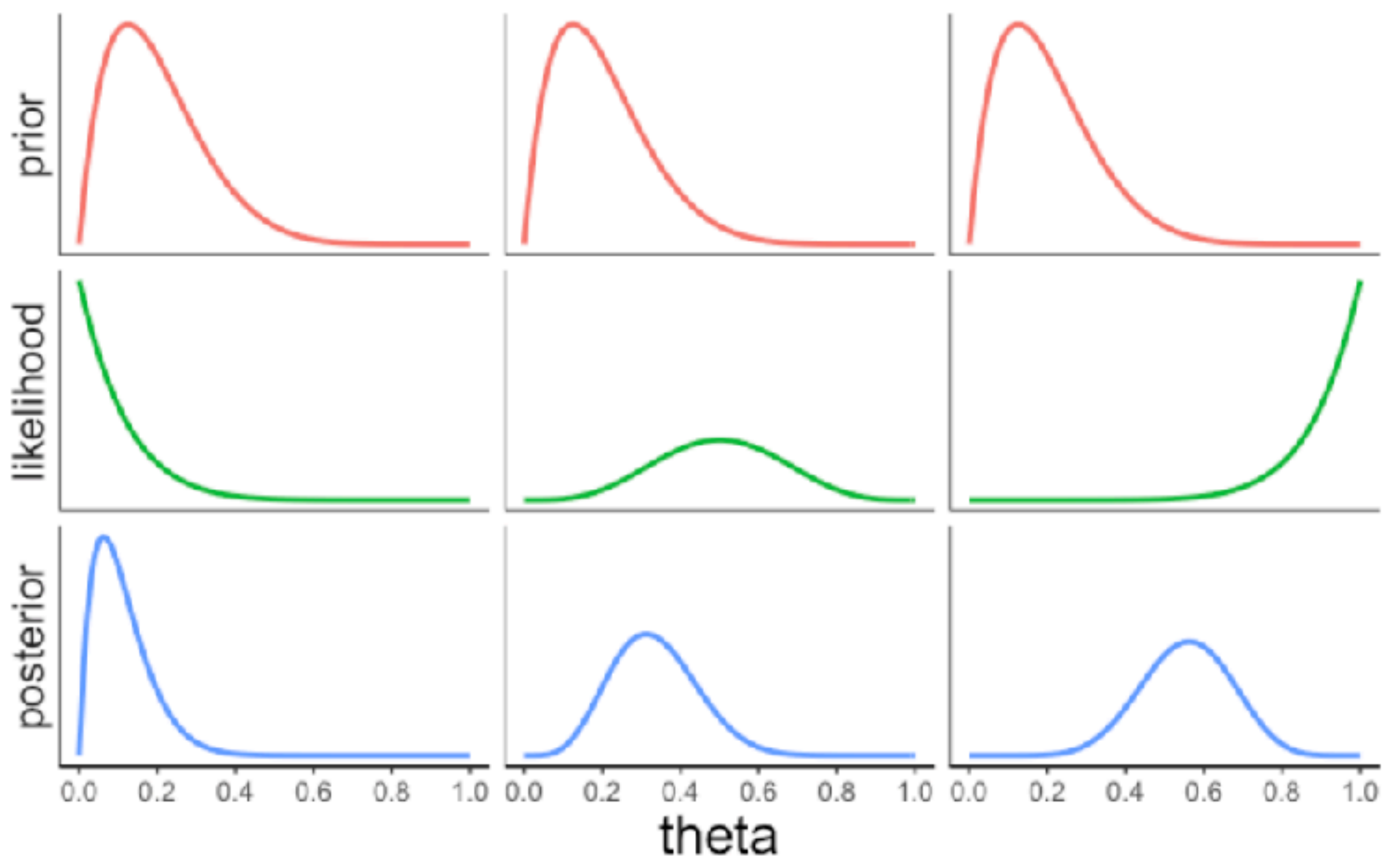
The effect of the **likelihood**

same prior, different data



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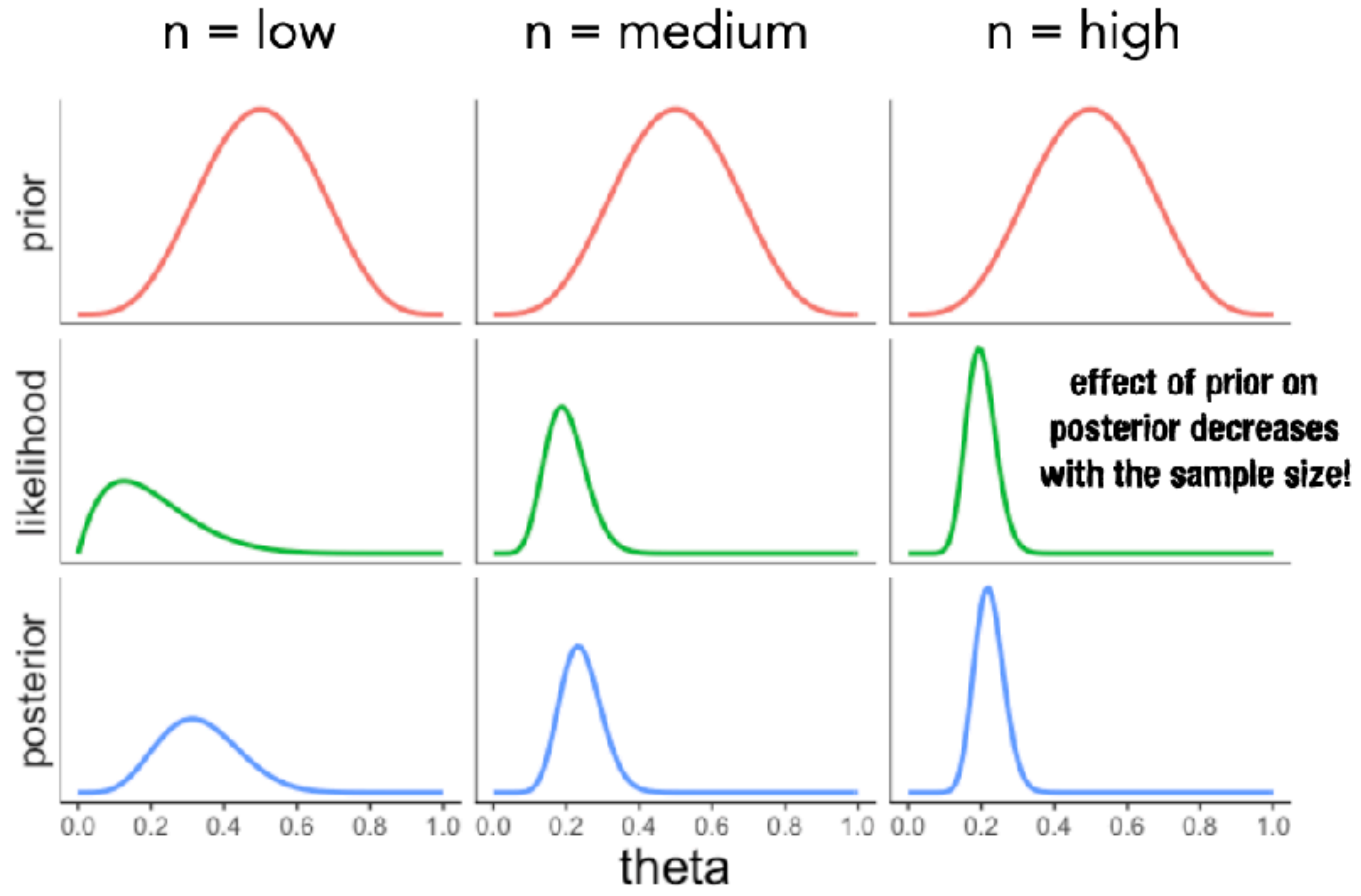
The effect of **sample size**



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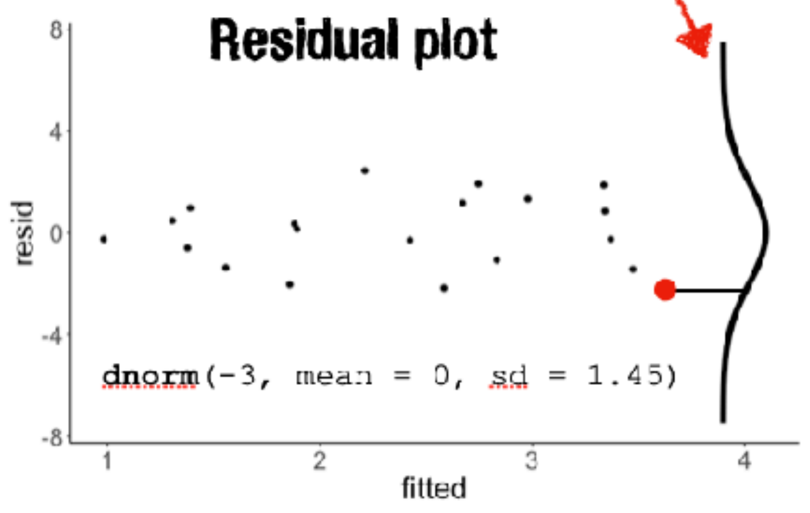
Quick recap: **likelihood**, prior, inference

Likelihood

Gaussian distribution

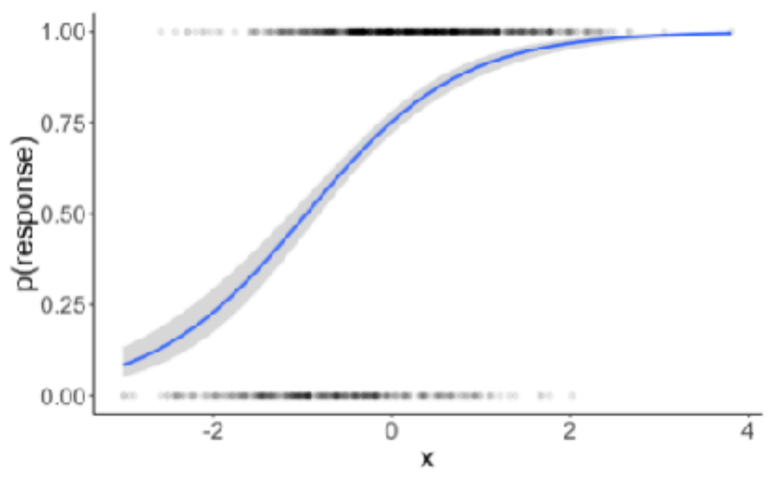
$$Y_i = b_0 + b_1 \cdot x_i + e_i$$

$$e_i \sim \mathcal{N}(0, \sigma)$$

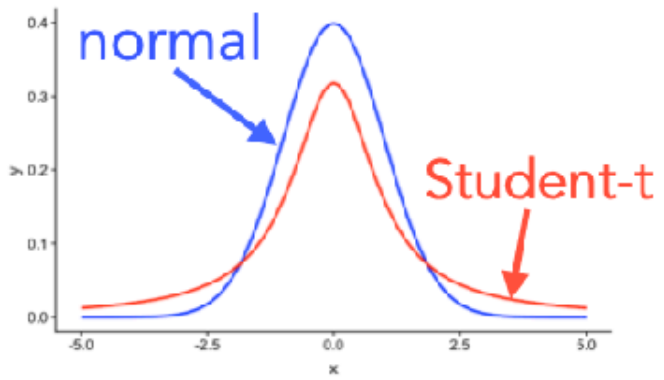


Binomial distribution

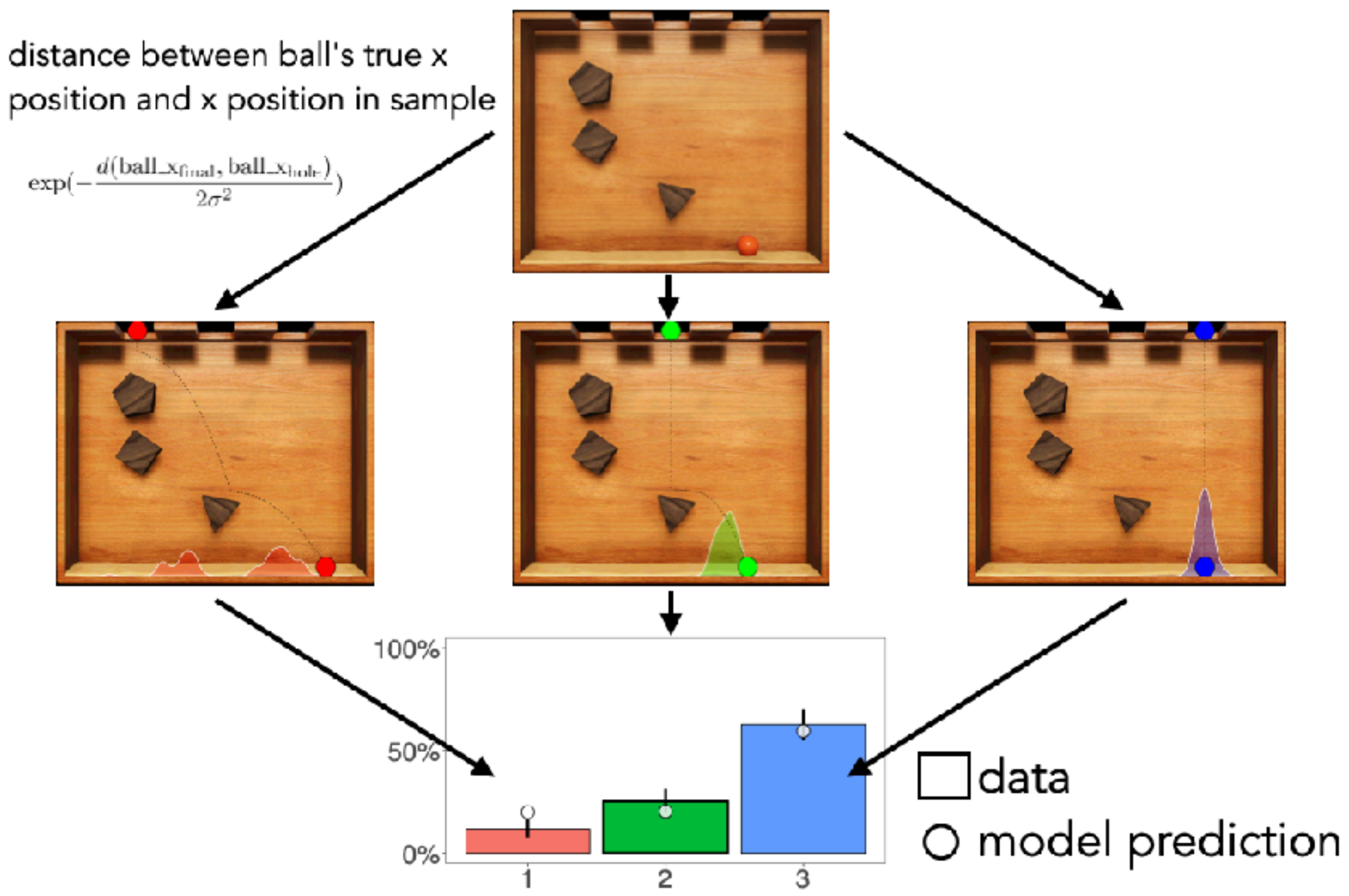
```
fit.glm = glm(formula = survived ~ 1 + fare,
              family = "binomial",
              data = df.titanic)
```



Likelihood



- **Bernoulli:**
  - binary data
  - a single trial
- **Poisson**: count of discrete events
- **Beta-binomial**: like binomial but probability of success may change across trials
- **Student-t:**
  - same as Normal
  - handles greater variability in the data (distribution has **fat tails**)
- ...

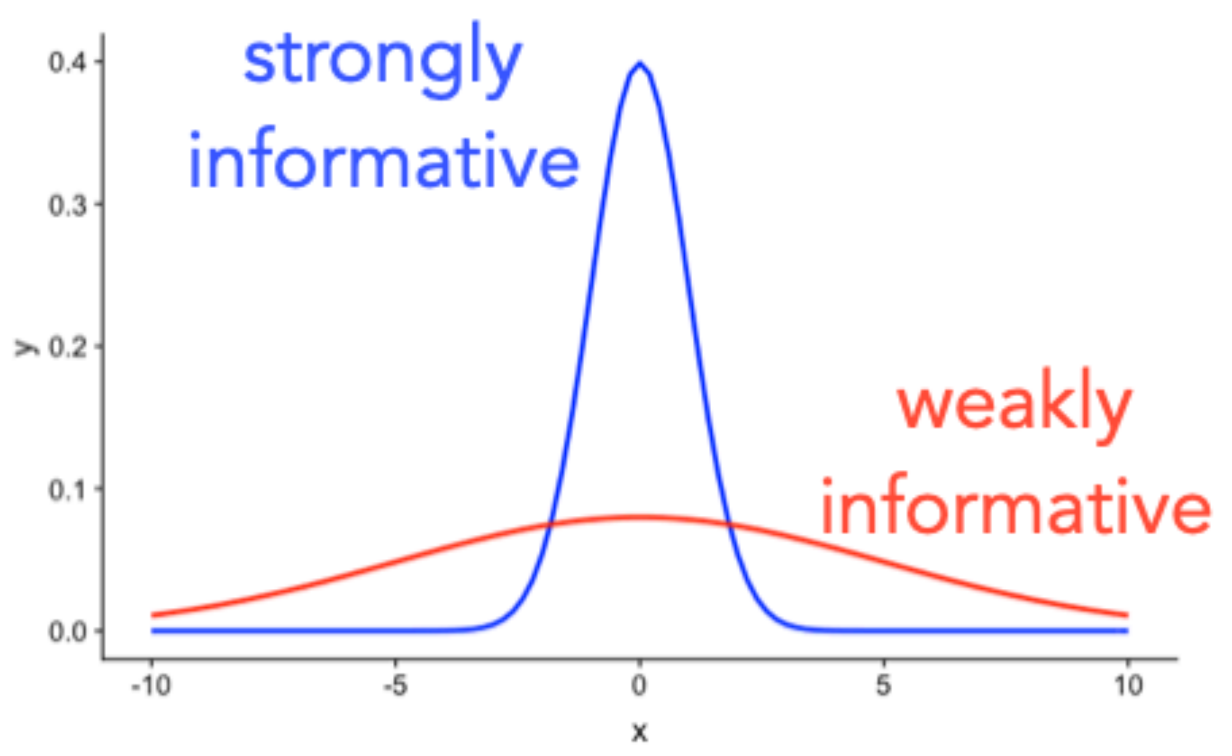


Quick recap: likelihood, **prior**, inference

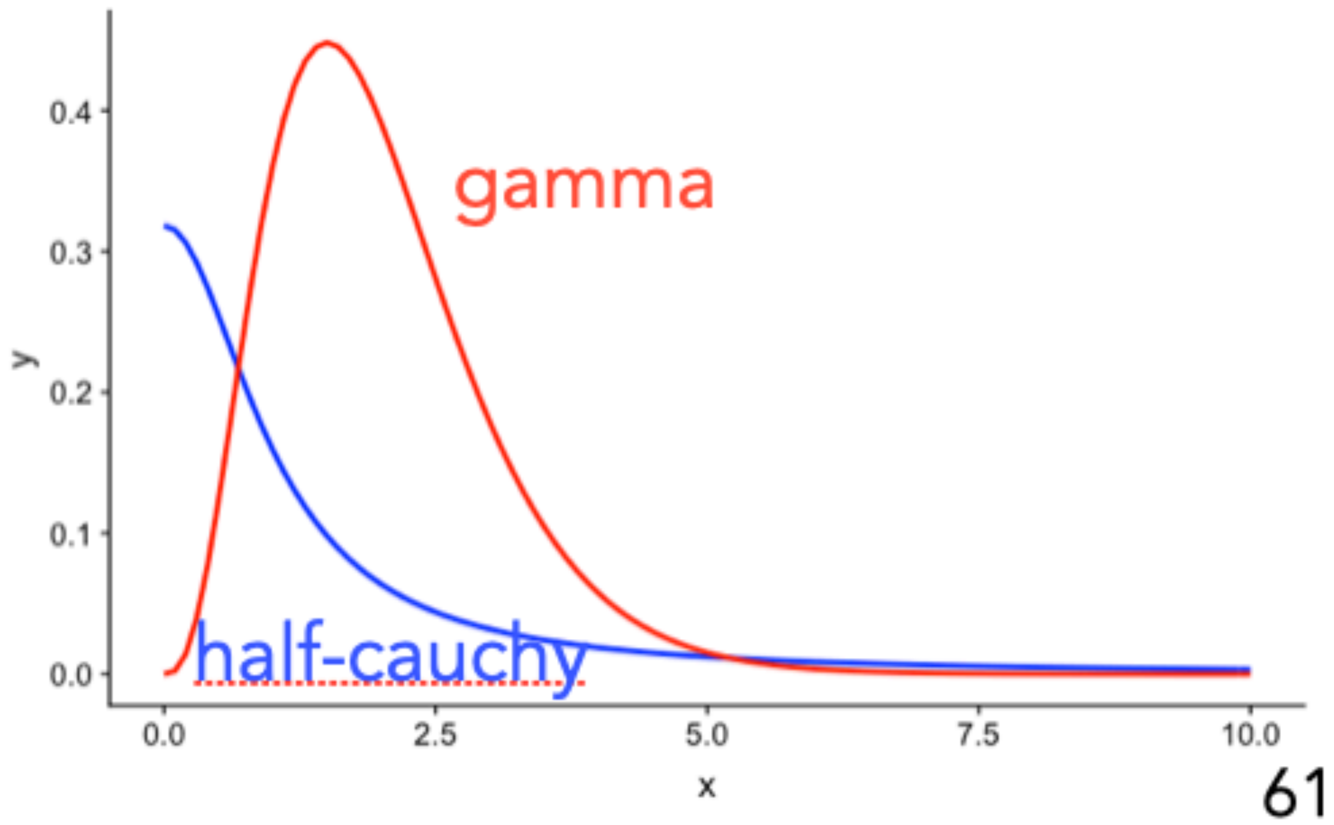
Prior

- **uniform**:
  - continuous or discrete
  - bounded between minimum and maximum
- **gaussian:**
  - sd determines how informative the prior is
- **gamma, half-cauchy**:
  - for parameters we know are positive

for **beta coefficients** in a regression



for **standard deviation** of the Gaussian



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Quick recap: likelihood, prior, **inference**

Likelihood Prior

Posterior

$$p(H \mid D) = \frac{p(D \mid H) \cdot p(H)}{p(D)}$$

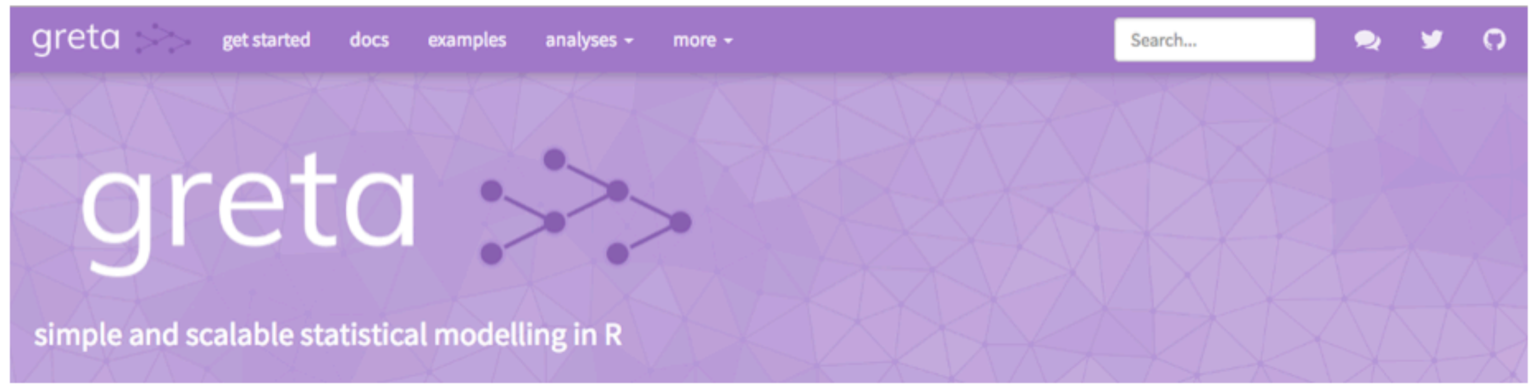
Normalizing constant

the devil is in the denominator ...

Quick recap: Doing Bayesian data analysis

Software packages

```r
library("greta")
```



- let's us write Bayesian models directly in R with a simple syntax
- uses Tensorflow to implement Hamiltonian Monte Carlo sampling (a fast inference algorithm ...)

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Model specification

```r
library("greta")
library("tidybayes")

# variables & priors
b0 = normal(0, 10)
b1 = normal(0, 10)
sd = cauchy(0, 3, truncation = c(0, Inf))

# linear predictor
mu = b0 + b1 * attitude$complaints

# observation model (likelihood)
distribution(attitude$rating) = normal(mu, sd)

# define the model
m = model(b0, b1, sd)
```

priors

linear combination

Gaussian likelihood

build the model

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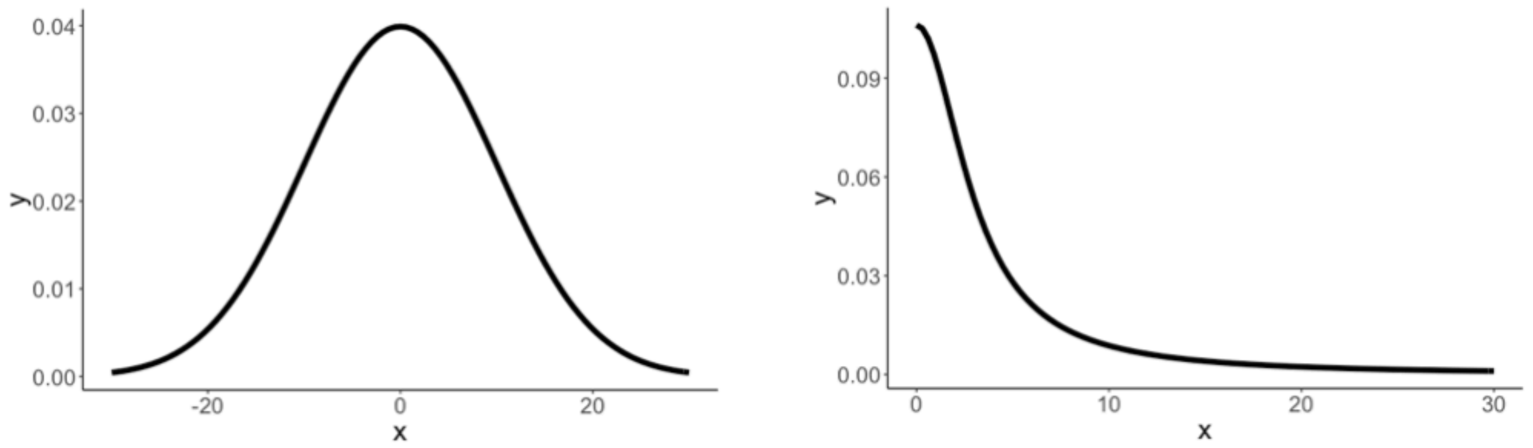
Priors



Gaussian prior on intercept and coefficient

Truncated Cauchy prior on the standard deviation

weakly informative priors (allow for a wide range of possible values)

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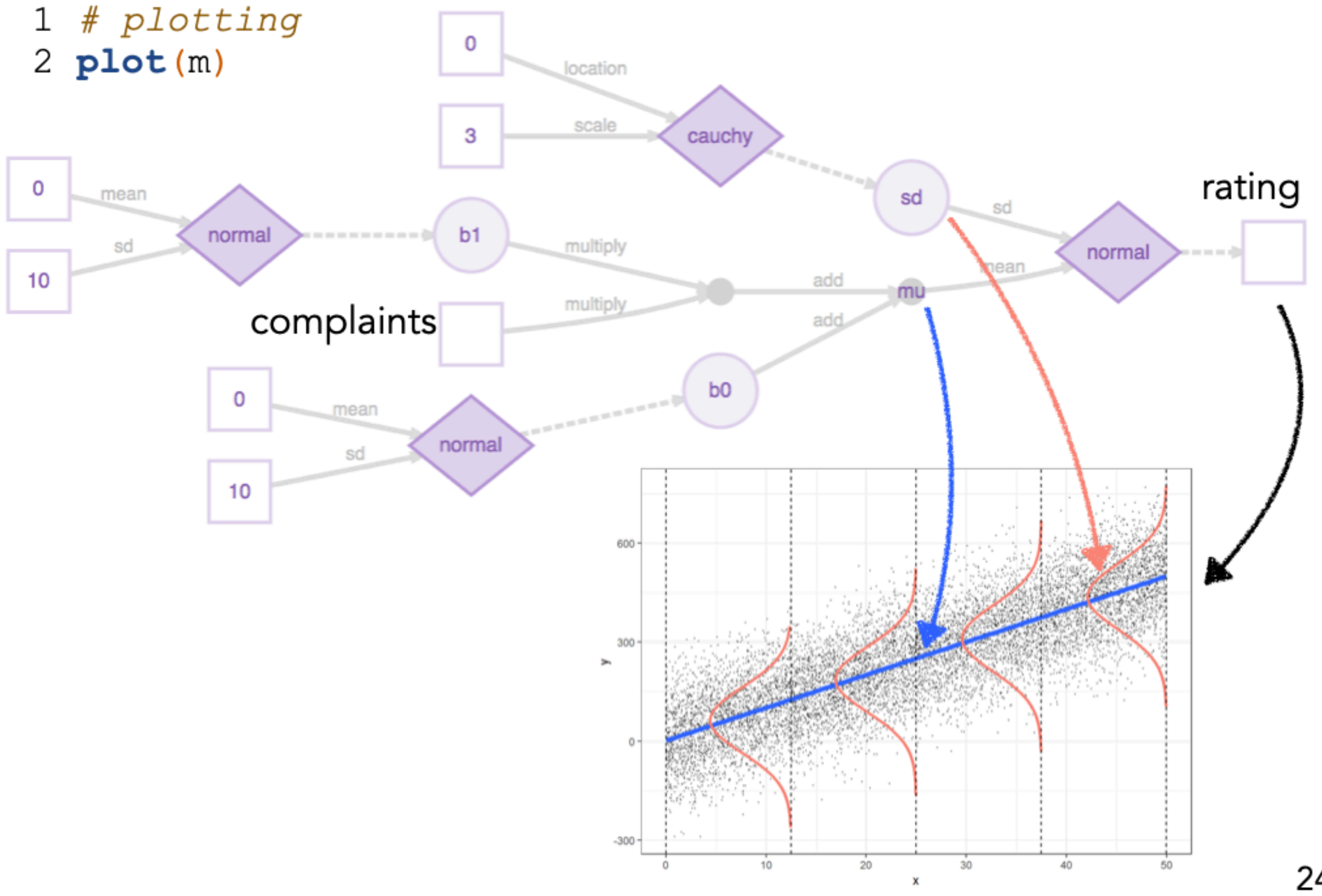
Graphical representation of the model

```r
# plotting
plot(m)
```



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Quick recap: Doing Bayesian data analysis

Inference via sampling

Markov Chain Monte Carlo inference

```
# sampling
draws = mcmc(m, n_samples = 1000)

# tidy up the draws
df.draws = tidy_draws(draws) %>%
  clean_names()
```

chain	iteration	draw	b0	b1	sd
1	1	1	6.08	0.87	7.60
1	2	2	1.12	0.95	7.66
1	3	3	-1.83	0.99	7.01
1	4	4	-4.23	1.02	6.64
1	5	5	3.26	0.87	7.96
1	6	6	-1.04	0.98	7.67
1	7	7	-0.83	0.97	10.12
1	8	8	-1.41	0.97	8.02
1	9	9	9.46	0.81	6.30
1	10	10	10.02	0.84	6.57

each of these is a solution for explaining the data

nice visualization of MCMC samplers

https://github.com/chi-feng/mcmc-demo

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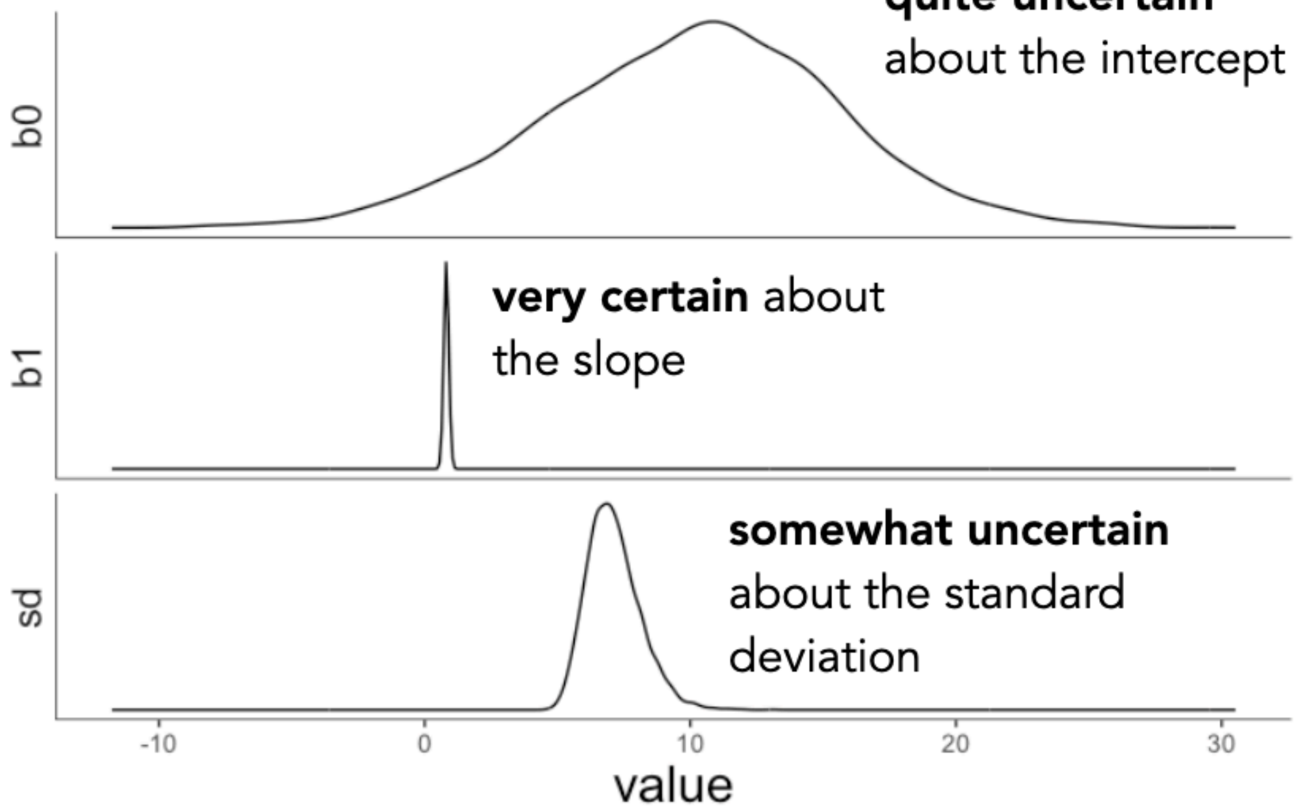
Visualize the posterior



quite uncertain about the intercept

very certain about the slope

somewhat uncertain about the standard deviation

This is the solution of a Bayesian analysis.
A posterior distribution over each parameter in our model.
We can use this to visualize model predictions, and to test hypotheses.

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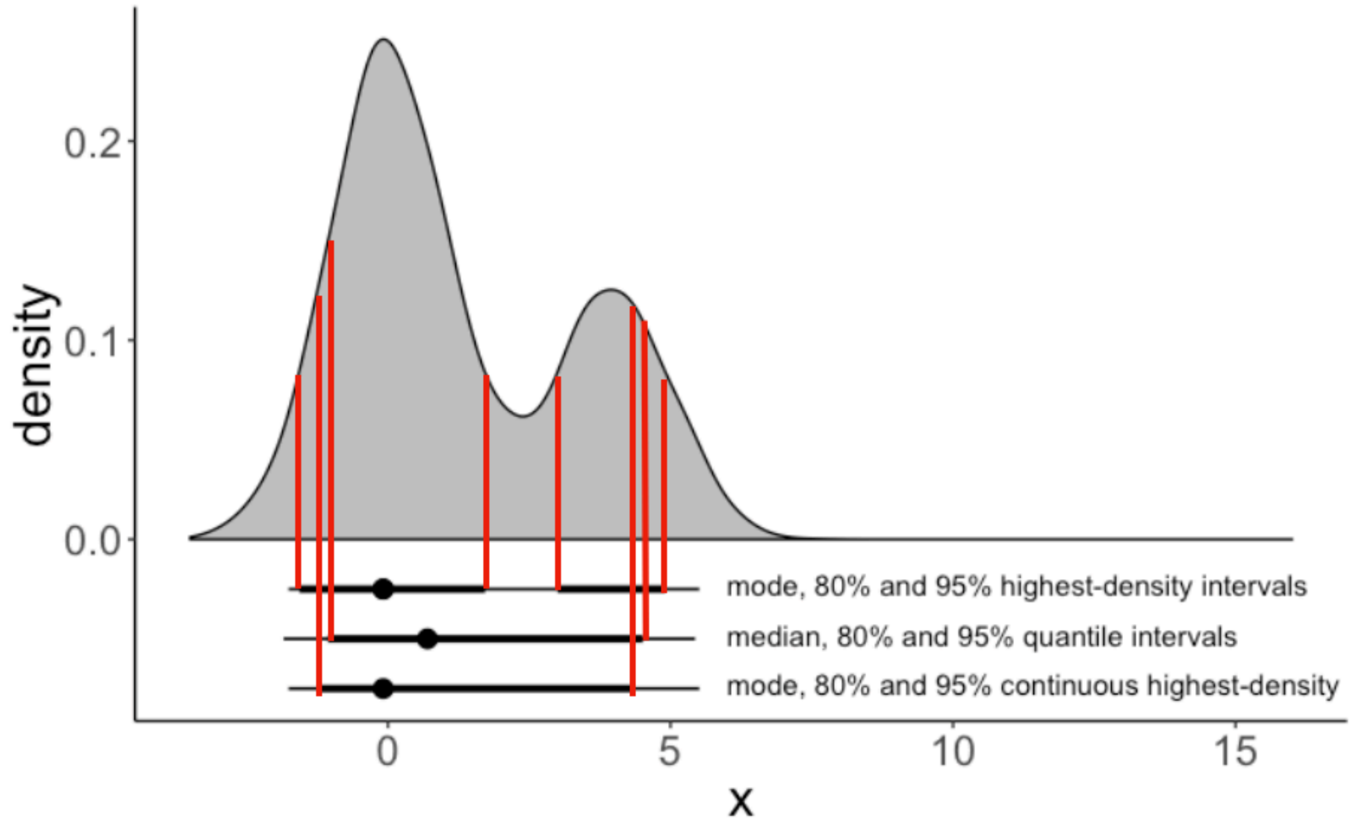
Different kinds of credible intervals



ways of summarizing the posterior distribution

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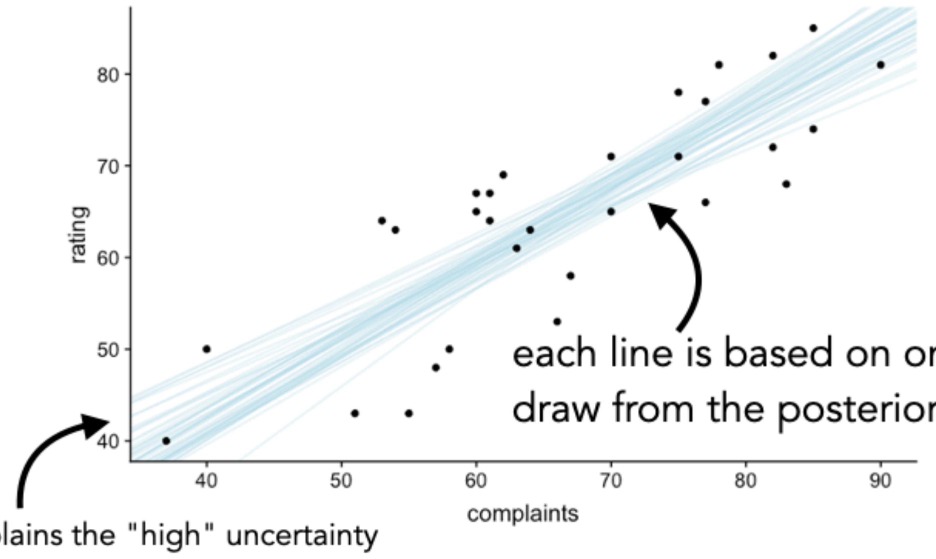
Visualize the model predictions

```
ggplot(data = df.attitude,
       mapping = aes(x = complaints,
                     y = rating)) +
  geom_abline(data = df.draws %>%
                sample_n(size = 50),
              aes(intercept = b0,
                  slope = b1),
              alpha = 0.3,
              color = "lightblue") +
  geom_point()
```



each line is based on one draw from the posterior

explains the "high" uncertainty about the intercept

chain	iteration	draw	b0	b1	sd
1	1	1	6.08	0.87	7.60
1	2	2	1.12	0.95	7.66
1	3	3	-1.83	0.99	7.01
1	4	4	-4.23	1.02	6.64
1	5	5	3.26	0.87	7.96
1	6	6	-1.04	0.98	7.67
1	7	7	-0.83	0.97	10.12
1	8	8	-1.41	0.97	8.02
1	9	9	9.46	0.81	6.30
1	10	10	10.02	0.84	6.57

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Paul Bürkner

Doing Bayesian data analysis with BRMS

Software packages

`library("brms")`

Bayesian regression modeling with Stan

- very powerful package that makes it easy to run Bayesian regression models
- we specify models using the same syntax we've already learned based on **`lm`**`()`, **`glm`**`()`, and **`lmer`**`()`
- `brms` turns this into Stan code and fits the model
- we can then use `tidybayes` to investigate the posterior

Software packages

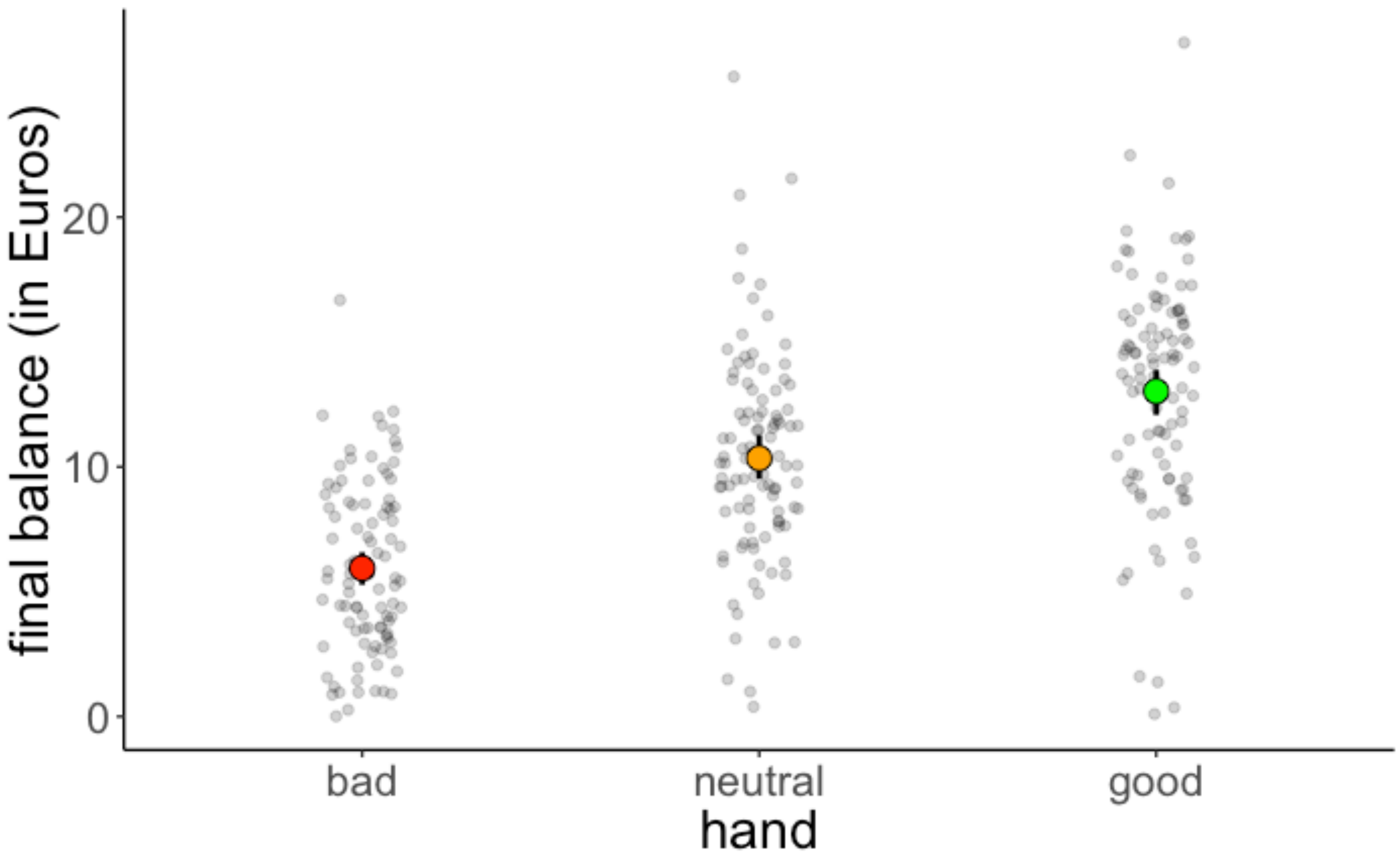
Stan

Stan® is a state-of-the-art platform for statistical modeling and high-performance statistical computation. Thousands of users rely on Stan for statistical modeling, data analysis, and prediction in the social, biological, and physical sciences, engineering, and business.

https://mc-stan.org/

Poker data

Poker data



Using `lm()`

```r
fit.lm_poker = lm(formula = balance ~ 1 + hand,
                  data = df.poker)

fit.lm_poker %>% summary()
```

```
Call:
lm(formula = balance ~ 1 + hand, data = df.poker)

Residuals:
     Min       1Q   Median       3Q      Max 
-12.9264  -2.5902  -0.0115   2.6573  15.2834 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   5.9415     0.4111  14.451  < 2e-16 ***
handneutral   4.4051     0.5815   7.576 4.55e-13 ***
handgood      7.0849     0.5815  12.185  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.111 on 297 degrees of freedom
Multiple R-squared:  0.3377,	Adjusted R-squared:  0.3332 
F-statistic:  75.7 on 2 and 297 DF,  p-value: < 2.2e-16
```

Using brm()

Cool!

```r
fit.brm_poker = brm(formula = balance ~ 1 + hand,
                    data = df.poker)

fit.brm_poker %>% summary()
```

```
 Family: gaussian
  Links: mu = identity; sigma = identity
Formula: balance ~ 1 + hand
   Data: df.poker (Number of observations: 300)
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup samples = 4000

Population-Level Effects:
            Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept       5.93      0.41     5.12     6.72 1.00     2986     2744
handneutral     4.41      0.58     3.30     5.55 1.00     3497     2903
handgood        7.10      0.58     5.99     8.29 1.00     3545     2932

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     4.12      0.17     3.81     4.46 1.00     3650     2921

Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample
is a crude measure of effective sample size, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Comparison between `lm()` and `brm()`

`lm()`

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.9415     0.4111  14.451  < 2e-16 ***
handneutral   4.4051     0.5815   7.576 4.55e-13 ***
handgood      7.0849     0.5815  12.185  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

`brm()`

```
Population-Level Effects:
            Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept       5.93      0.41     5.12     6.72 1.00     2986     2744
handneutral     4.41      0.58     3.30     5.55 1.00     3497     2903
handgood        7.10      0.58     5.99     8.29 1.00     3545     2932
```

almost identical results!

What about the priors?

```r
fit.brm_poker = brm(formula = balance ~ 1 + hand,
                    data = df.poker)
```

By default, `brms` uses weakly informative priors for the model parameters.

There are quite a few other defaults, let's take a look under the hood ...

"Full" specification of the model

```
fit.brm_poker_full = brm(
  formula = balance ~ 1 + hand,
  family = "gaussian",
  data = df.poker,
  prior = c(
    prior(normal(0, 10), class = "b", coef = "handgood"),
    prior(normal(0, 10), class = "b", coef = "handneutral"),
    prior(student_t(3, 3, 10), class = "Intercept"),
    prior(student_t(3, 0, 10), class = "sigma")
  ),
  inits = list(
    list(Intercept = 0, sigma = 1, handgood = 5, handneutral = 5),
    list(Intercept = -5, sigma = 3, handgood = 2, handneutral = 2),
    list(Intercept = 2, sigma = 1, handgood = -1, handneutral = 1),
    list(Intercept = 1, sigma = 2, handgood = 2, handneutral = -2)
  ),
  iter = 4000,
  warmup = 1000,
  chains = 4,
  file = "cache/brm_poker_full",
  seed = 1
)
```

likelihood

priors

initialization

how many runs in the inference chain

how long for the warmup

how many chains

save the model result

make reproducible

fitting Bayesian models takes some time, so storing results is key

Turned into Stan code

```
// generated with brms 2.7.0
functions {
}
data {
  int<lower=1> N;  // total number of observations
  vector[N] Y;  // response variable
  int<lower=1> K;  // number of population-level effects
  matrix[N, K] X;  // population-level design matrix
  int prior_only;  // should the likelihood be ignored?
}
transformed data {
  int Kc = K - 1;
  matrix[N, K - 1] Xc;  // centered version of X
  vector[K - 1] means_X;  // column means of X before centering
  for (i in 2:K) {
    means_X[i - 1] = mean(X[, i]);
    Xc[, i - 1] = X[, i] - means_X[i - 1];
  }
}
parameters {
  vector[Kc] b;  // population-level effects
  real temp_Intercept;  // temporary intercept
  real<lower=0> sigma;  // residual SD
}
transformed parameters {
}
model {
  vector[N] mu = temp_Intercept + Xc * b;
  // priors including all constants
  target += normal_lpdf(b[1] | 0, 10);
  target += normal_lpdf(b[2] | 0, 10);
  target += student_t_lpdf(temp_Intercept | 3, 3, 10);
  target += student_t_lpdf(sigma | 3, 0, 10)
    - 1 * student_t_lccdf(0 | 3, 0, 10);
  // likelihood including all constants
  if (!prior_only) {
    target += normal_lpdf(Y | mu, sigma);
  }
}
generated quantities {
  // actual population-level intercept
  real b_Intercept = temp_Intercept - dot_product(means_X, b);
}
```

- probabilistic programming language
- flexible construction of Bayesian models
- ports have been written for R, Python, Julia, ...
- implements a fast inference algorithm

Results

posterior samples

b_Intercept	b_handneutral	b_handgood	sigma
5.97	4.27	7.48	3.94
5.11	5.25	7.40	3.91
7.03	3.78	5.80	4.48
5.72	4.18	7.25	4.00
6.01	4.44	6.15	4.57
5.94	4.69	6.72	4.36
6.39	3.84	6.40	3.92
5.24	5.15	7.69	4.16
6.12	4.51	7.20	4.14
6.43	3.71	6.37	4.13
5.85	5.01	7.32	4.00
6.51	3.58	6.62	3.95
5.85	4.45	7.62	4.17
5.80	5.45	6.36	4.10
5.48	5.51	7.22	3.99

⋮

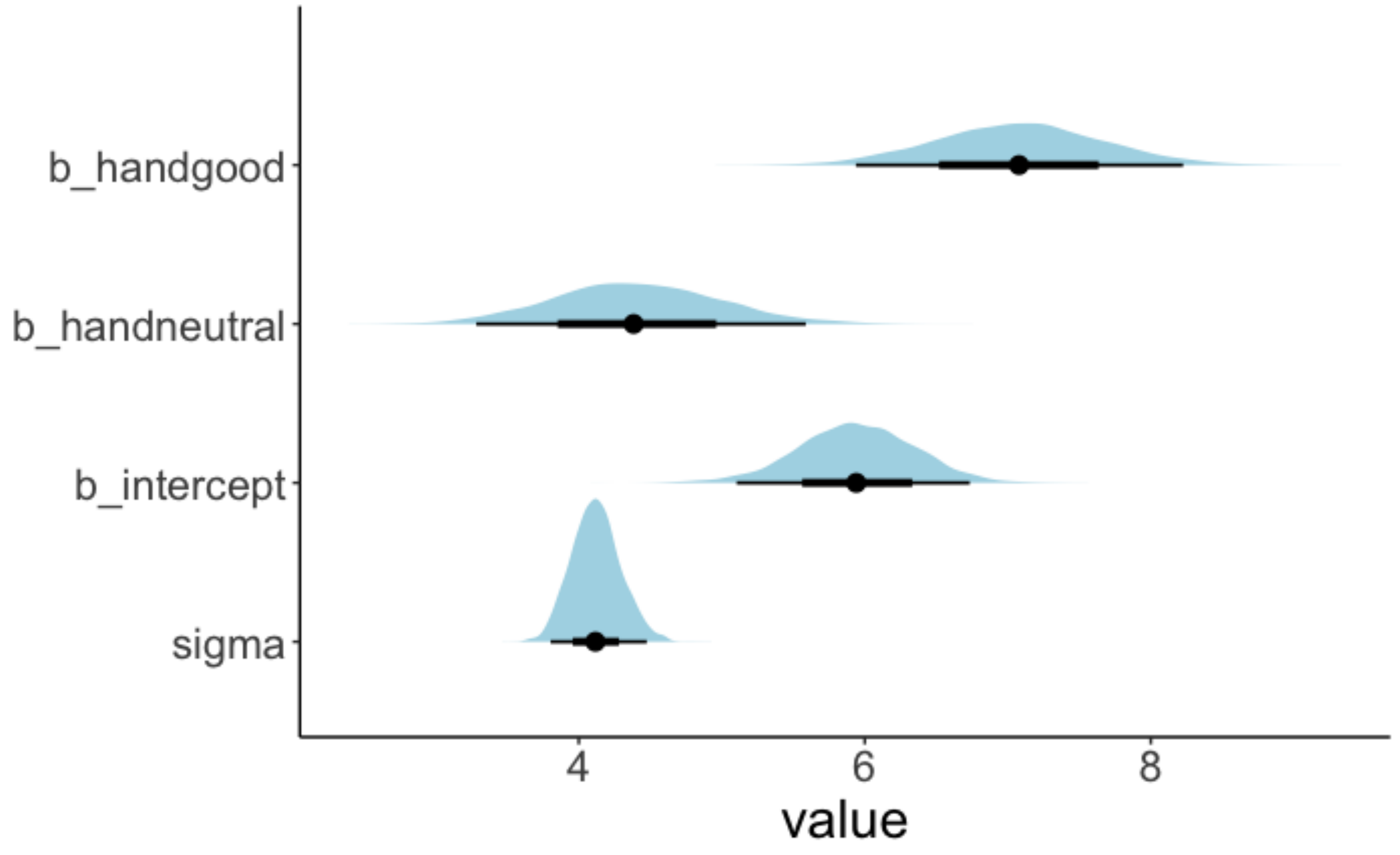
summary of posterior

parameter	lower	mode	upper
b_handgood	5.97	7.07	8.27
b_handneutral	3.21	4.43	5.51
b_intercept	5.17	5.95	6.77
sigma	3.81	4.12	4.47

maximum a posteriori

MAP estimate and 95% highest density interval

visualization



Testing hypotheses

Results

posterior samples

b_Intercept	b_handneutral	b_handgood	sigma
5.97	4.27	7.48	3.94
5.11	5.25	7.40	3.91
7.03	3.78	5.80	4.48
5.72	4.18	7.25	4.00
6.01	4.44	6.15	4.57
5.94	4.69	6.72	4.36
6.39	3.84	6.40	3.92
5.24	5.15	7.69	4.16
6.12	4.51	7.20	4.14
6.43	3.71	6.37	4.13
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6.51	3.58	6.62	3.95
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⋮

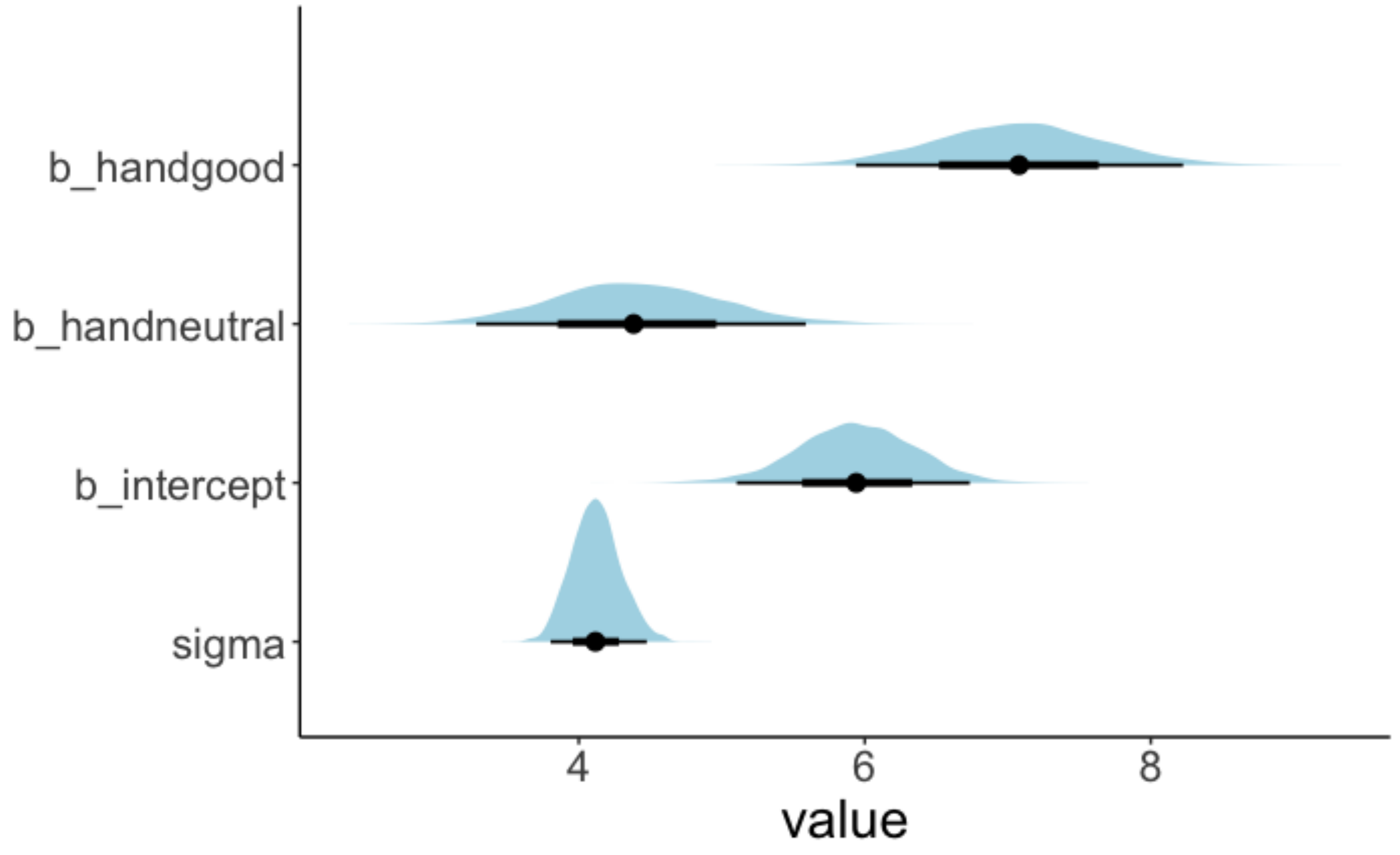
summary of posterior

parameter	lower	mode	upper
b_handgood	5.97	7.07	8.27
b_handneutral	3.21	4.43	5.51
b_intercept	5.17	5.95	6.77
sigma	3.81	4.12	4.47

maximum a posteriori

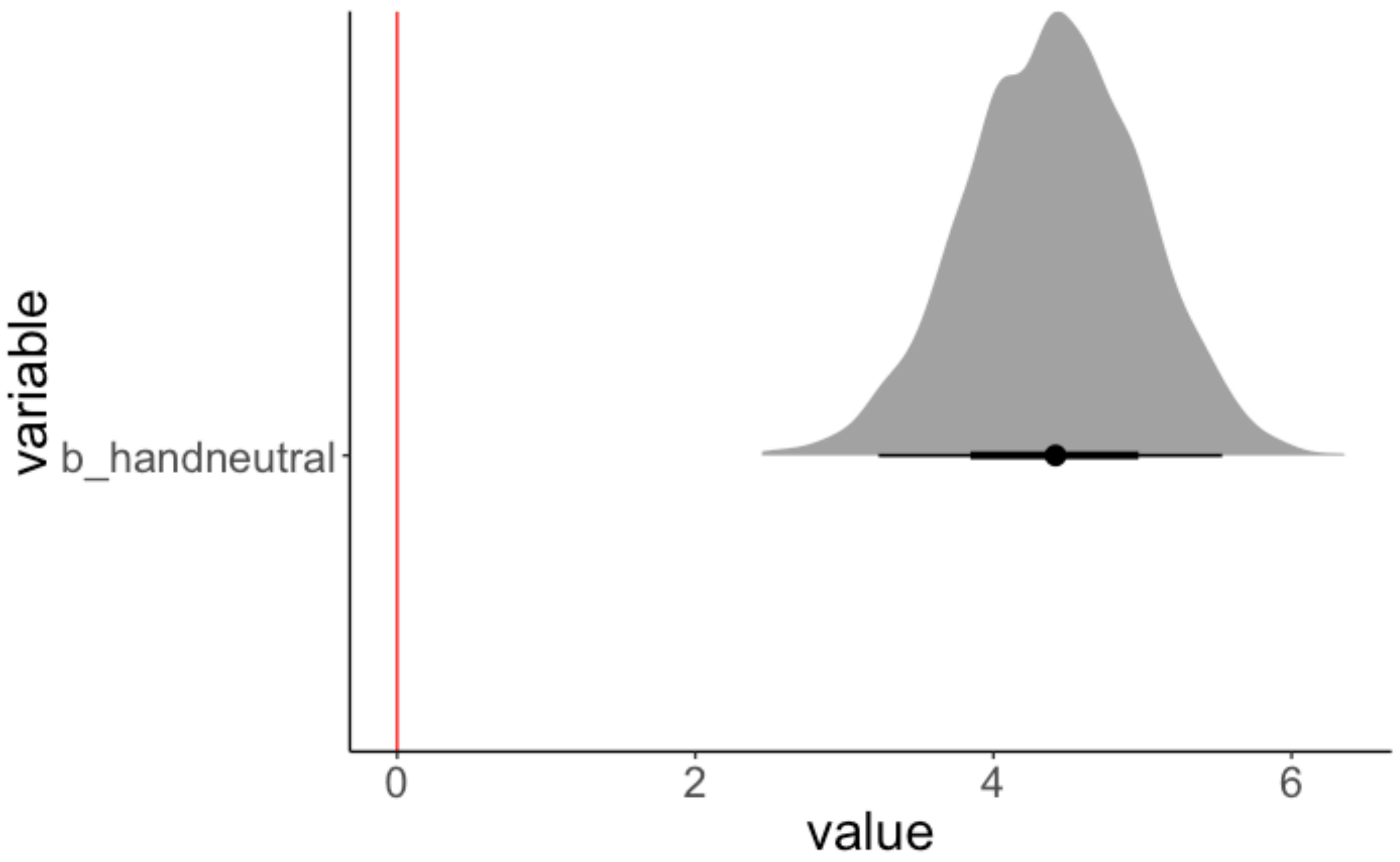
MAP estimate and 95% highest density interval

visualization



Asking questions based on the posterior

Do neutral hands earn more money than bad hands?



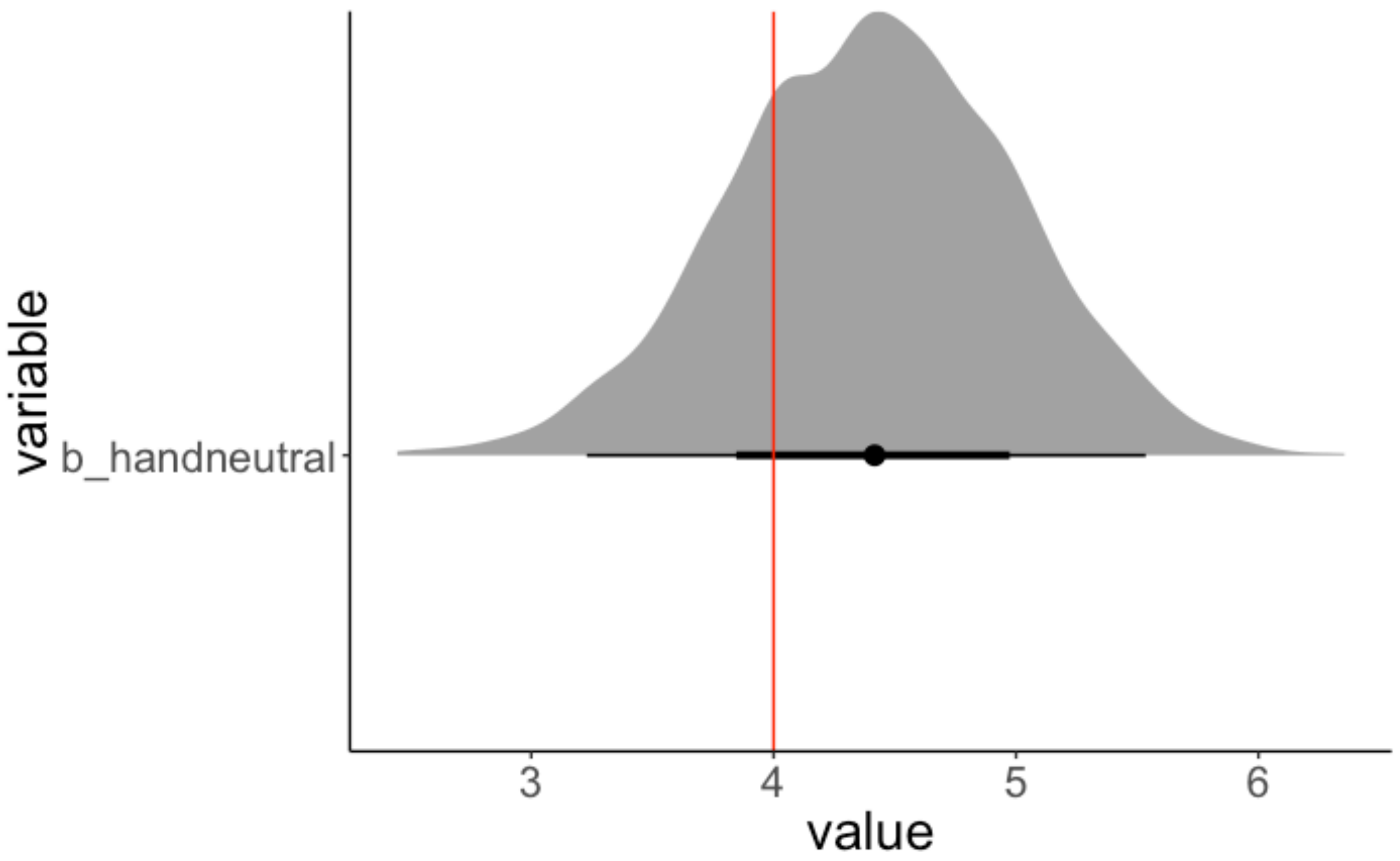
What's the probability that handneutral is less than 0?

```
hypothesis(fit.brm,
           hypothesis = "handneutral < 0")
```

p = 0

Asking questions based on the posterior

Do neutral hands earn much more money than bad hands?



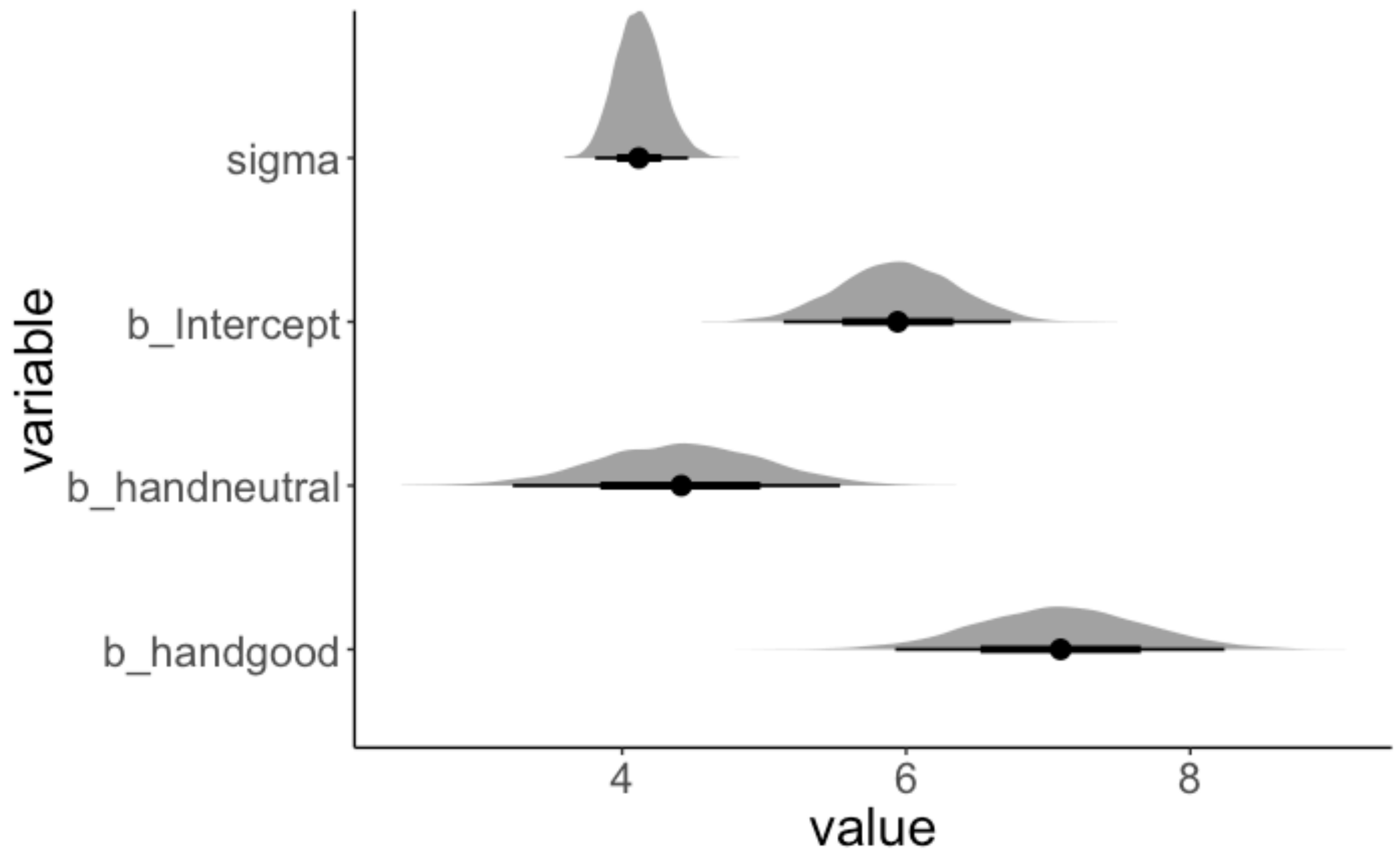
What's the probability that handneutral is **more than 4**?

```r
hypothesis(fit.brm,
           hypothesis = "handneutral > 4")
```

$p = 0.75$

Asking questions based on the posterior

Do good hands make twice as much as bad hands?

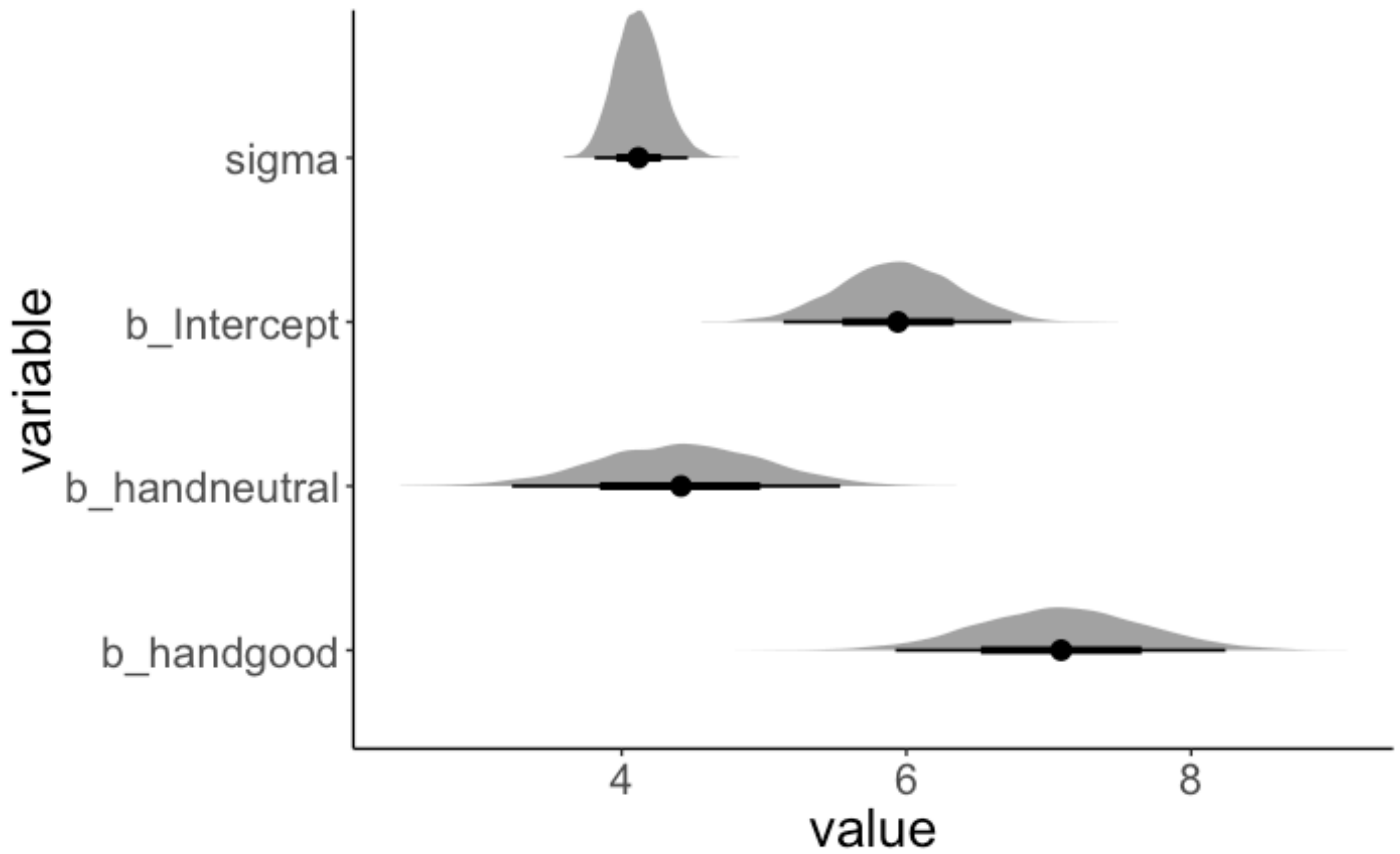



```r
hypothesis(fit.brm,
           hypothesis = "handgood + Intercept > 2 * Intercept")
```

$p = 0.89$

Asking questions based on the posterior

Are neutral hands worse than bad and good hands combined?




```r
hypothesis(fit.brm,
           hypothesis = "Intercept + handneutral < (Intercept + Intercept + handgood) / 2")
```

p = 0.04

Testing hypothesis

```
df.hypothesis = fit.brm %>%
  posterior_samples() %>%
  clean_names() %>%
  select(starts_with("b_")) %>%
  mutate(neutral = b_intercept + b_handneutral,
         bad_good_average = (b_intercept + b_intercept + b_handgood)/2,
         hypothesis = neutral < bad_good_average)
```

samples from the posterior

b_intercept	b_handneutral	b_handgood	neutral	bad_good_average	hypothesis
6.07	4.10	7.20	10.17	9.67	FALSE
6.06	4.44	6.95	10.49	9.53	FALSE
5.88	5.00	6.73	10.87	9.24	FALSE
5.85	4.78	6.18	10.63	8.94	FALSE
5.86	4.46	7.68	10.32	9.70	FALSE

```
df.hypothesis %>%
  summarize(p = sum(hypothesis)/n())
```

$p = 0.04$

Testing hypotheses

Having a posterior distribution allows us to ask questions about the data in a very flexible way!

The "emmeans" package is your friend!

```
1 fit.brm_poker %>%
2   emmeans(specs = consec ~ hand)
```

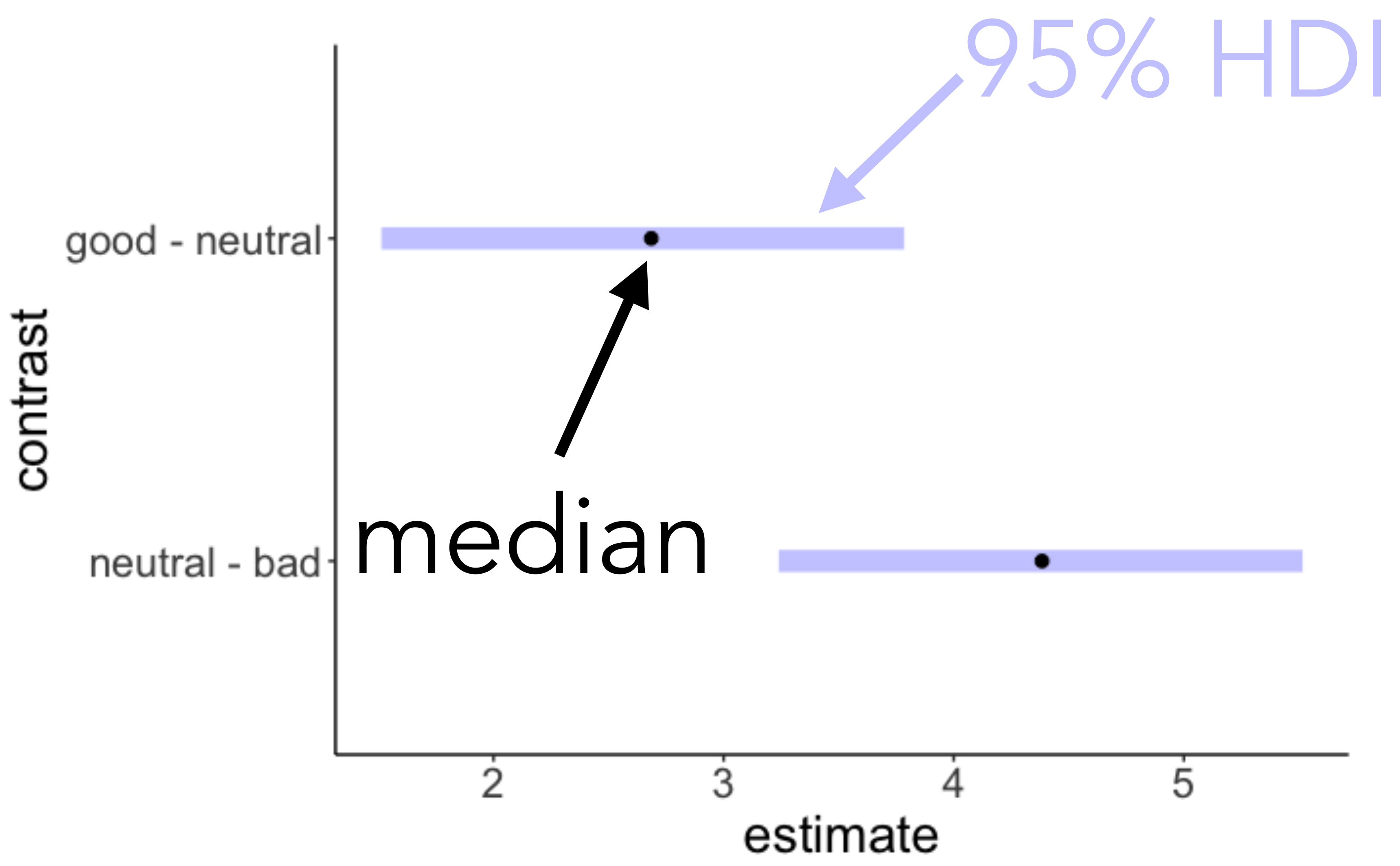
estimated median for each group

contrasts

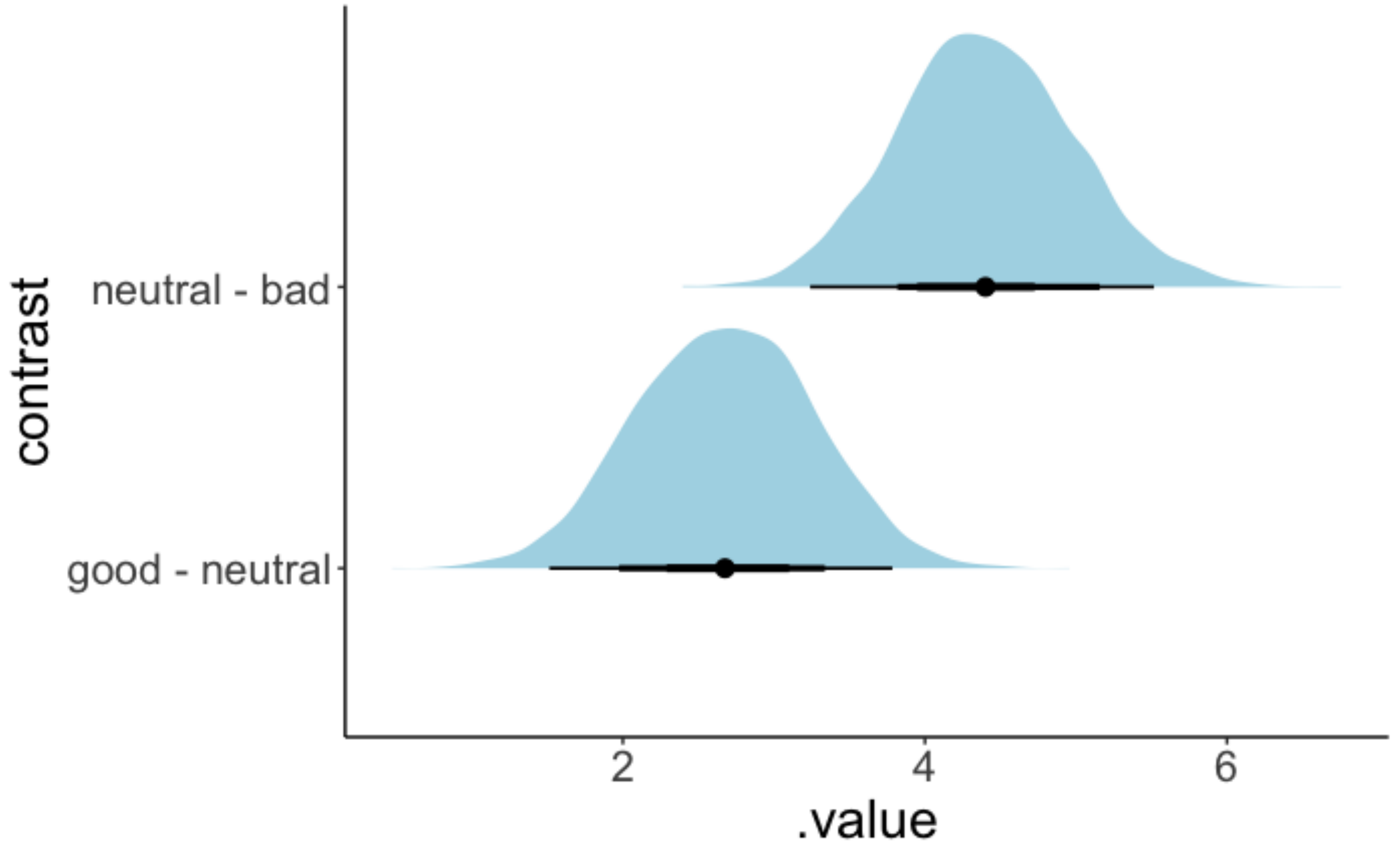
```
$emmeans
 hand     emmean lower.HPD upper.HPD
 bad        5.94      5.16      6.78
 neutral   10.34      9.55     11.15
 good      13.02     12.22     13.82

Point estimate displayed: median
HPD interval probability: 0.95

$contrasts
 contrast       estimate lower.HPD upper.HPD
 neutral - bad      4.38      3.24      5.52
 good - neutral     2.69      1.51      3.78

Point estimate displayed: median
HPD interval probability: 0.95
```

Visualizing the contrasts

```r
fit.brm_poker %>%
  emmeans(specs = consec ~ hand) %>%
  pluck("contrasts") %>%
  plot()
```




```r
fit.brm_poker %>%
  emmeans(specs = consec ~ hand) %>%
  pluck("contrasts") %>%
  gather_emmeans_draws() %>%
  ggplot(mapping = aes(y = contrast,
                       x = .value)) +
  stat_halfeye(fill = "lightblue",
               point_interval = mean_hdi,
               .width = c(0.5, 0.75, 0.95))
```



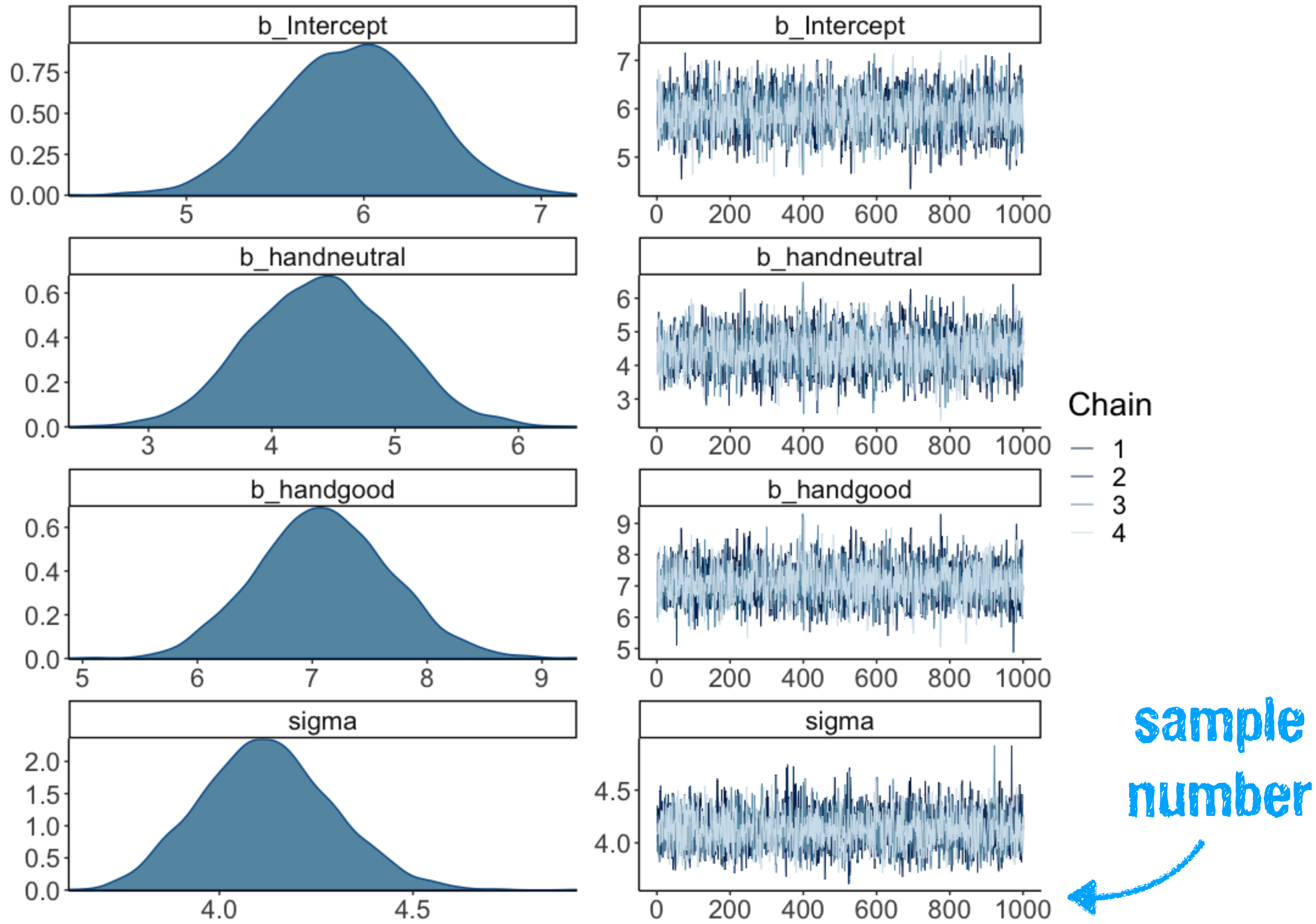
mean, 50% HDI, 75% HDI, 95% HDI

Model evaluation

1. Check whether inference worked

Can we trust the model results?

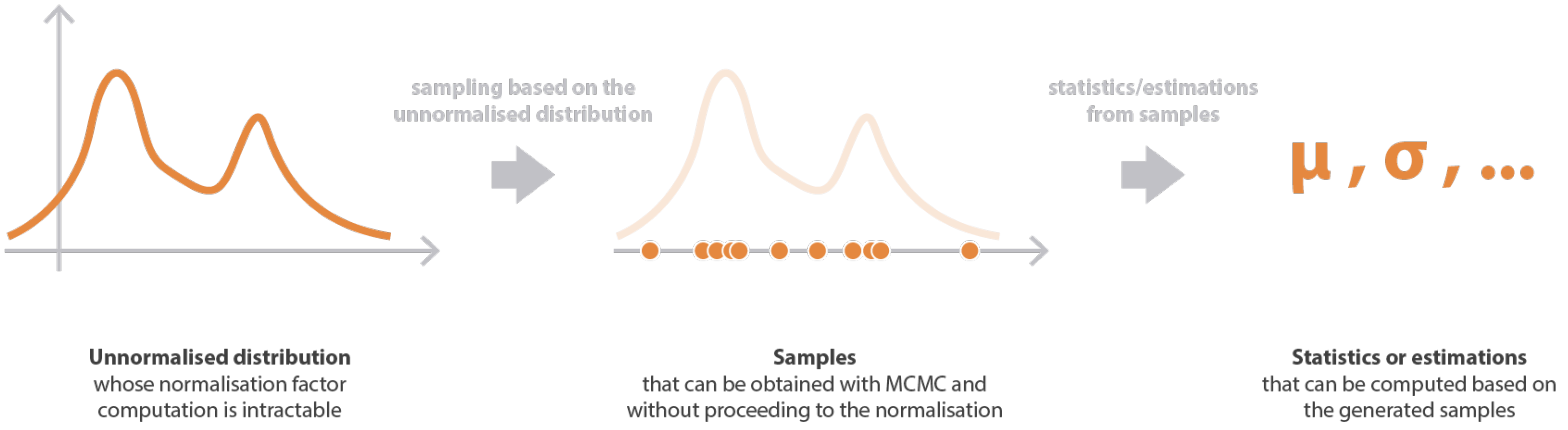
```r
plot(fit.brm_poker)
```



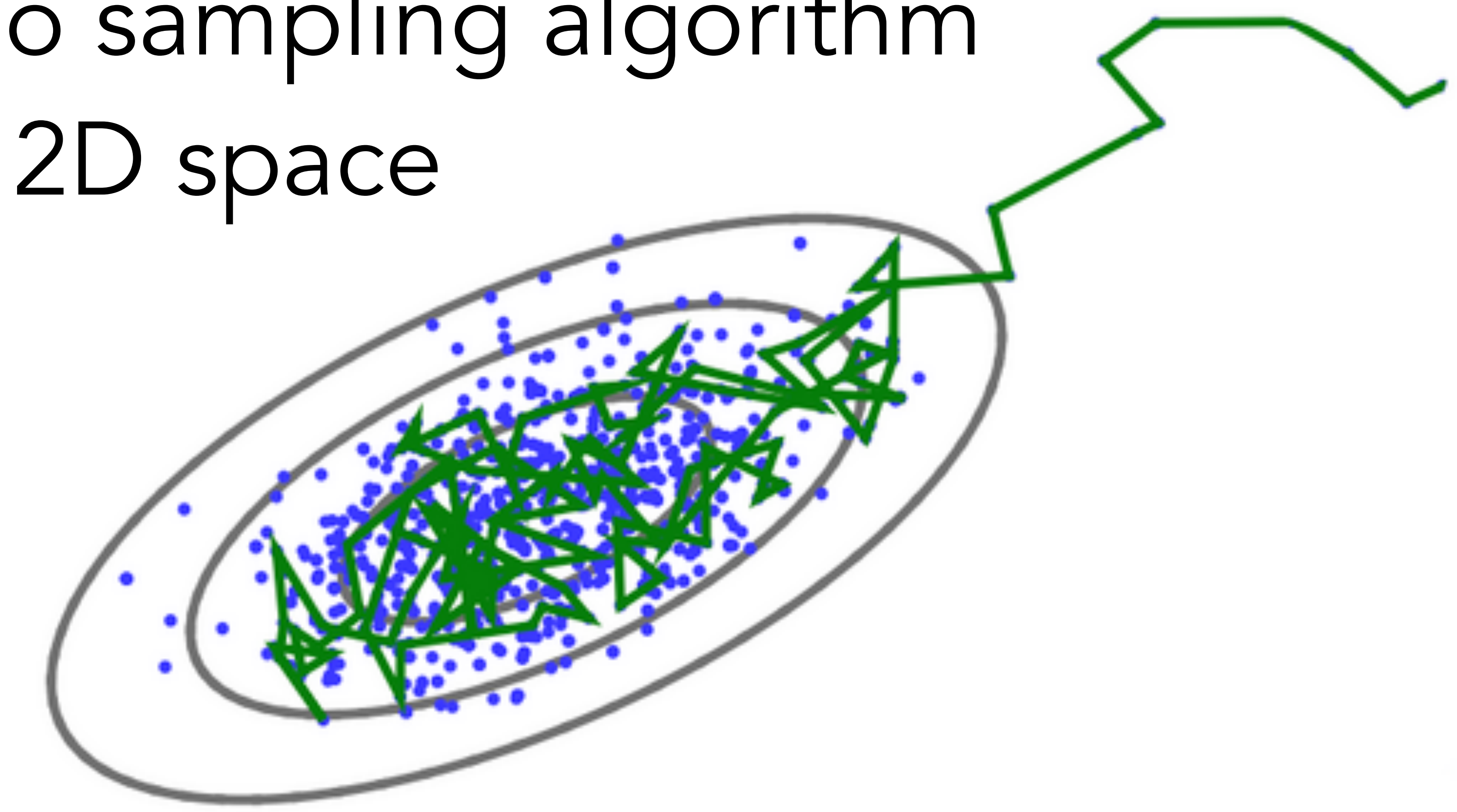
https://chi-feng.github.io/mcmc-demo/

Can we trust the model results?

Inference via Markov Chain Monte Carlo (MCMC)



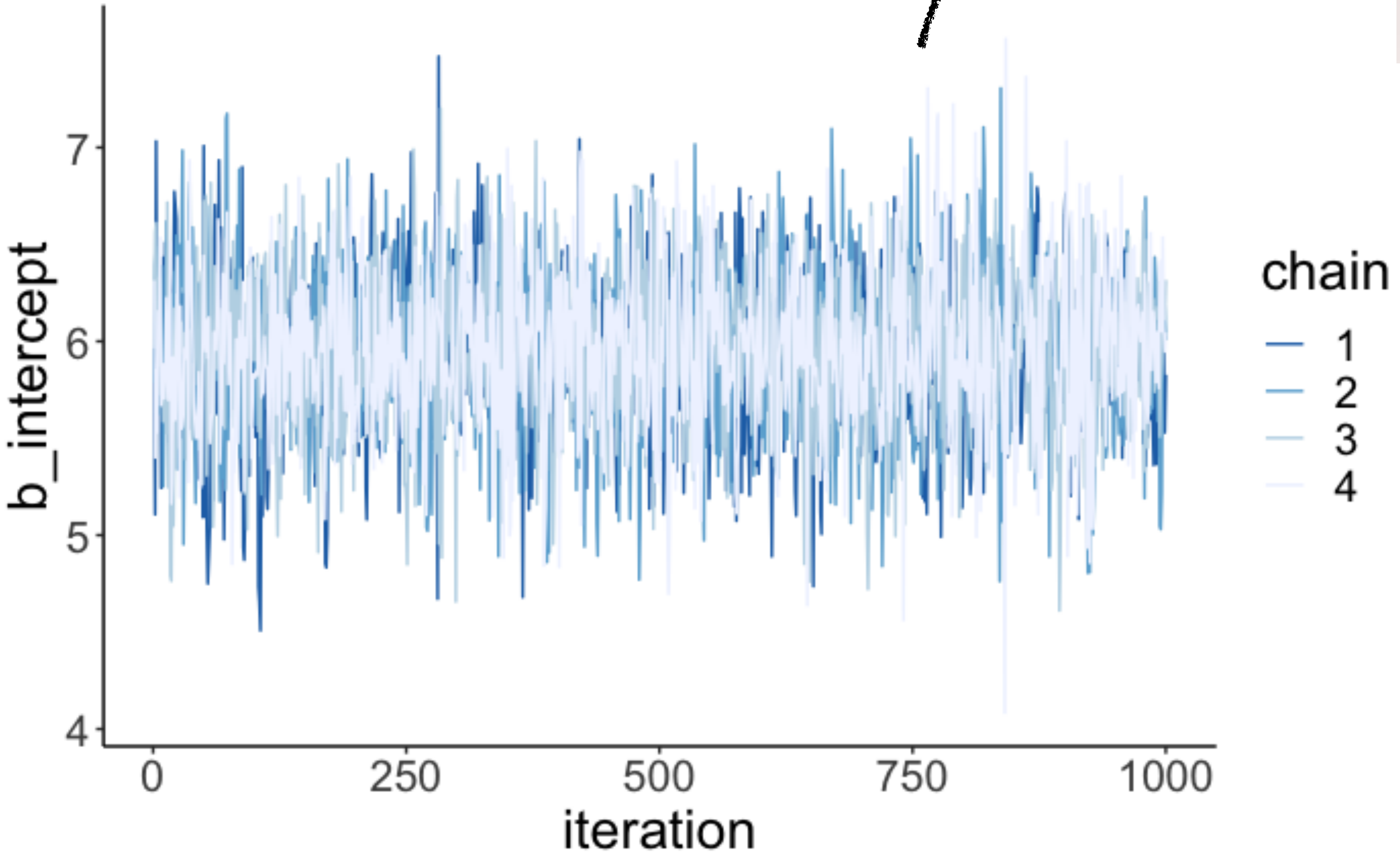
Markov Chain Monte Carlo sampling algorithm in a 2D space



goal: draw **independent** samples from the posterior distribution

Can we trust the model results?

looks like a fuzzy caterpillar



Stats twitter chimes in ...

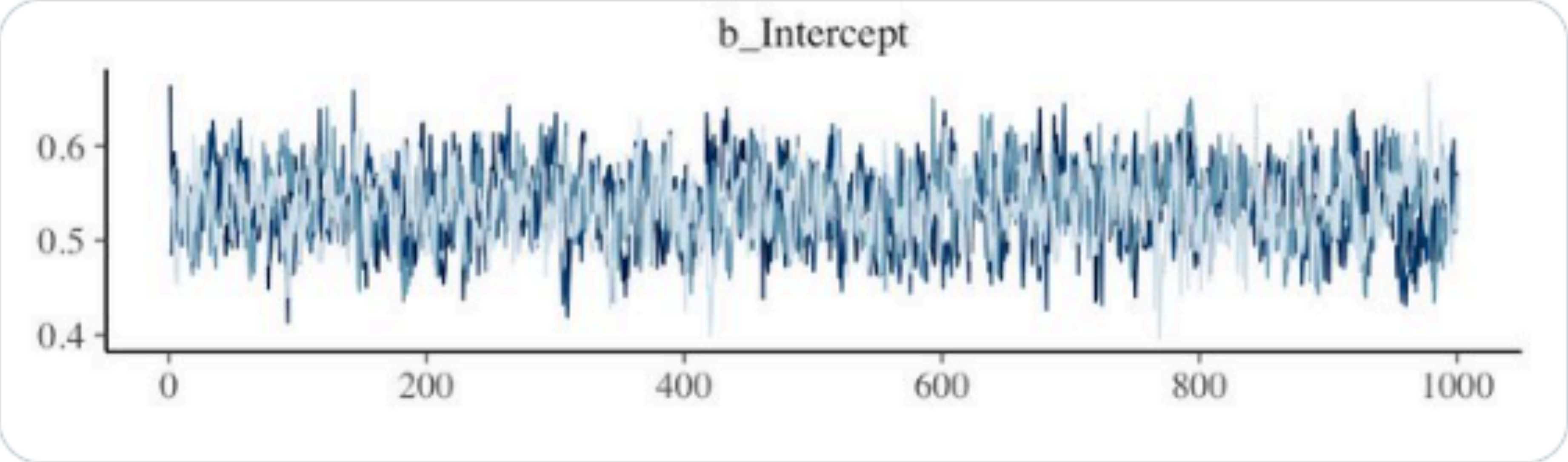
Andrew Heiss
@andrewheiss

love that "looking for fuzzy caterpillars" is like a legitimate analytical strategy

Chelsea Parlett-Pelleriti
@ChelseaParlett

Do your chains just flow?
Do they sample to and fro?
Do they mix together well?
Is your R-hat small, or no?

Are your trace plots looking killer,
like a fuzzy caterpillar?
Do your chains just flow?



When things don't work out

only two data points!

incredibly wide uniform priors

10000000000

```
df.data = tibble(y = c(-1, 1))

fit.brm_wrong = brm(data = df.data,
                    family = gaussian,
                    formula = y ~ 1,
                    prior = c(prior(uniform(-1e10, 1e10), class = Intercept),
                              prior(uniform(0, 1e10), class = sigma)),
                    inits = list(list(Intercept = 0, sigma = 1),
                                 list(Intercept = 0, sigma = 1)),
                    iter = 4000,
                    warmup = 1000,
                    chains = 2,
                    file = "cache/brm_wrong")
```

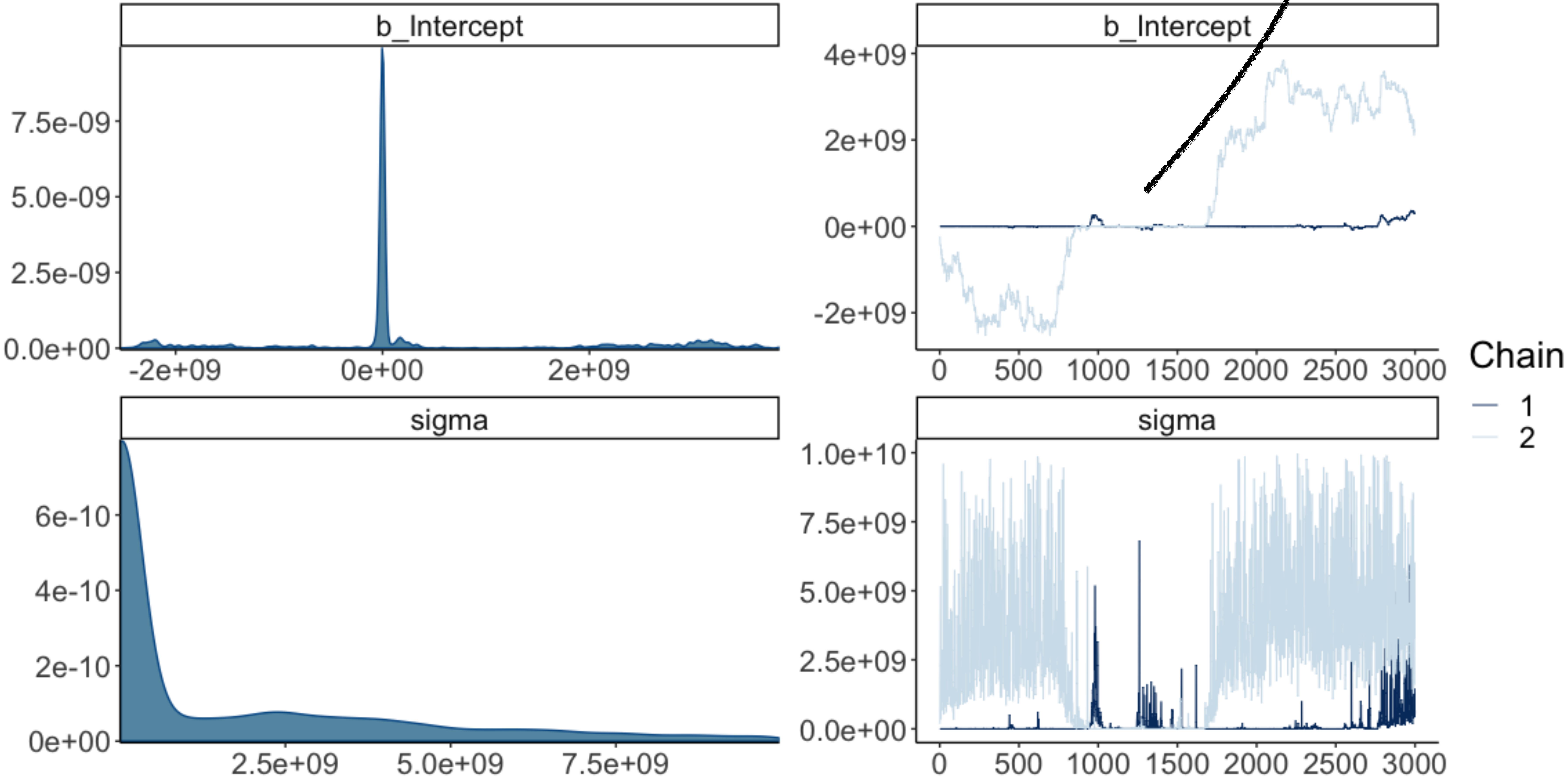
When things don't work out

```
summary(fit.brm_wrong)
```

```
The model has not converged (some Rhats are > 1.1). Do not analyse the results!
We recommend running more iterations and/or setting stronger priors.There were 1203 divergent
transitions after warmup. Increasing adapt_delta above 0.8 may help.
See http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup Family: gaussian
  Links: mu = identity; sigma = identity
Formula: y ~ 1
   Data: df.data (Number of observations: 2)
Samples: 2 chains, each with iter = 4000; warmup = 1000; thin = 1;
         total post-warmup samples = 6000

Population-Level Effects:
              Estimate     Est.Error       l-95% CI      u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept 357550121.58 1416057299.71 -2244033111.47 3333594132.43 1.78        3       24

Family Specific Parameters:
           Estimate     Est.Error l-95% CI      u-95% CI Rhat Bulk_ESS Tail_ESS
sigma 1524412740.64 2392424321.98 21668.93 8317582240.06 1.40        4       41

Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample
is a crude measure of effective sample size, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

When things don't work out

doesn't look like a
fuzzy caterpillar



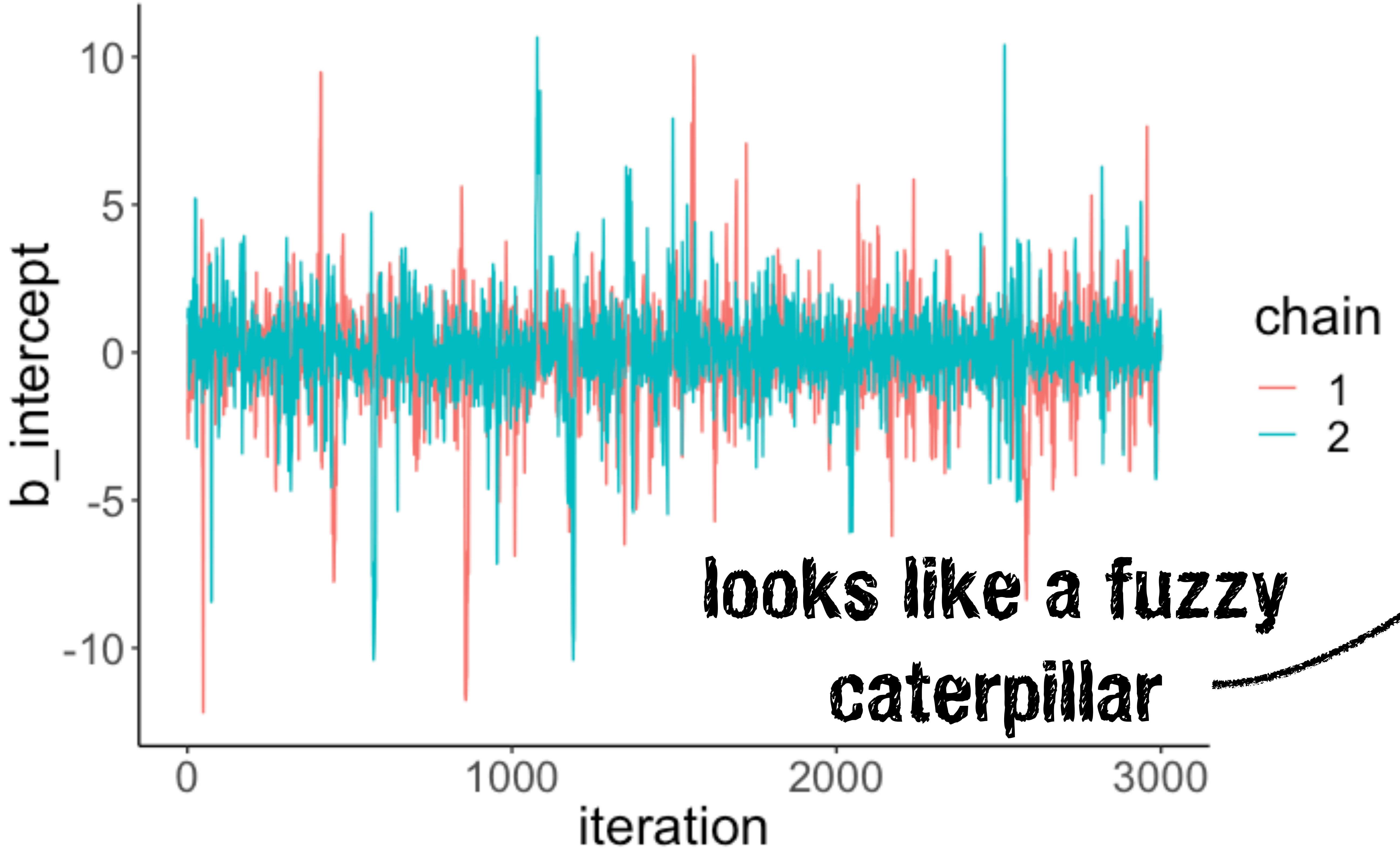
Having somewhat informative priors fixes things

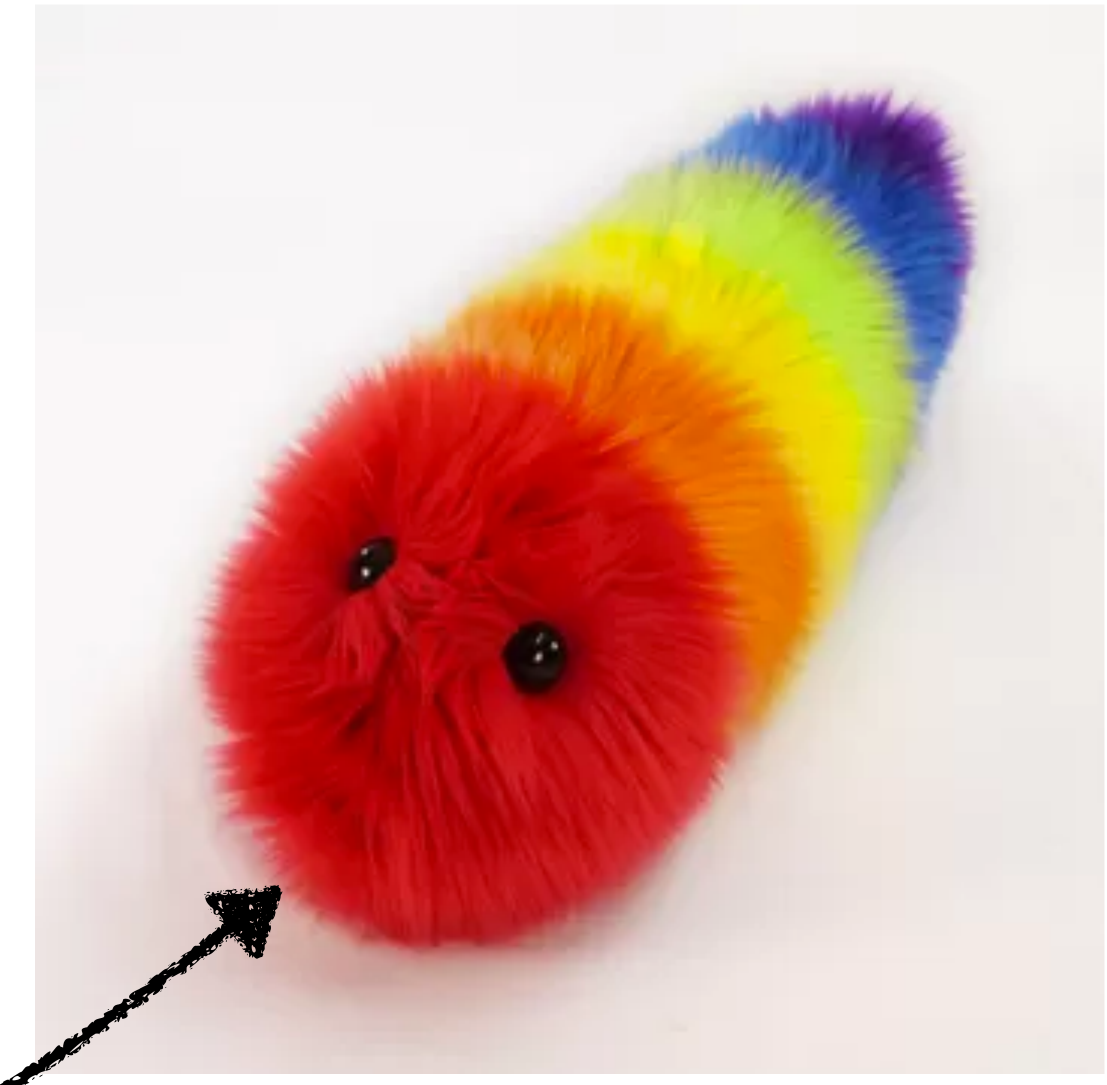
```r
fit.brm_right = brm(data = df.data,
                    family = gaussian,
                    formula = y ~ 1,
                    prior = c(prior(normal(0, 10), class = Intercept), # more reasonable priors
                              prior(cauchy(0, 1), class = sigma)),
                    iter = 4000,
                    warmup = 1000,
                    chains = 2,
                    seed = 1,
                    file = "cache/brm_right")
```

more reasonable priors

```
Family: gaussian
  Links: mu = identity; sigma = identity
Formula: y ~ 1
   Data: list(y = c(-1, 1)) (Number of observations: 2)
Samples: 2 chains, each with iter = 4000; warmup = 1000; thin = 1;
         total post-warmup samples = 6000

Population-Level Effects:
          Estimate Est.Error l-95% CI u-95% CI Eff.Sample Rhat
Intercept    -0.06      1.72    -3.78     3.27       1033 1.00

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI Eff.Sample Rhat
sigma     2.21      6.99     0.61     6.92       1006 1.00

Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample
is a crude measure of effective sample size, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Having somewhat informative priors fixes things





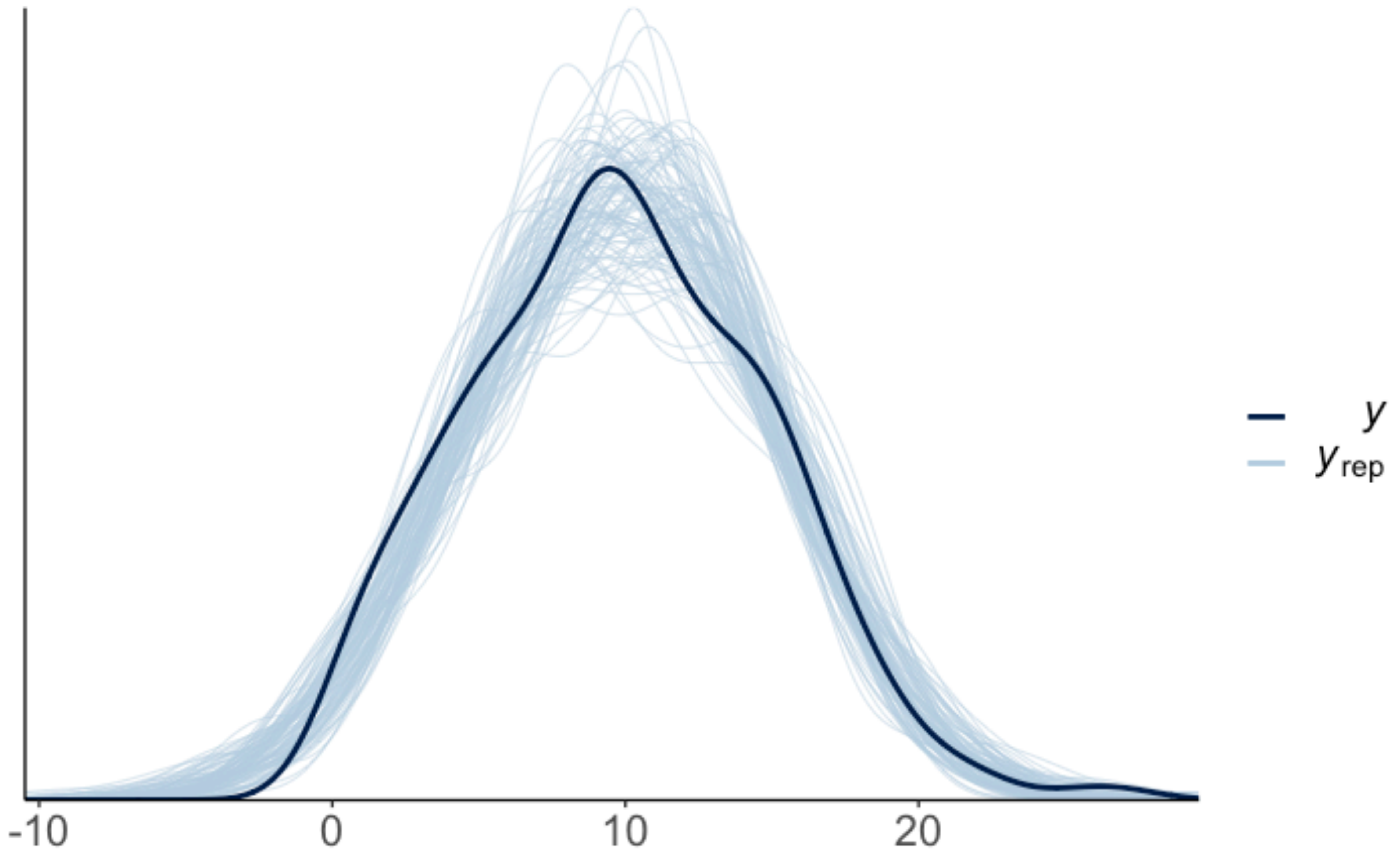
if things go wrong:

- set more informative priors
- run more warm-up samples
- adjust the sampling algorithm as suggested via the `control` argument

2. Visualize model predictions

Posterior predictive check

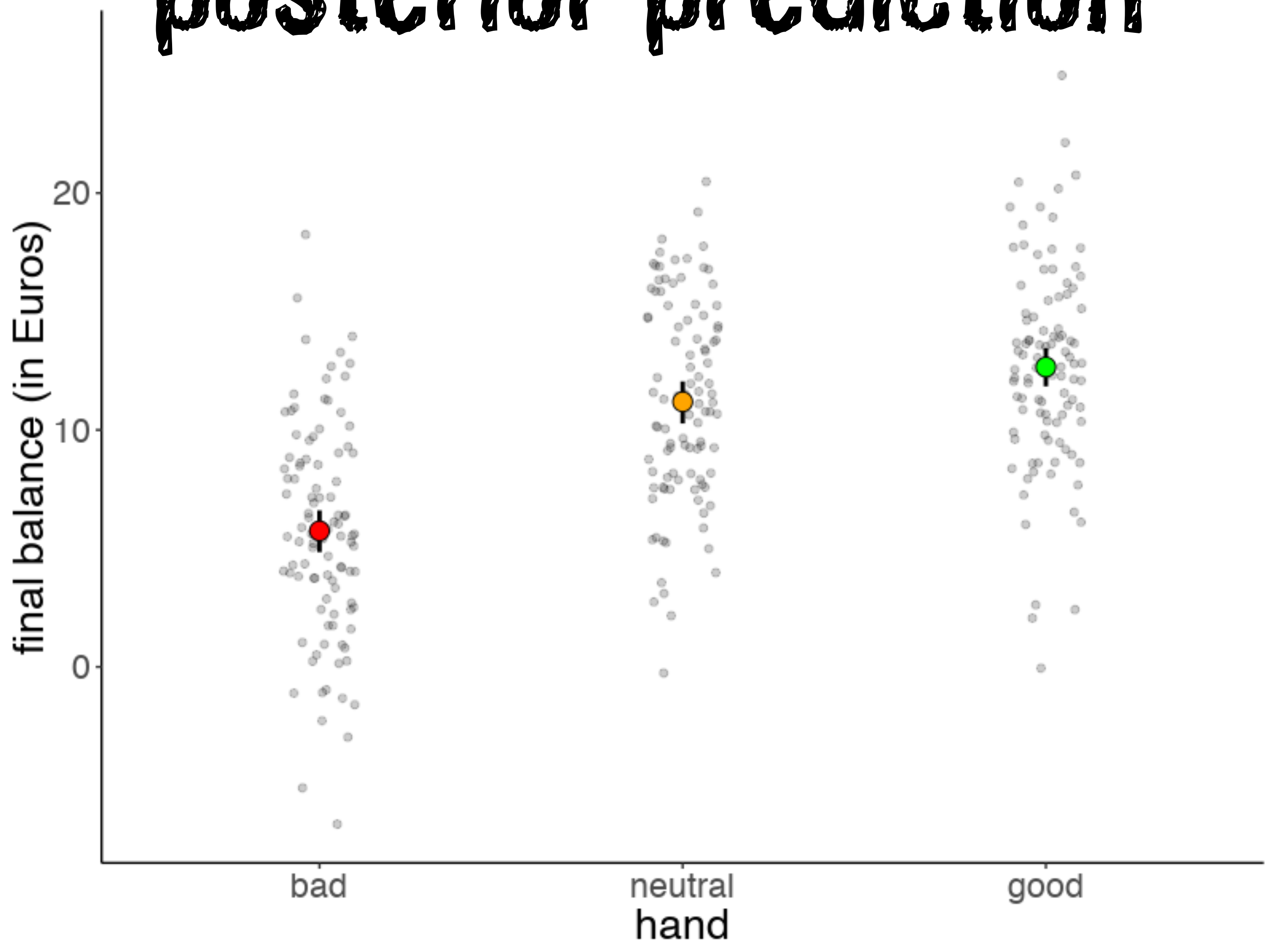
```
pp_check(fit.brm, nsamples = 100)
```



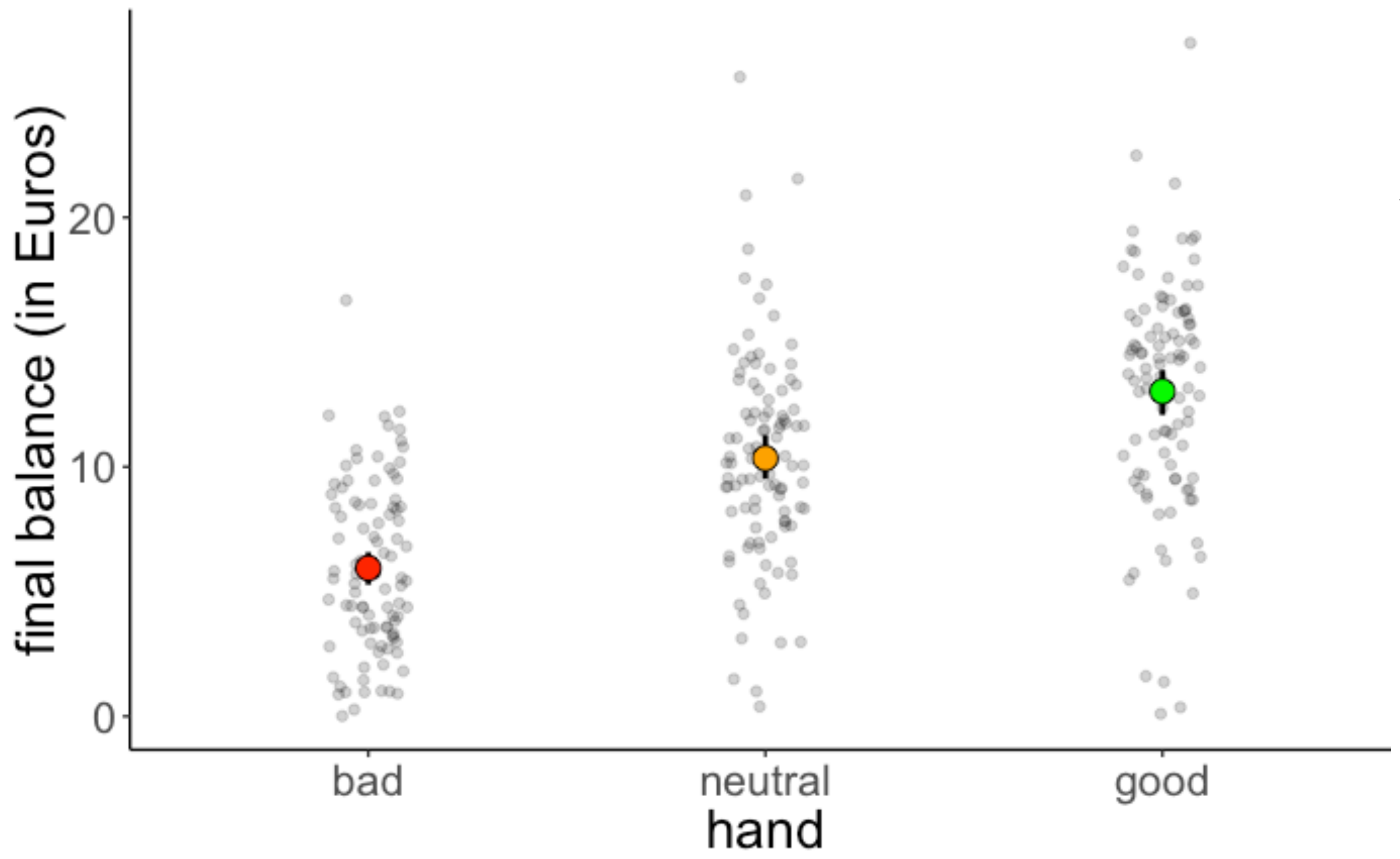
The model accurately captures the distribution of the response variable

Posterior predictive check

posterior prediction



original data

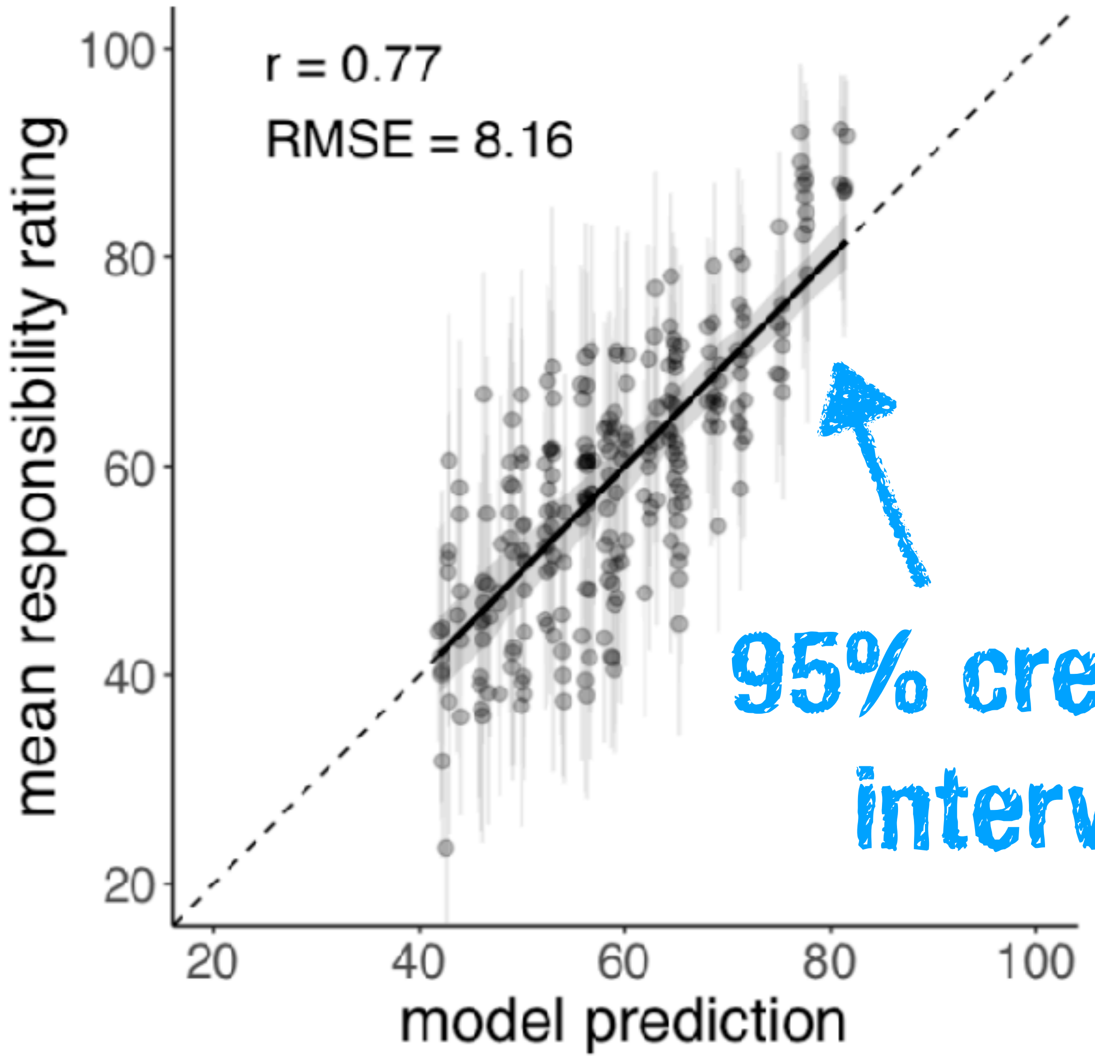


take a look at course notes for how to make these

Reporting results

Reporting results

Plots



95% credible interval

Tables

Table 1

Estimates of the mean, standard error, and 95% HDIs of the different predictors in the Bayesian mixed effects model. Note: n_causes = number of causes.

`responsibility ~ 1 + surprise + pivotality + n_causes + (1 + surprise + pivotality + n_causes | participant)`

term	estimate	std.error	lower 95% HDI	upper 95% HDI
intercept	59.94	3.25	54.70	65.22
surprise	21.68	4.57	14.17	29.23
pivotality	13.52	1.82	10.47	16.53
n_causes	-5.72	0.50	-6.55	-4.90

model formula

parameter estimates

can also include Random Effects

Text

We computed a Bayesian mixed effects model with random intercepts and slopes to predict participants' responsibility judgments (see Table 1). Figure 6b shows a scatter plot of the model predictions and participants' responsibility judgments for the full set of 170 scenarios (with 250 judgments). Overall, the model predicts participants' responsibility judgments well with r = .77 and RMSE = 8.16. Table 1 shows the estimates of the different predictors. As can be seen, none of the predictors' 95% HDIs overlap with 0.[1]

[1]For any statistical claim, we report the mean of the posterior distribution together with the 95% highest-density interval (HDI). All Bayesian models were written in Stan (Carpenter et al., 2017) and accessed with the brms package (Bürkner, 2017) in R (R Core Team, 2019).

Some more examples

Recipe for Bayesian analysis with `brms`

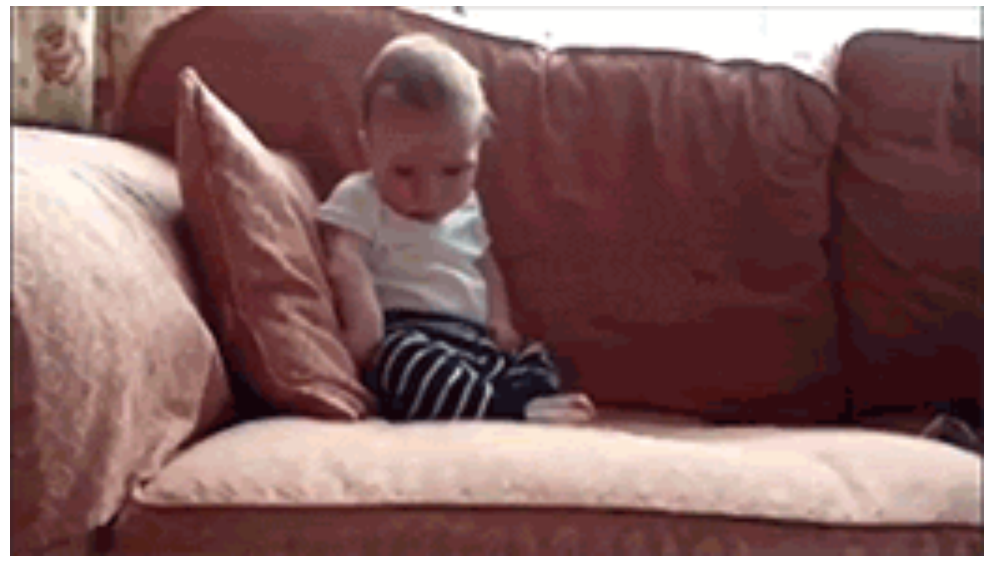
visualization is everywhere!

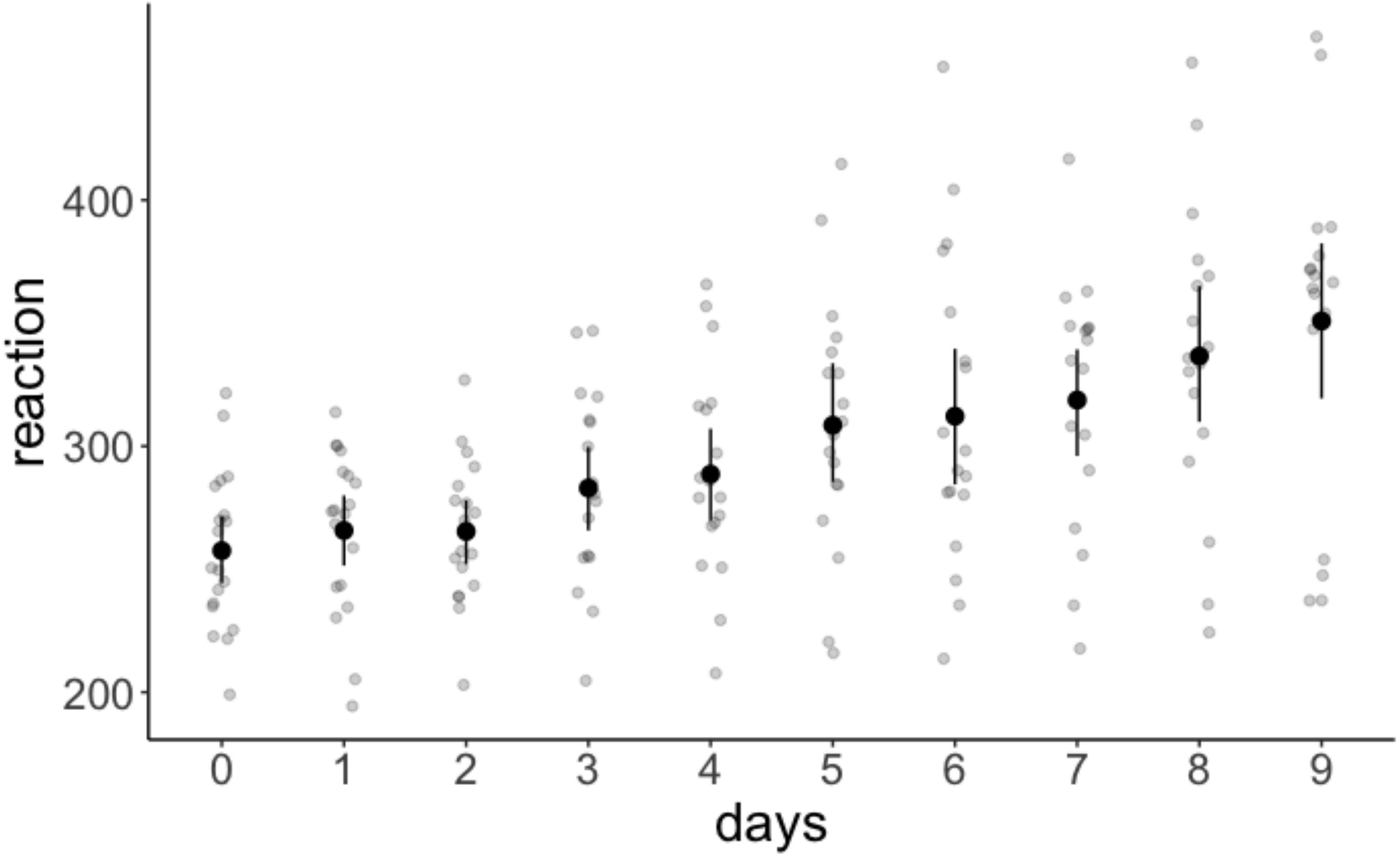
1. Visualize the data
2. Specify and fit the model
3. Model evaluation
   a) Did the inference work?
   b) Visualize model predictions
4. Interpret the model parameters
5. Test specific hypotheses
6. Report results

Sleep data

1. Visualize the data

Feeling sleepy?

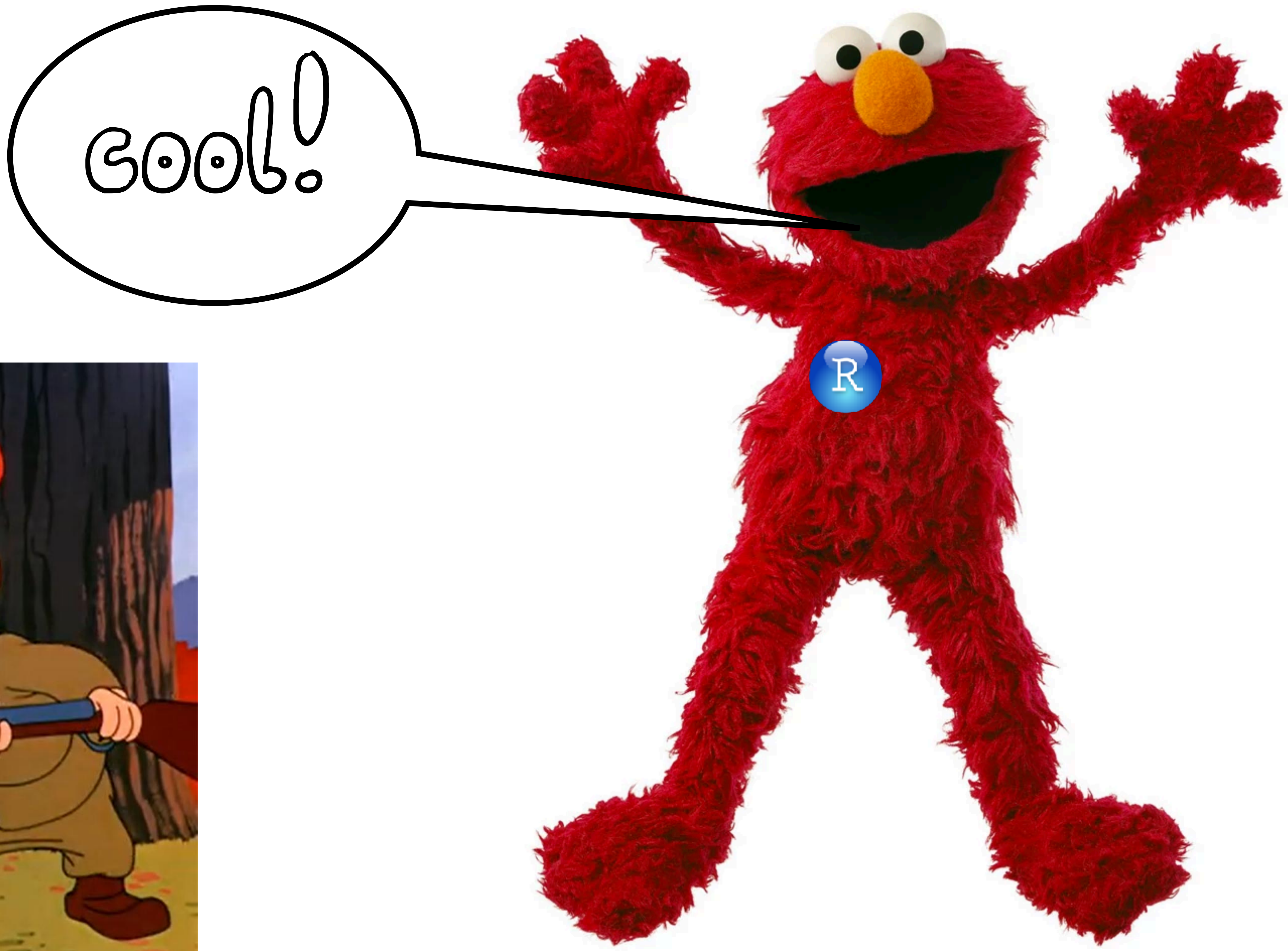




2. Specify and fit the model

1. Specify and fit the model

```r
fit.brm_sleep = brm(formula = reaction ~ 1 + days + (1 + days | subject),
                    data = df.sleep,
                    seed = 1,
                    file = "cache/brm_sleep")
```



3. Model evaluation

a) Did the inference work?

```r
fit.brm_sleep %>%
  summary()
```

Rhat of 1.00 is good!

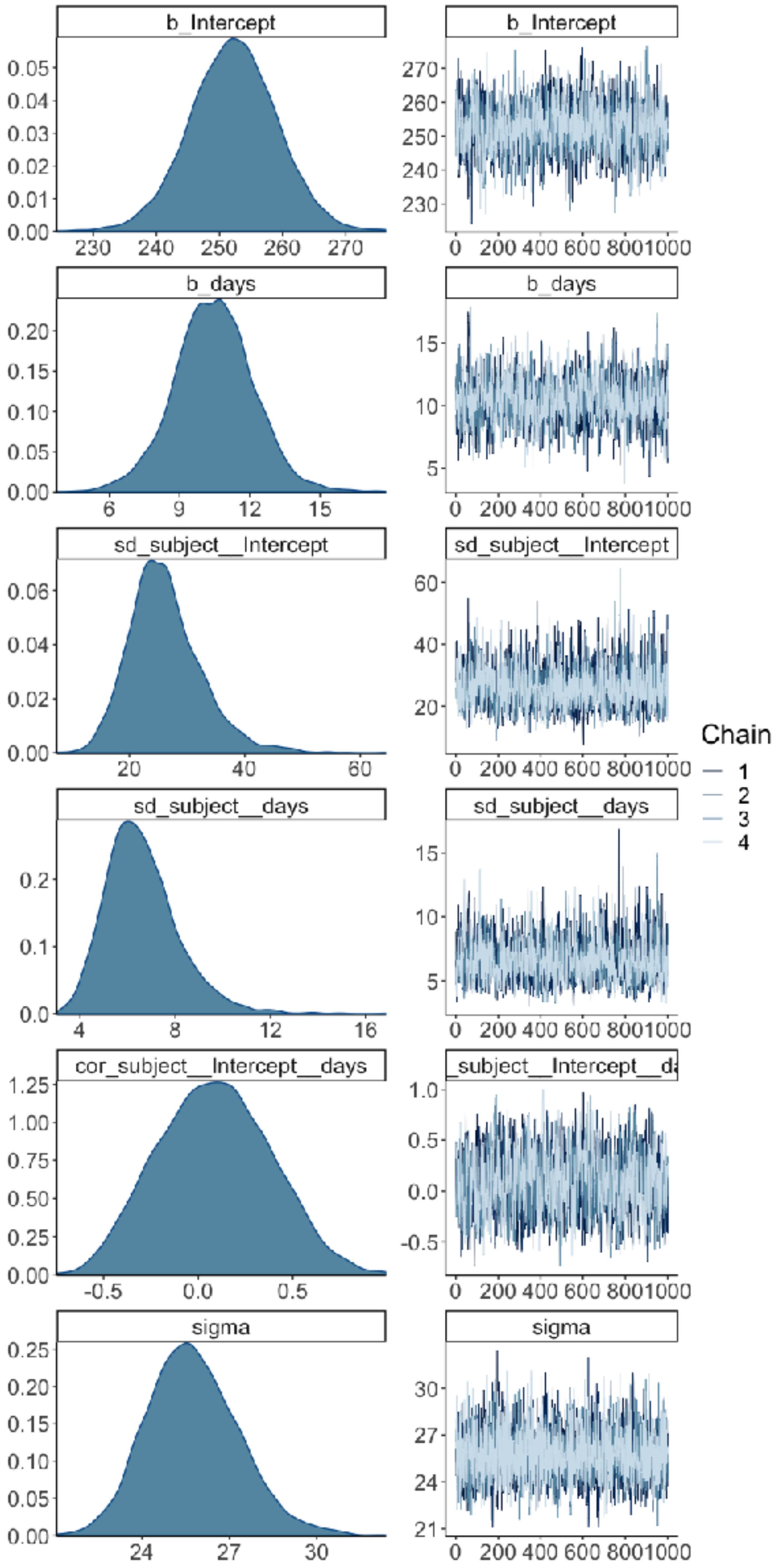
```
Family: gaussian
  Links: mu = identity; sigma = identity
Formula: reaction ~ 1 + days + (1 + days | subject)
   Data: df.sleep (Number of observations: 183)
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup samples = 4000

Group-Level Effects:
~subject (Number of levels: 20)
                    Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)          26.18      6.25    15.65    40.54 1.00     1879     2463
sd(days)                6.59      1.53     4.14    10.13 1.00     1145     1625
cor(Intercept,days)     0.09      0.29    -0.46     0.67 1.00      993     1526

Population-Level Effects:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept   252.18      6.86   238.47   265.42 1.00     1826     2766
days         10.46      1.69     7.13    13.78 1.00     1203     1782

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma    25.77      1.57    22.93    29.14 1.00     3864     2773

Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample
is a crude measure of effective sample size, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Roughly speaking, the effective sample size (**ESS**) of a quantity of interest captures how many independent draws contain the same amount of information as the dependent sample obtained by the MCMC algorithm.

https://mc-stan.org/misc/warnings.html

a) Did the inference work?

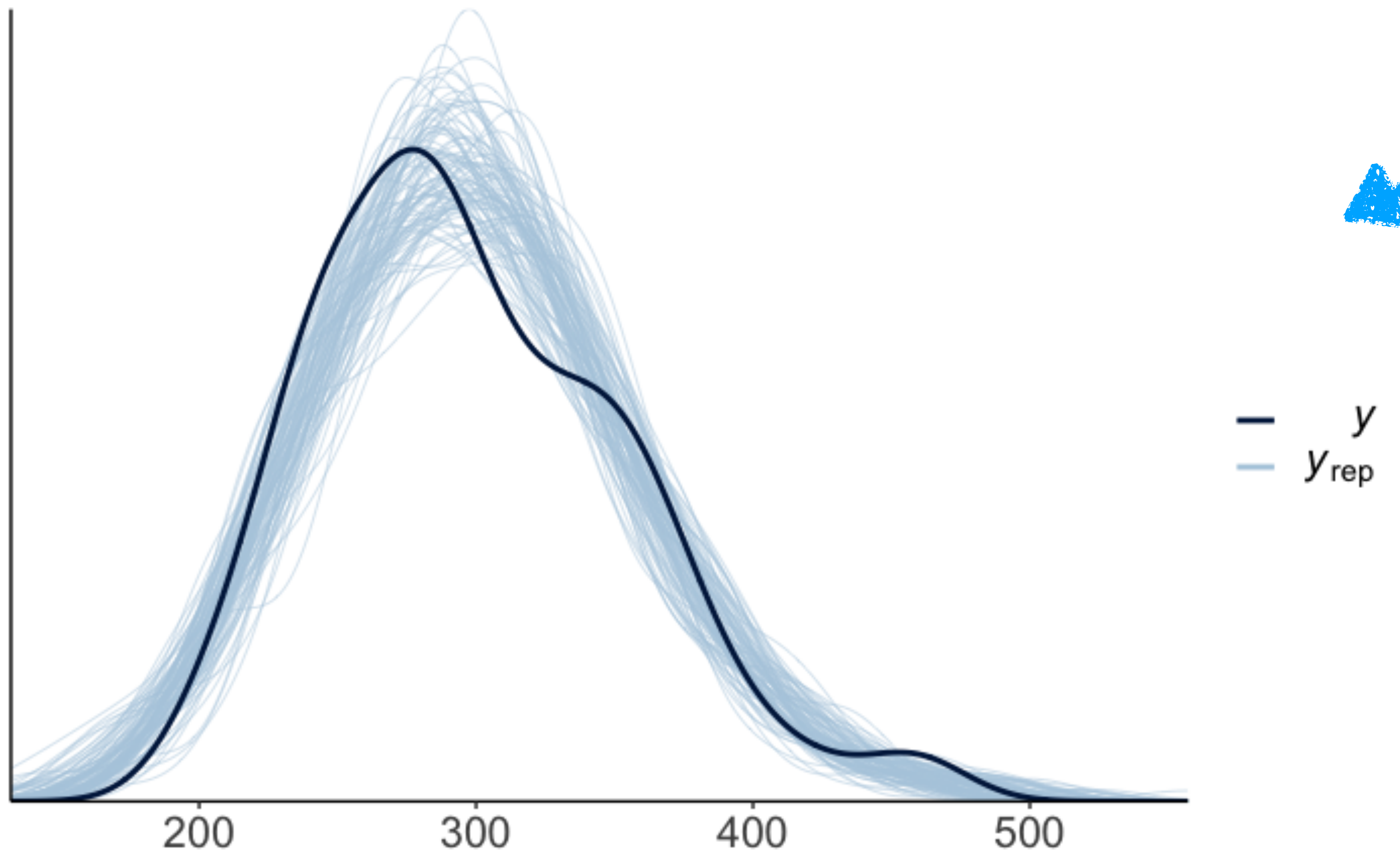
```r
fit.brm_sleep %>%
  plot(N = 6)
```



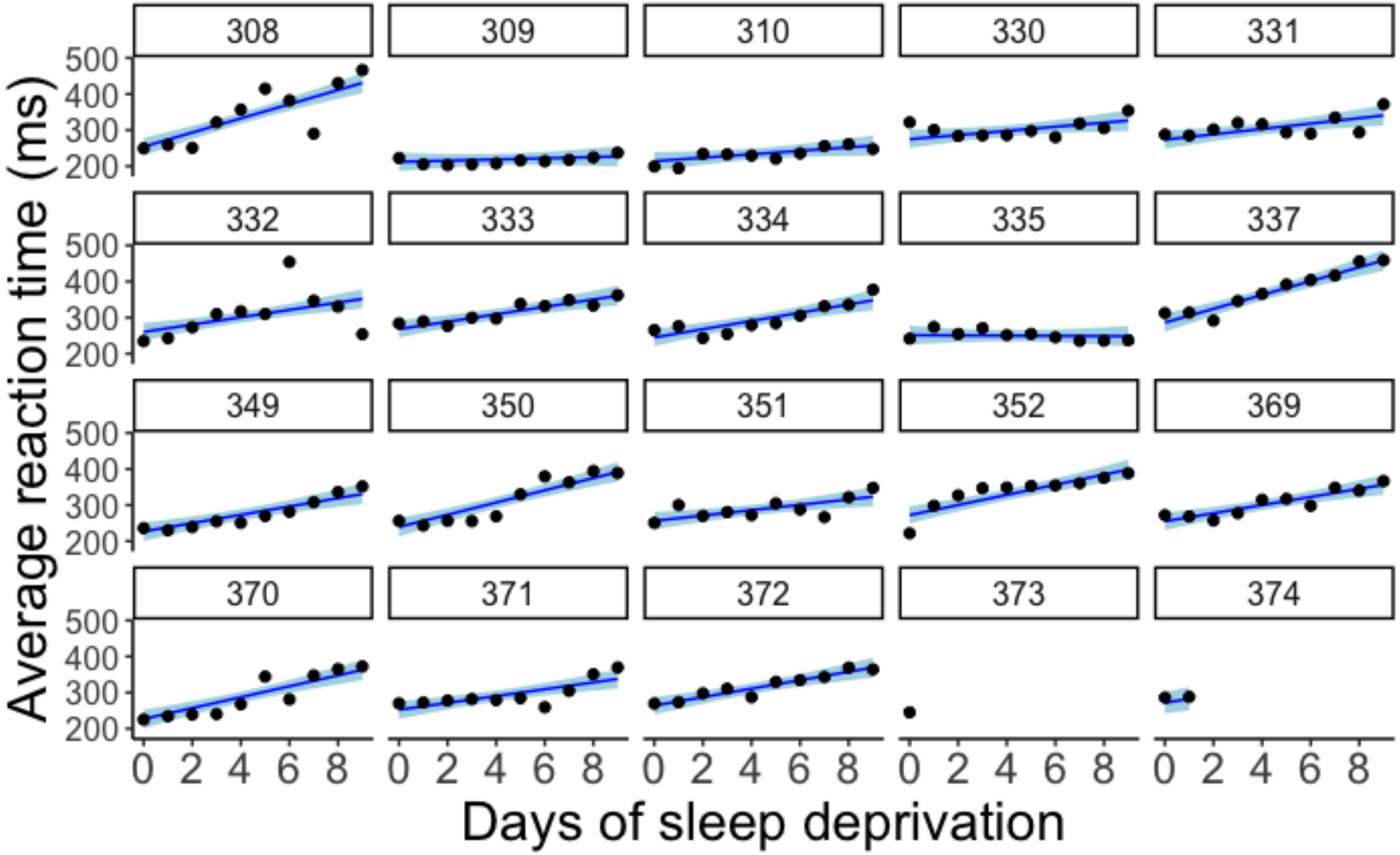
these look good!

also looks good!

```r
pp_check(fit.brm_sleep,
         nsamples = 100)
```

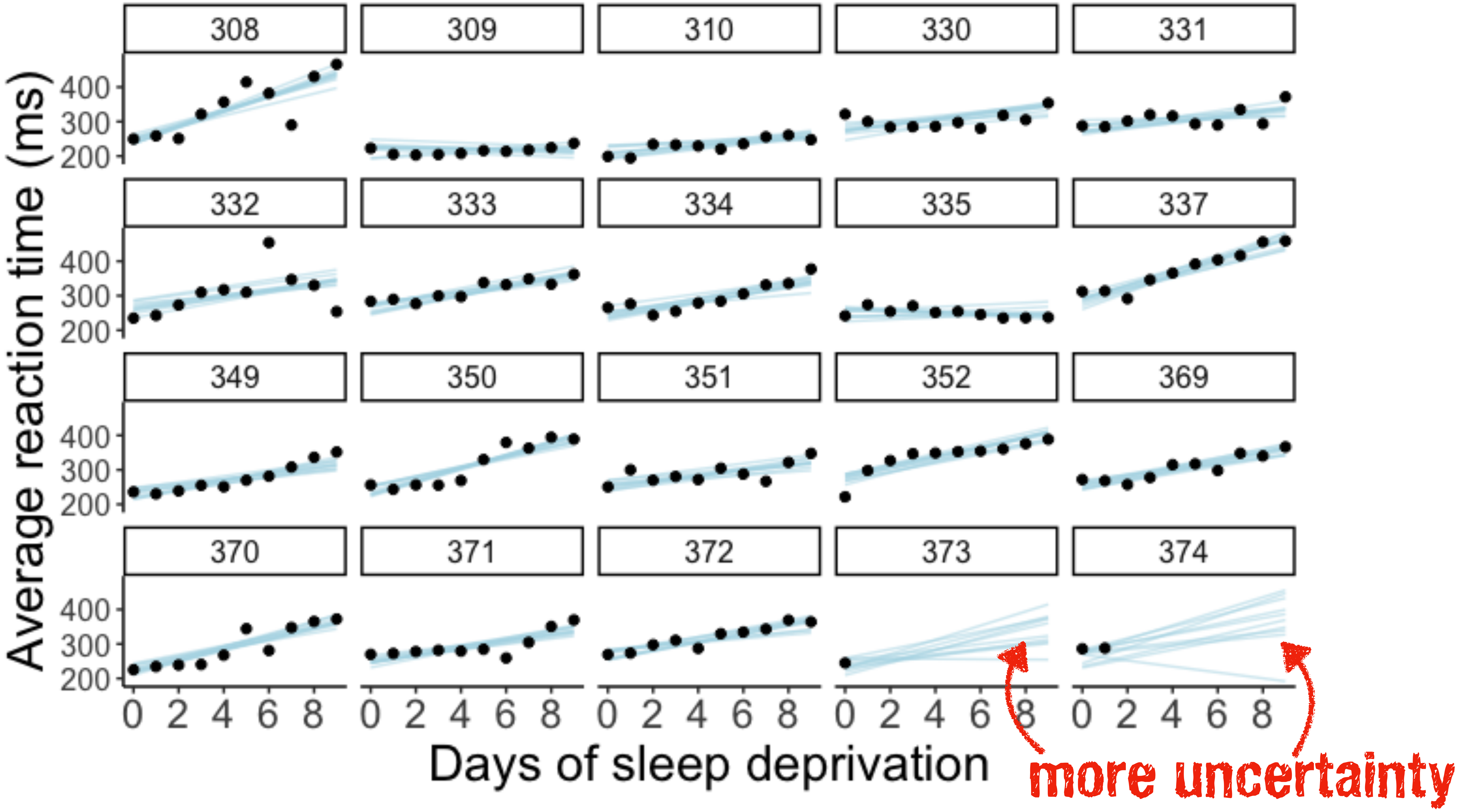


b) Visualize the model predictions



regression lines with 95% highest density intervals

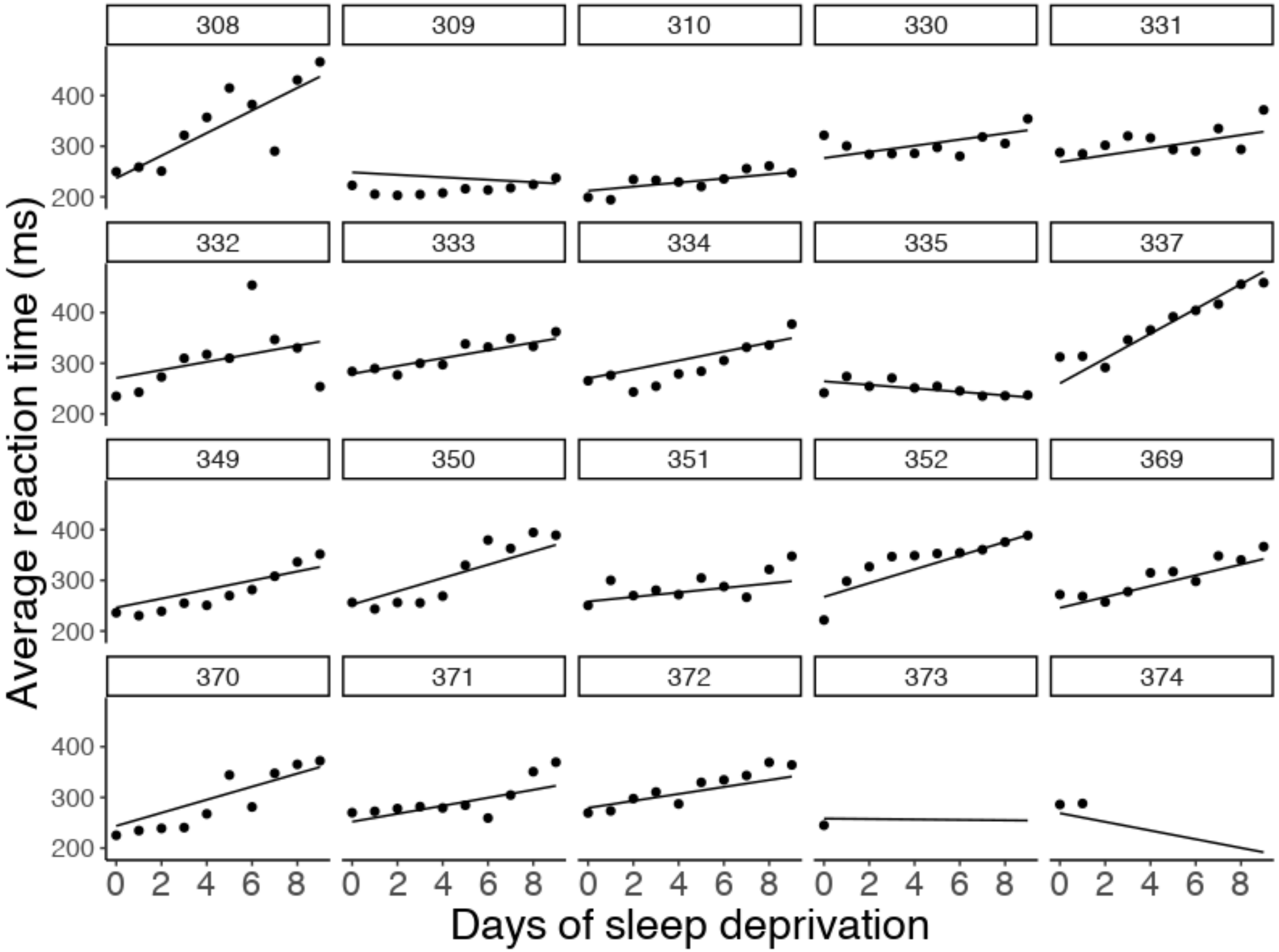
b) Visualize the model predictions



10 random samples from the posterior distribution

b) Visualize the model predictions

if you're feeling fancy



4. Interpret the model parameters

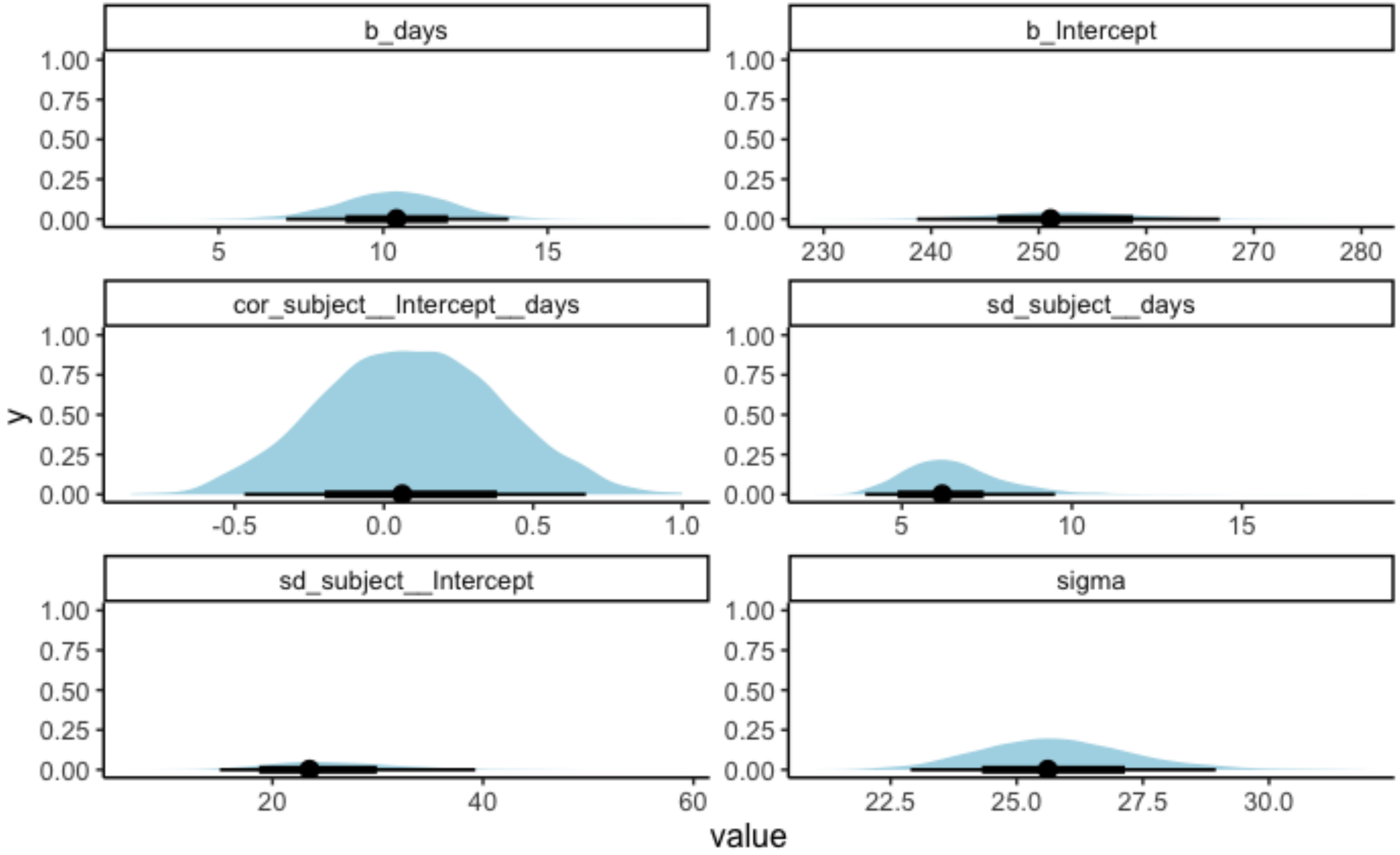
4. Interpret the model parameters

```r
fit.brm_sleep %>%
  tidy(conf.method = "HPDinterval")
```

95% highest density interval

effect	component	group	term	estimate	std.error	conf.low	conf.high
fixed	cond	NA	(Intercept)	252.39	7.00	238.69	266.82
fixed	cond	NA	days	10.34	1.72	7.05	13.81
ran_pars	cond	subject	sd__(Intercept)	26.14	6.37	15.00	39.27
ran_pars	cond	subject	sd__days	6.55	1.54	3.92	9.50
ran_pars	cond	subject	cor__(Intercept).days	0.09	0.30	-0.47	0.68
ran_pars	cond	Residual	sd__Observation	25.80	1.54	22.90	28.95

4. Interpret the model parameters



Posterior distribution for most parameters

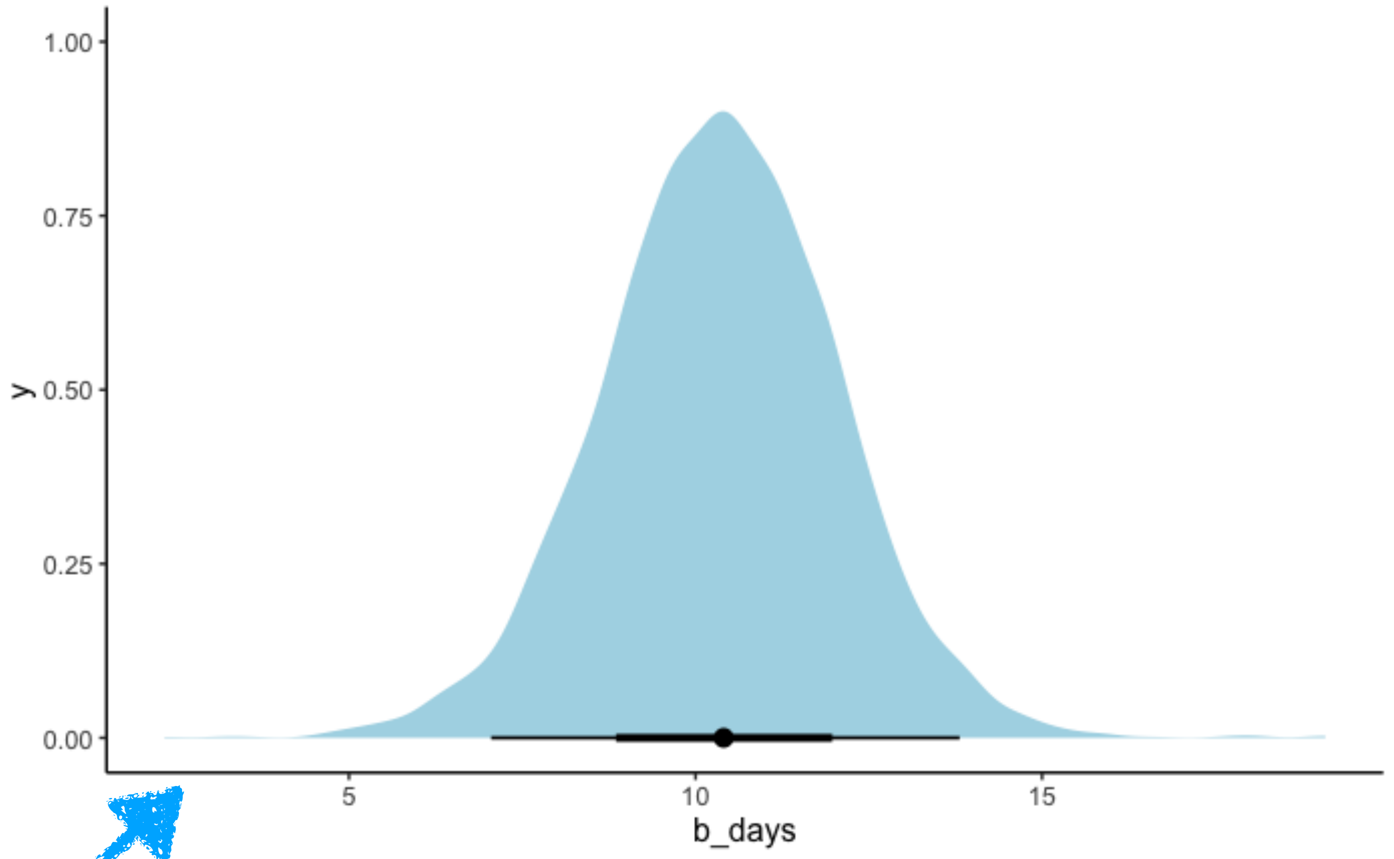
5. Test specific hypotheses

5. Test specific hypotheses

Did reaction times increase with the number of days of sleep deprivation?

```r
fit.brm_sleep %>%
  summary()
```

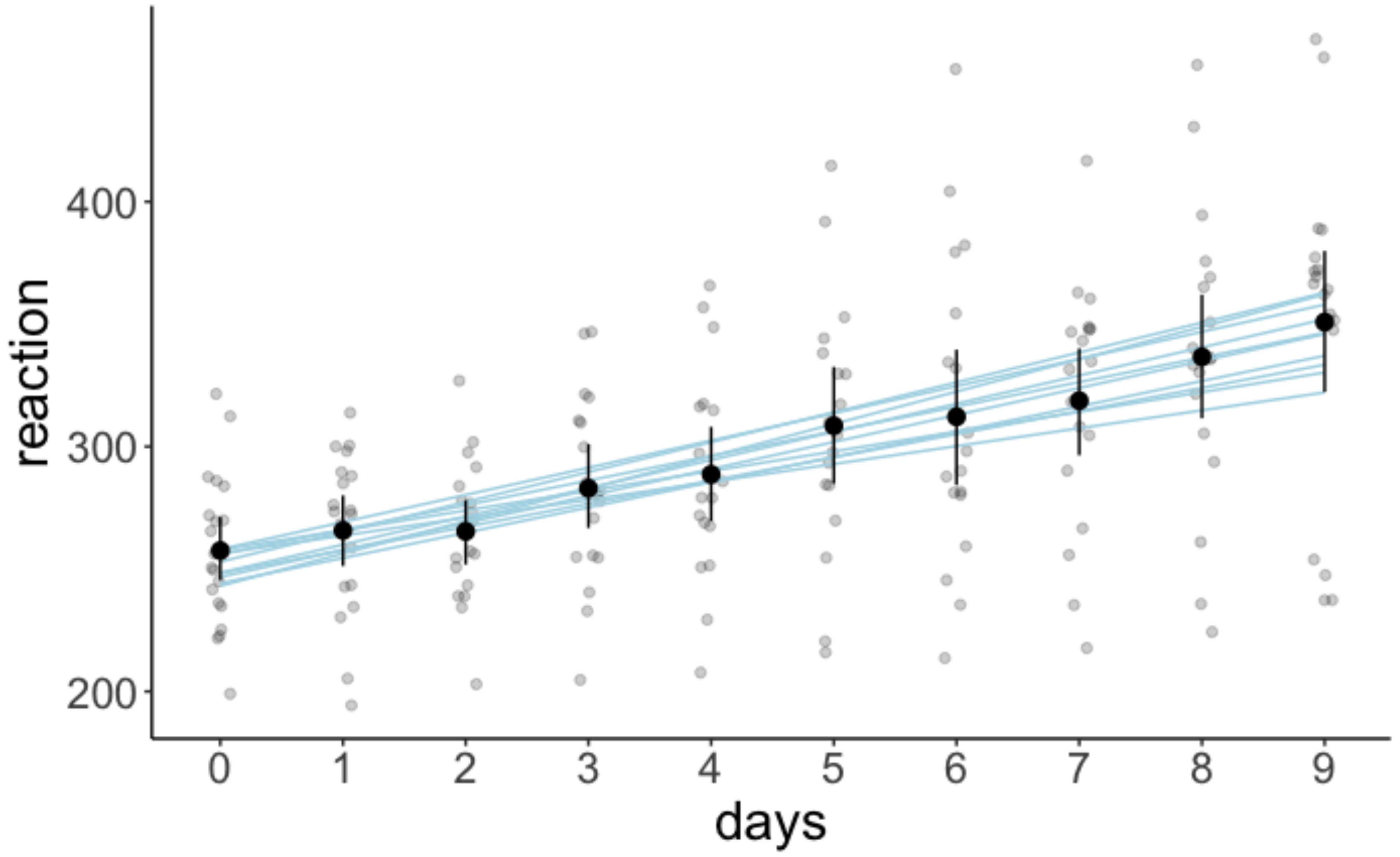
```
Family: gaussian
  Links: mu = identity; sigma = identity
Formula: reaction ~ 1 + days + (1 + days | subject)
   Data: df.sleep (Number of observations: 183)
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup samples = 4000

Group-Level Effects:
~subject (Number of levels: 20)
                    Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)          26.18      6.25    15.65    40.54 1.00     1879     2463
sd(days)                6.59      1.53     4.14    10.13 1.00     1145     1625
cor(Intercept,days)     0.09      0.29    -0.46     0.67 1.00      993     1526

Population-Level Effects:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept   252.18      6.86   238.47   265.42 1.00     1826     2766
days         10.46      1.69     7.13    13.78 1.00     1203     1782

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma    25.77      1.57    22.93    29.14 1.00     3864     2773

Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample
is a crude measure of effective sample size, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```



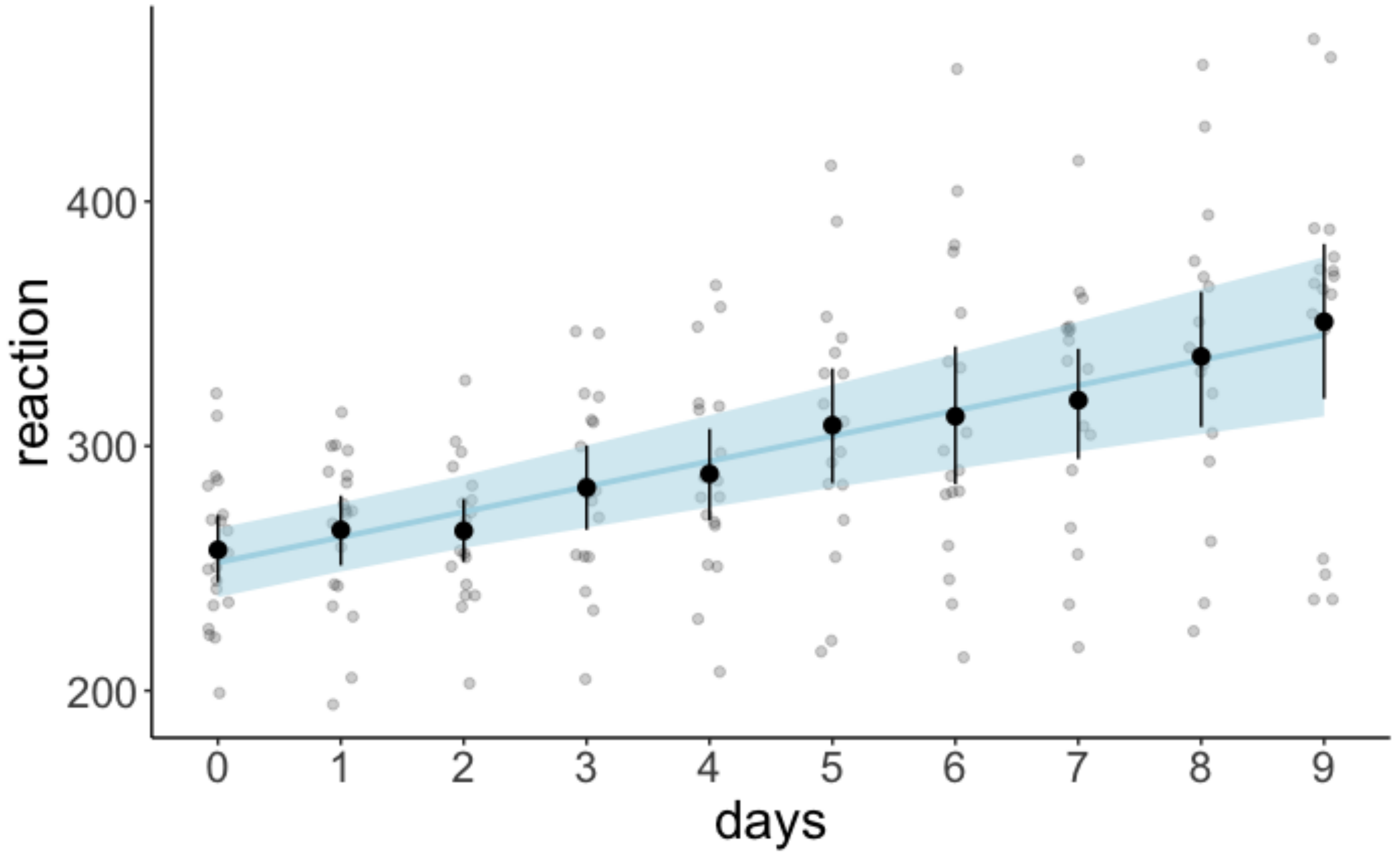
6. Report results

6. Report results

10 draws from the posterior



credible intervals



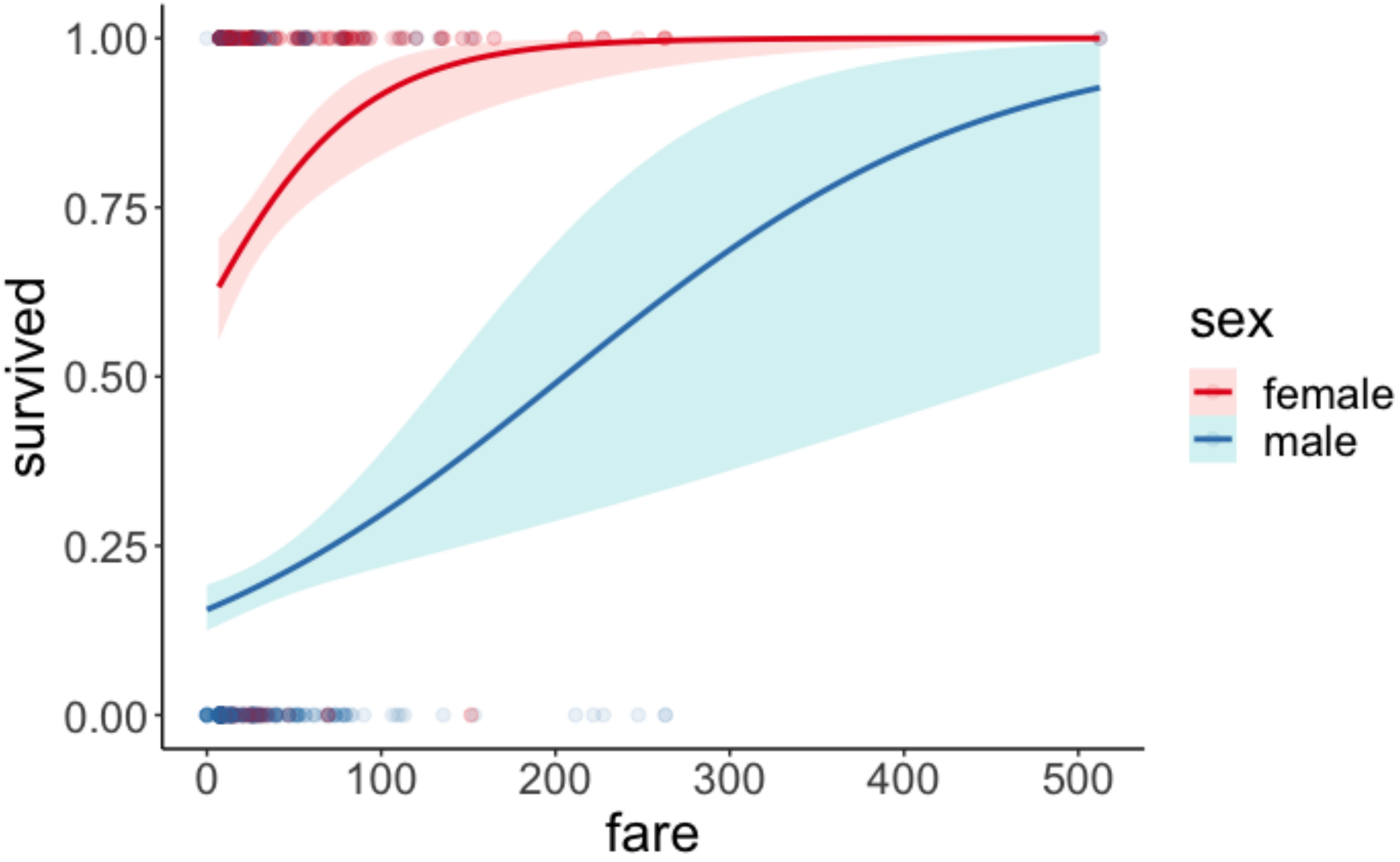
With each day of sleep deprivation, the reaction time increased by 10.5ms (95% HDI: 7.13, 13.78).

Recipe for Bayesian analysis with `brms`

1. Visualize the data
2. Specify and fit the model
3. Model evaluation
   a) Did the inference work?
   b) Visualize model predictions
4. Interpret the model parameters
5. Test specific hypotheses
6. Report results

Titanic data

1. Visualize the data


1.00
0.75
0.50
0.25
0.00
survived
0
100
200
300
400
500
fare
sex
female
male

2. Specify and fit the model

1. Specify and fit the model

```
fit.brm_titanic = brm(formula = survived ~ 1 + fare * sex,
                      family = "bernoulli",
                      data = df.titanic,
                      file = "cache/brm_titanic",
                      seed = 1)
```

just need to
change the family

3. Model evaluation

a) Did the inference work?

```
fit.brm_titanic %>%
  summary()
```

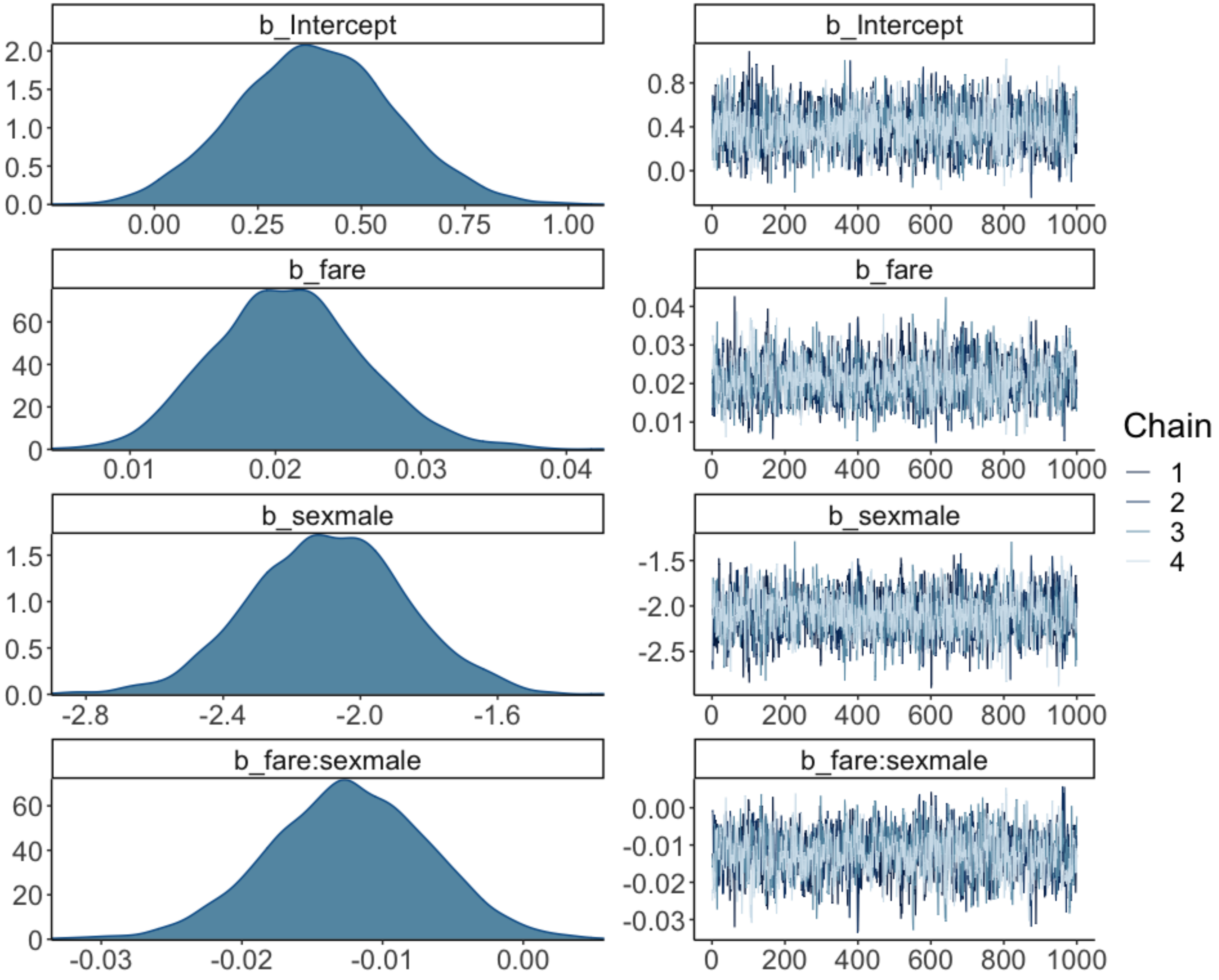
```
 Family: bernoulli
  Links: mu = logit
Formula: survived ~ 1 + fare * sex
   Data: df.titanic (Number of observations: 891)
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup samples = 4000

Population-Level Effects:
             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept        0.39      0.19     0.03     0.76 1.00     2010     2625
fare             0.02      0.01     0.01     0.03 1.00     1545     2124
sexmale         -2.09      0.23    -2.54    -1.65 1.00     1754     1984
fare:sexmale    -0.01      0.01    -0.02    -0.00 1.00     1479     2041

Samples were drawn using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

looks good

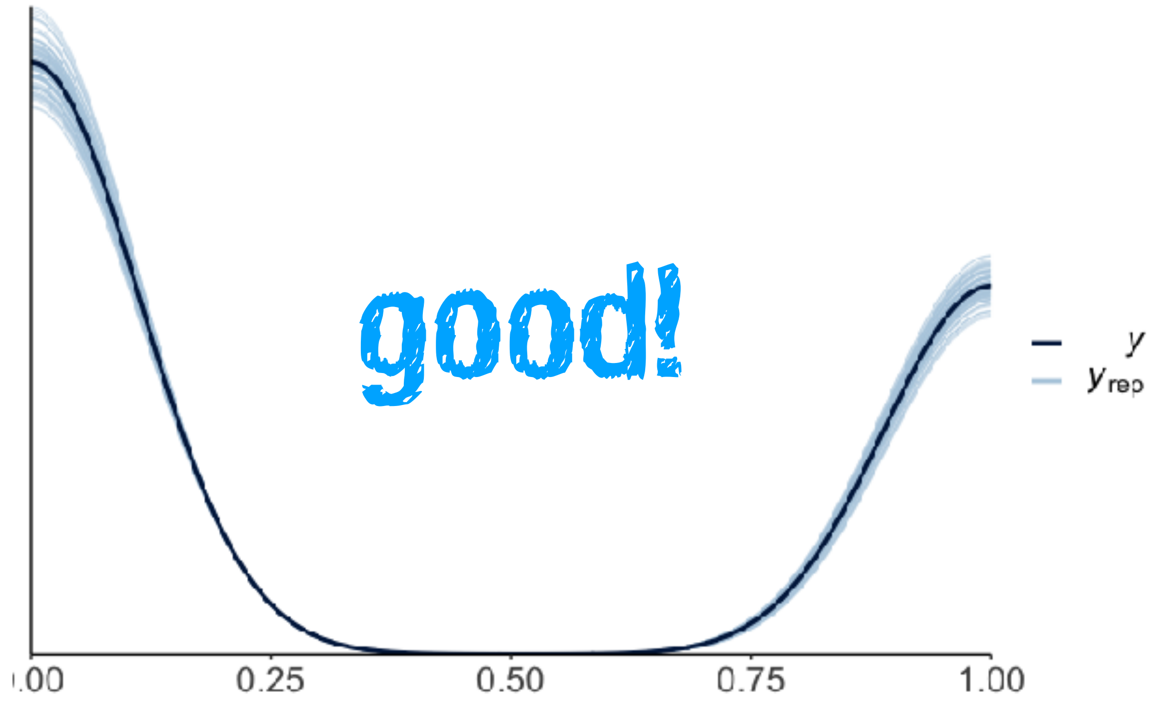
https://mc-stan.org/misc/warnings.html

a) Did the inference work?

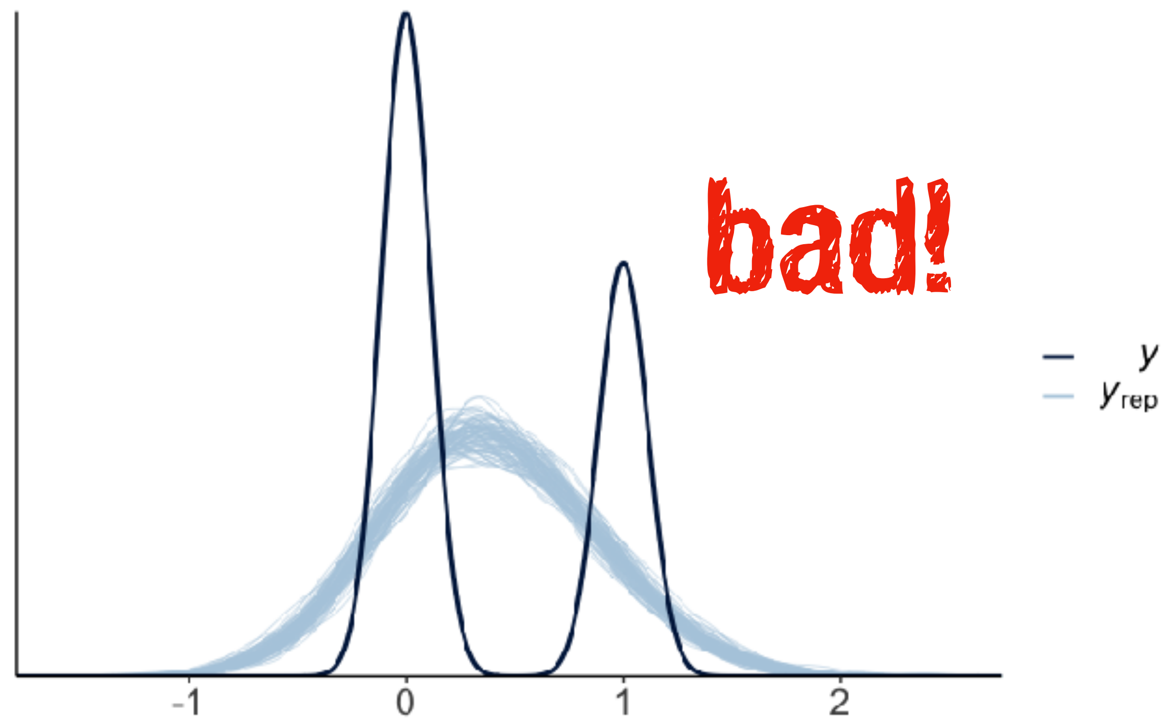
```
fit.brm_titanic %>%
  plot()
```



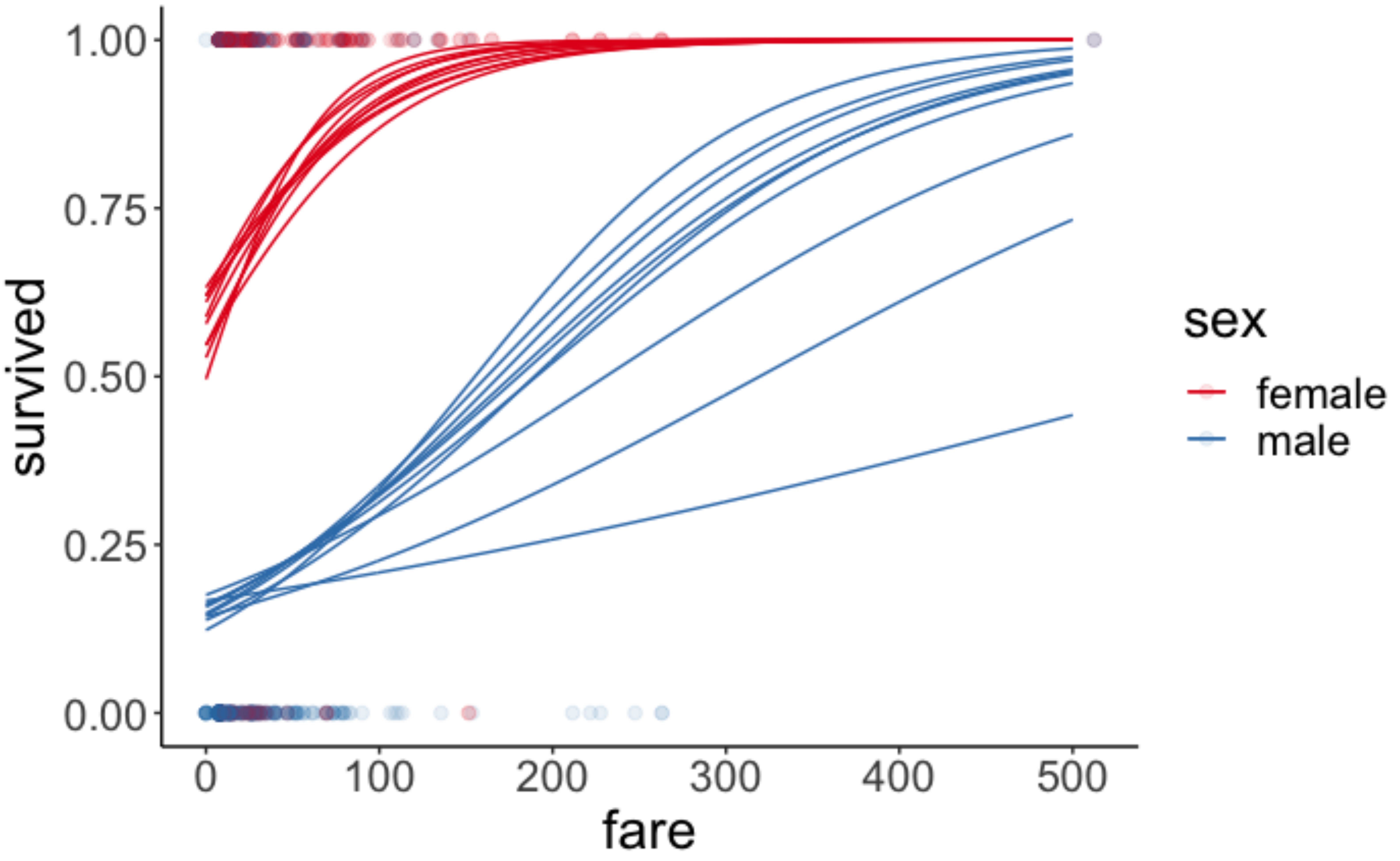

```
pp_check(fit.brm_titanic,
         nsamples = 100)
```



model with Gaussian family



b) Visualize the model predictions



4. Interpret the model parameters

4. Interpret the model parameters

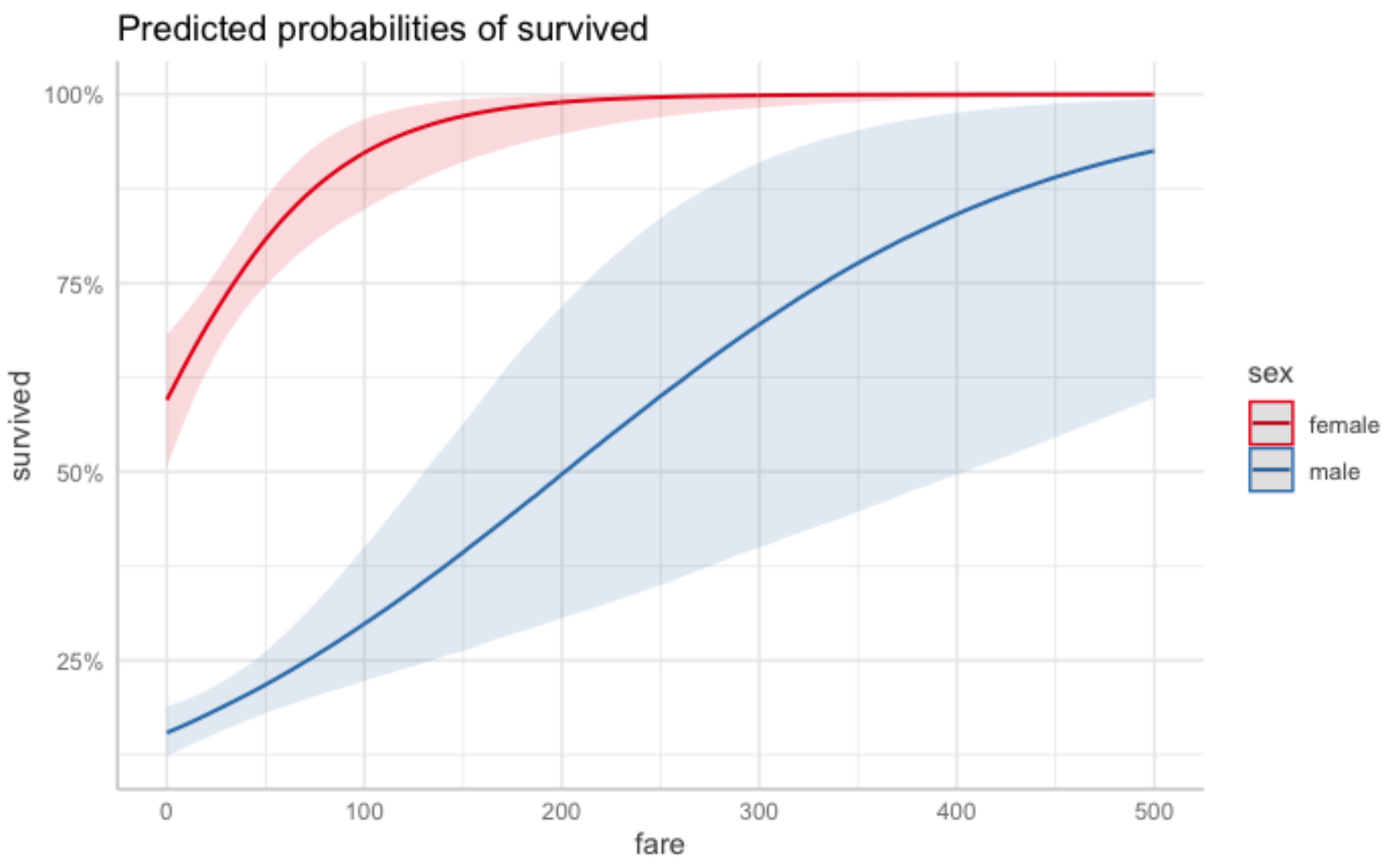
```
 Family: bernoulli
  Links: mu = logit
Formula: survived ~ 1 + fare * sex
   Data: df.titanic (Number of observations: 891)
Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup samples = 4000

Population-Level Effects:
             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept        0.39      0.19     0.03     0.76 1.00     2010     2625
fare             0.02      0.01     0.01     0.03 1.00     1545     2124
sexmale         -2.09      0.23    -2.54    -1.65 1.00     1754     1984
fare:sexmale    -0.01      0.01    -0.02    -0.00 1.00     1479     2041

Samples were drawn using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

log odds

4. Interpret the model parameters

```r
fit.brm_titanic %>%
  ggpredict(terms = c("fare [0:500]", "sex"))
```

```
# Predicted probabilities of survived
# x = fare

# sex = female

  x | Predicted |       95% CI
------------------------------
  0 |      0.60 | [0.51, 0.68]
 83 |      0.89 | [0.82, 0.95]
167 |      0.98 | [0.93, 1.00]
250 |      1.00 | [0.97, 1.00]
333 |      1.00 | [0.99, 1.00]
500 |      1.00 | [1.00, 1.00]

# sex = male

  x | Predicted |       95% CI
------------------------------
  0 |      0.15 | [0.12, 0.19]
 83 |      0.27 | [0.21, 0.35]
167 |      0.43 | [0.28, 0.62]
250 |      0.60 | [0.35, 0.84]
333 |      0.75 | [0.43, 0.94]
500 |      0.93 | [0.60, 0.99]
```



5. Test specific hypotheses

5. Test specific hypotheses

Were women more likely to survive than men?

```
fit.brm_titanic %>%
  emmeans(specs = pairwise ~ sex,
          type = "response")
```

$$\frac{\left(\frac{p_f}{1-p_f}\right)}{\left(\frac{p_m}{1-p_m}\right)}$$

```
NOTE: Results may be misleading due to involvement in interactions
$emmeans
 sex    response lower.HPD upper.HPD
 female    0.743      0.69     0.795
 male      0.194      0.16     0.225

Point estimate displayed: median
Results are back-transformed from the logit scale
HPD interval probability: 0.95

$contrasts
 contrast      odds.ratio lower.HPD upper.HPD
 female / male       12.1      8.39      16.6

Point estimate displayed: median
Results are back-transformed from the log odds ratio scale
HPD interval probability: 0.95
```

5. Test specific hypotheses

Was the effect of fare on survival different for men vs women?

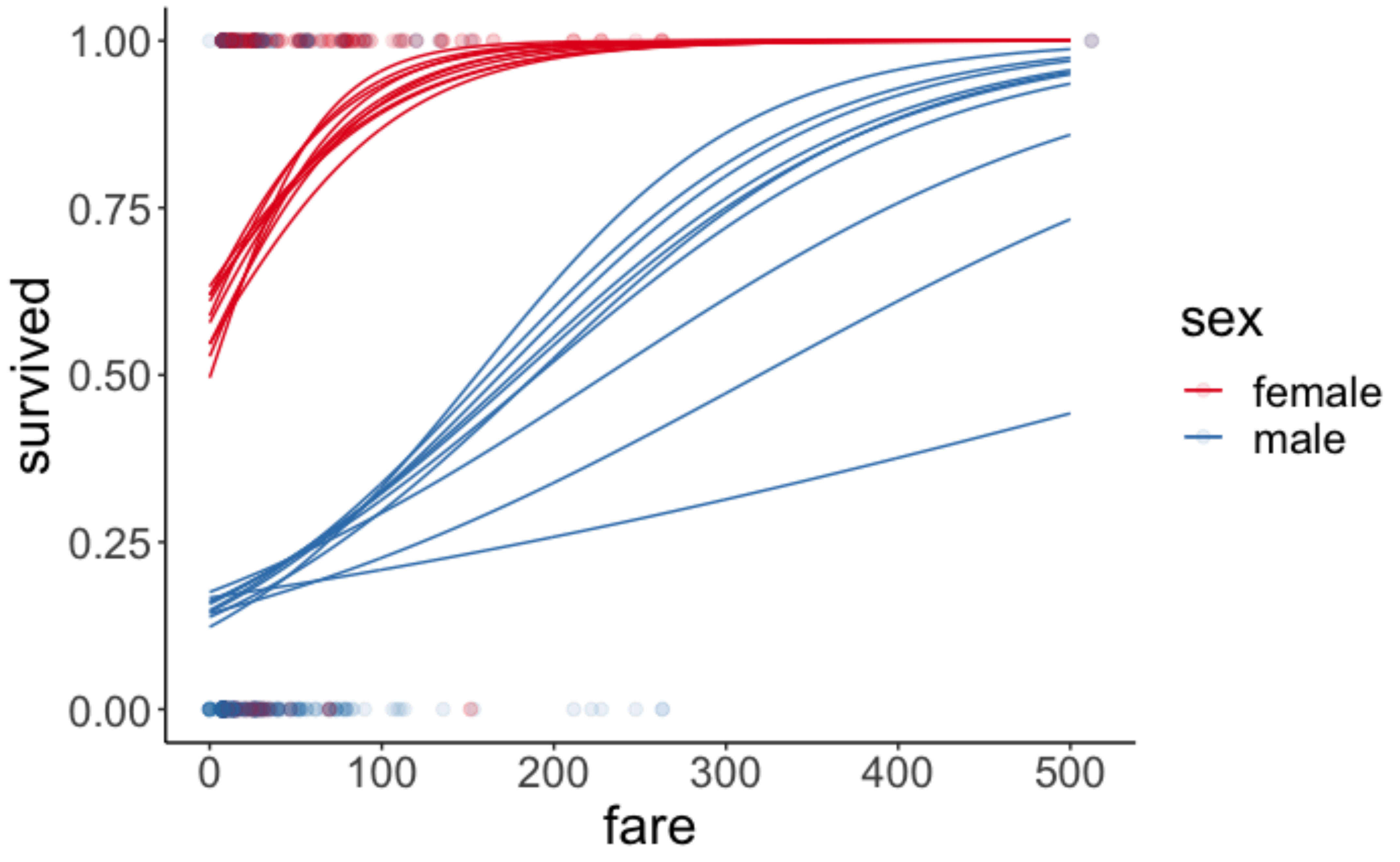
```r
fit.brm_titanic %>%
  emtrends(specs = pairwise ~ sex,
           var = "fare")
```

```
$emtrends
 sex      fare.trend lower.HPD upper.HPD
 female      0.02083   0.01129    0.0316
 male        0.00845   0.00385    0.0135

Point estimate displayed: median
HPD interval probability: 0.95

$contrasts
 contrast        estimate lower.HPD upper.HPD
 female - male     0.0124  0.000884    0.0232

Point estimate displayed: median
HPD interval probability: 0.95
```

the chance of survival increased more with fare for female than male passengers

6. Report results

6. Report results



Female passengers were more likely to survive (74.3%) than male passengers (19.4%). The estimated odds ratio of survival for female vs. male passengers was 12.1 [8.4, 16.6].

The chance of survival increased more with fare for female compared to male passengers. The difference in slopes on the log odds scale was 0.01 [0, 0.02].

Plan for today

- Quick recap
- Doing Bayesian data analysis **with BRMS**
  - Testing hypotheses
  - Model evaluation
  - Reporting results
- Some more examples
  - Sleep data
  - Titanic data
- Going beyond

Feedback N

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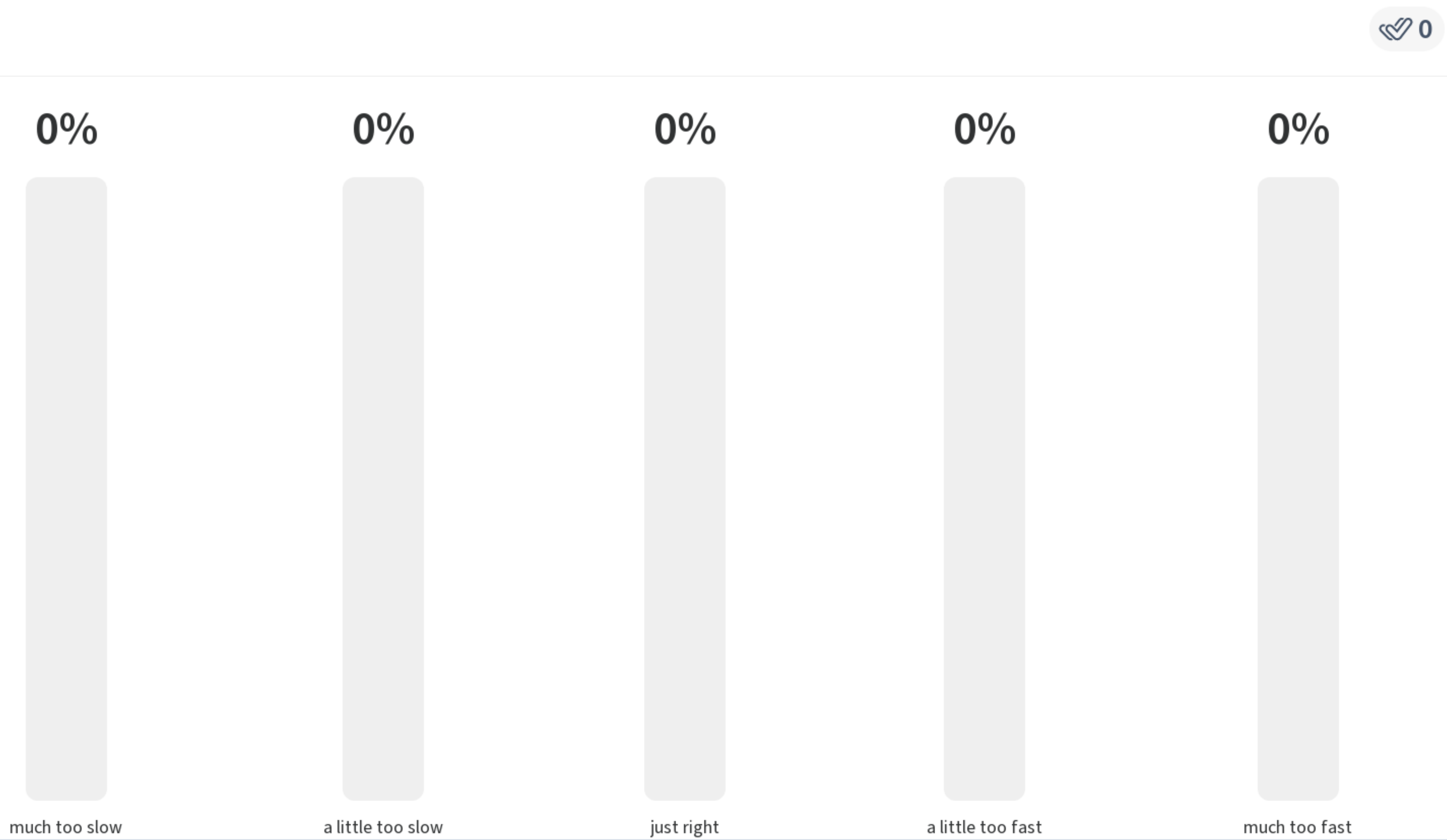
0
0%
0%
0%
0%
0%
much too slow
a little too slow
just right
a little too fast
much too fast

Thanks